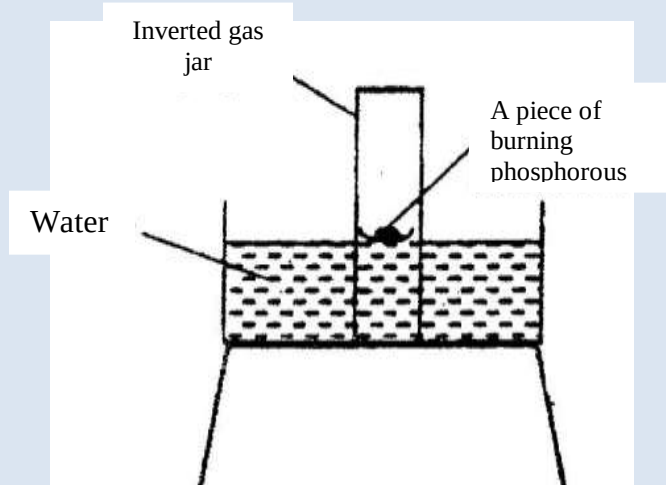


KCSE 2006 CHEM PP1 QUESTIONS

1. (a) What is meant by isomerism? (1mk)
(b) Draw and name two isomers of butane. (2 mks)
2. The diagram below represent a set-up that was used to show that part of air is used during burning.



- a) Given that phosphorus used was in excess, draw a diagram of the set-up at the end of the experiment (when there was no further observable change). (1mk)
- b) Suggest one modification that should be made on the apparatus if the percentage of the air used is to be determined. (1mk)
3. 60cm^3 of oxygen gas diffused through a porous partition in 50 seconds. How long would it take 60cm^3 of sulphur (IV) oxide gas to diffuse through the same partition under the same conditions? (S= 32.0, O = 16.0) (3mks)
4. a) Complete the nuclear equation below. (1mk)
- $$\begin{array}{ccc} 37 & & 37 \\ \text{A} \dots\dots\dots & \text{B} \\ 18 & & 17 \end{array}$$
- (b) State one:
- (i) Use of radioisotopes in agriculture (1mk)
- (ii) Danger associated with exposure of human beings to radioisotopes (1 mk)

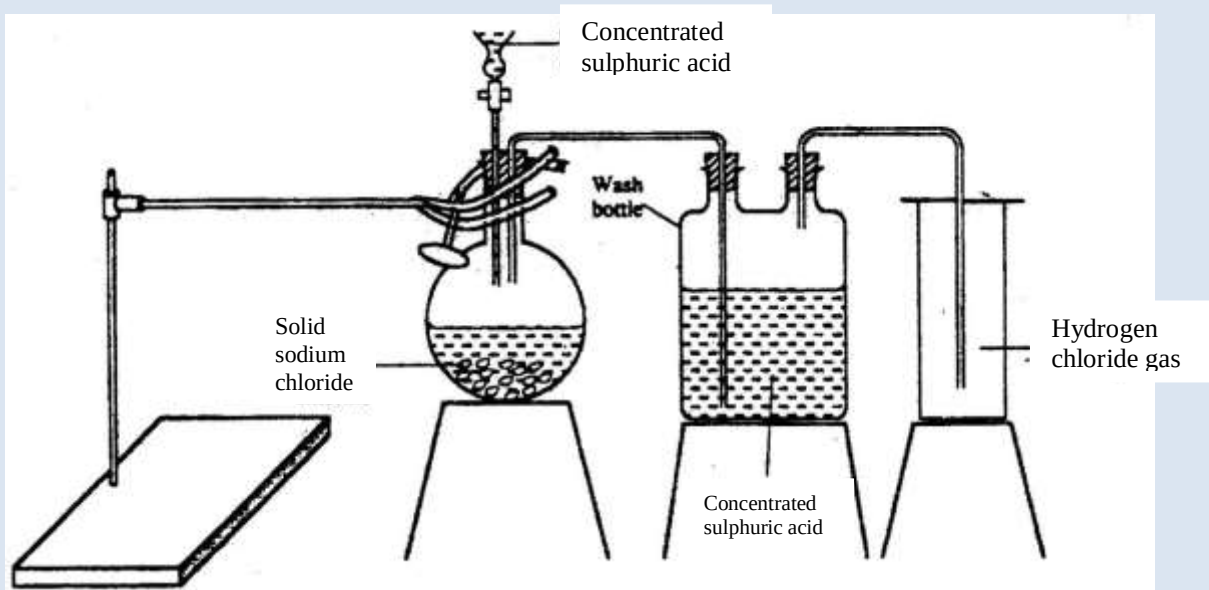
KCSE CHEM PAPER 1 PAST PAPERS MR CHEPKWONY

5. The atomic numbers of elements C and D are 19 and 9 respectively. State and explain the electrical conductivity of the compound CD in:
- (a) Solid state (1 ½ mks)
 - (b) aqueous state. (1 ½ mks)
6. In an experiment to study the properties of concentrated sulphuric acid, a mixture of the acid and wood charcoal was heated in a boiling tube.
- (a) write the equation of the reaction that took place in the boiling tube. (1mk)
 - (b) Using oxidation numbers, show that reduction and oxidation reactions took place in the boiling tube (2 mk)
7. A group of compounds called chlorofluorocarbons have a wide range of uses but they also have harmful effects on the environment.
State one:
- a) Use of chlorofluorocarbons (1 mk)
 - b) Harmful effect of chlorofluorocarbons on the environments. (1 mk)
8. When 94.5g of hydrated barium hydroxide, $\text{Ba}(\text{OH})_2 \cdot n\text{H}_2\text{O}$ were heated to constant mass, 51.3g of anhydrous barium hydroxide were obtained.
Determine the empirical formula of the hydrated barium hydroxide. (3 mks)
9. At 200C, NO_2 and N_2O_4 gases exist in equilibrium as shown in the equation below
- $$\underset{\text{Brown}}{2\text{NO}_{2(\text{g})}} \rightleftharpoons \underset{\text{pale yellow}}{\text{N}_2\text{O}_{4(\text{g})}}; \Delta H = -\text{ve}$$
- State and explain the observation that would be made when;
- a) A syringe containing the mixture at 200C is immersed in ice-cold water (1 ½ mks)
 - b) the volume of the gaseous mixture in a syringe is reduced. (1 ½ mks)
- 10 Name the process which takes place when:
- a) Solid carbon (IV) oxide (dry ice) changes directly into gas (1 mk)
 - b) A red litmus paper turns white when dropped into chlorine water. (1 mk)
 - c) Propane gas molecules are converted into a giant molecule (1 mk)
11. a) water from a town in Kenya is suspected to contain chloride ions but not sulphate ions.
Describe how the presence of the chloride ions in the water can be shown. (1 mk)

b) State one advantage of drinking hard water rather than soft water. (1 mk)

12. The table below shows the relative atomic masses and the percentage abundance of the isotopes L1 of element L.

13. The diagram below represents the set-up that was used to prepare and collect hydrogen chloride gas in the laboratory.



a) State the purpose of concentrated sulphuric acid in the wash bottle. (1 mk)

b) Write an equation for the reaction between dry hydrogen chloride gas and heated iron (1 mk)

c) Hydrogen chloride gas is dissolved in water to make hydrochloric acid. State one use of hydrochloric acid.

14. Below is a list of oxides.
 MgO , N_2O , K_2O , CaO , and Al_2O_3 .
Select:

a) A neutral oxide (1 mk)

b) A highly water soluble basic oxide (1 mk)

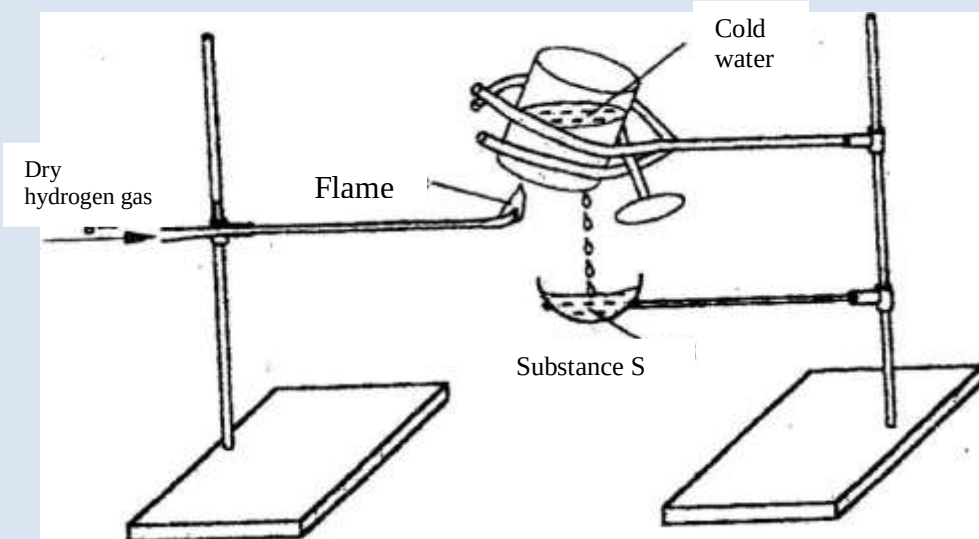
c) An oxide which can react with both sodium hydroxide solution and dilute hydrochloric acid. (1 mk)

15. Study the standard reduction potential given and answer the questions that follow.
(The letters are not the actual symbols of the elements).

	E° (volts)
$M^{2+}_{(aq)} + 2e \rightarrow M_{(s)}$	-0.76
$N^{2+}_{(aq)} + 2e \rightarrow N_{(s)}$	-2.37
$P^{+}_{(aq)} + e \rightarrow P_{(s)}$	+0.80
$Q^{2+}_{(aq)} + 2e \rightarrow Q_{(s)}$	-0.14

- a) The standard reduction potential for $Fe^{2+}_{(aq)}$ is -0.44 volts. Select the element which would best protect iron from rusting. (1 mk)
- b) Calculate the E° value for the cell represented as $M_{(s)} / M^{2+}_{(aq)} // P^{+}_{(aq)} / P_{(s)}$. (2mks)
- 16 When hydrogen sulphide gas was bubbled into an aqueous solution of iron (III) chloride, a yellow precipitate was deposited. (1mk)
- a) The standard reduction potential for $Fe^{2+}_{(aq)}$ is -0.44 volts. Select the element which would best protect iron from rusting. (1 mk)
- b) Write an equation for the reaction that took place. (1 mk)
- c) What type of reaction was undergone by hydrogen by hydrogen sulphide in this reaction? (1mk)
17. The first step in the industrial manufacture of nitric acid is the catalytic oxidation of ammonia gas.
- (a) What is the name of the catalyst used? (1 mk)
- (b) Write the equation for the catalytic oxidation of ammonia gas. (1mk)
- (c) Nitric acid is used to make ammonium nitrate, state two uses of ammonium nitrate. (1 mk)

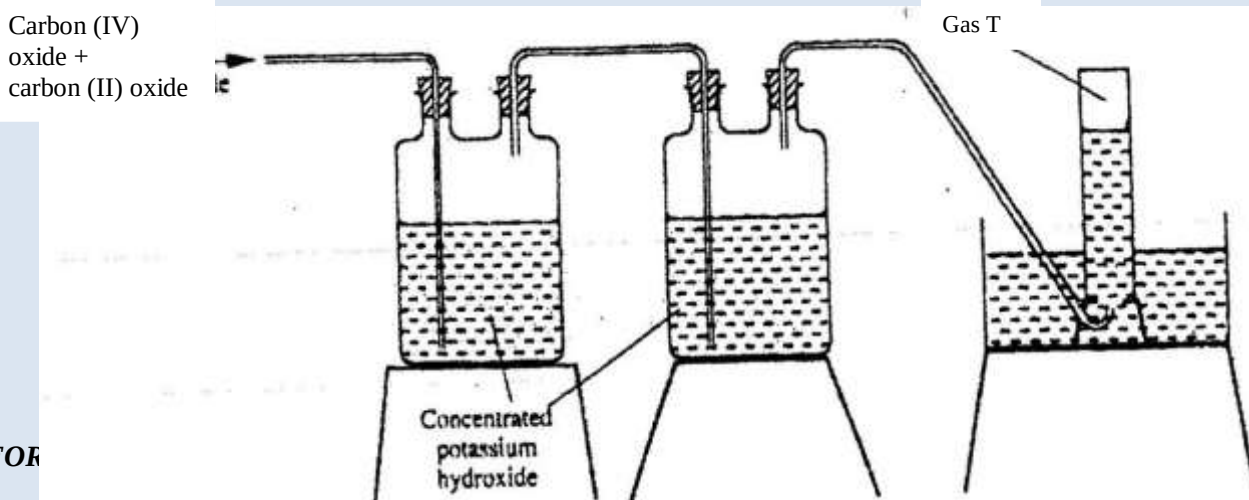
18. Study the diagram below and answer the question that follows.



Describe one chemical test that can be carried out to identify substance s. (2 mks)

19. a) starting from solid magnesium hydroxide. (1mk)
 b) Give one use of magnesium hydroxide. (1 mk)
20. a) Distinguish between a covalent bond and a co-ordinate bond (2 mks)
 b) Draw a diagram to show bonding in an ammonium ion. (1mk)
21. (a) Explain why the metals magnesium and aluminium are good conductors of electricity. (1 mk)
 b) Other than cost, give two reasons why aluminium is used for making electric cables while magnesium is not (2mks)

22. The diagram below represents part of a set – up used to prepare and collect gas T.



Concentrate
potassium
hydroxide

- a) Name two reagents that are reacted to produce both carbon (IV) oxide and carbon(II) oxide. (1mk)
- b) Write the equation for the reaction which takes place in the wash bottles. (1mk)
- c) Give a reason why carbon(II) oxide is not easily detected. (1mk)

23. Explain why the boiling point of ethanol is higher than that of hexane.
(Relative molecular mass of ethanol is 46 while that of hexane is 86)

24. a) Complete the table below to show the colour of the given indicator in

Indicator	Colour in	
	Acid solution	Basic solution
Methyl orange		Yellow
Phenolphthalein	Colourless	

- b) How does the P^H value of 0.1 M potassium hydroxide solution compare with that of 0.1M aqueous ammonia? Explain. (2mks)

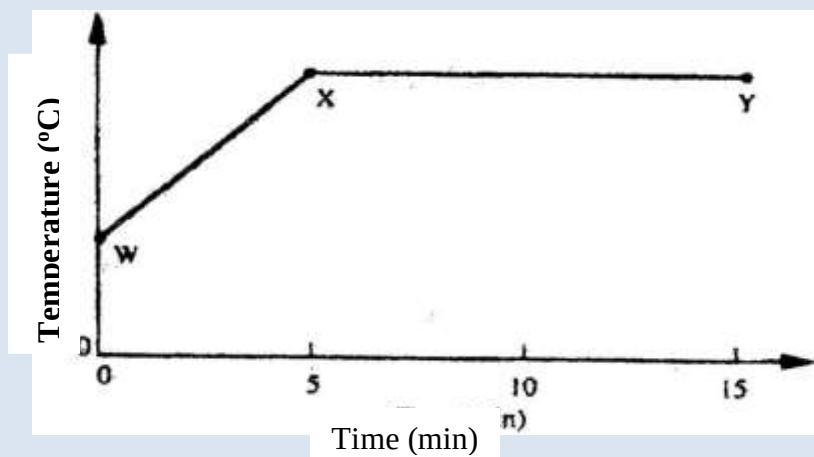
25. Study the properties of substances V1 to V4 in the table below and answer the questions that follow.

Substance	Solubility in water	Solubility	Melting Point($^{\circ}C$)	Boiling point($^{\circ}C$)
V1	Insoluble	Soluble	-30	250
V2	Insoluble	Insoluble	1535	3000
V3	Insoluble	Soluble	16.8	44.8
V4	Insoluble	Soluble	75	320

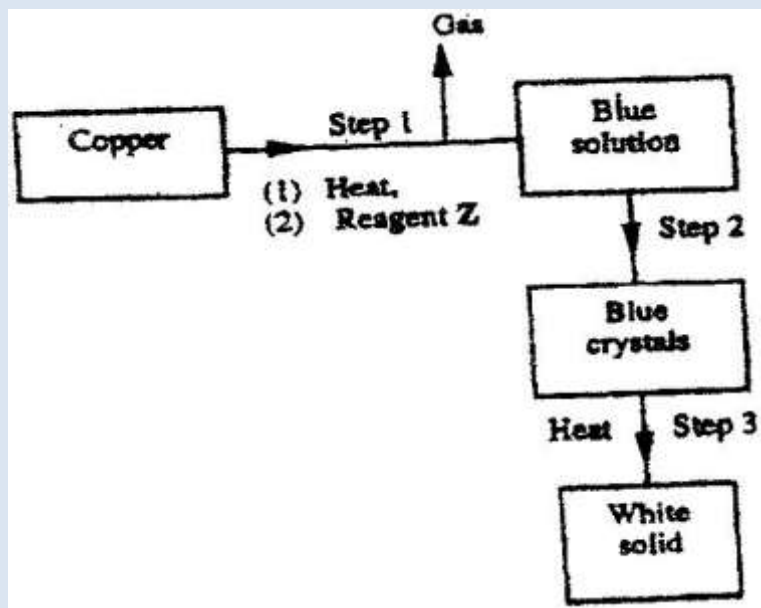
- a) Which of the substances are liquids at $240^{\circ}C$?
- b) Describe how a mixture containing V2 and V4 can be separated (2mks)

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26. The graph below shows a curve obtained when water at 20°C was heated for 15 minutes.

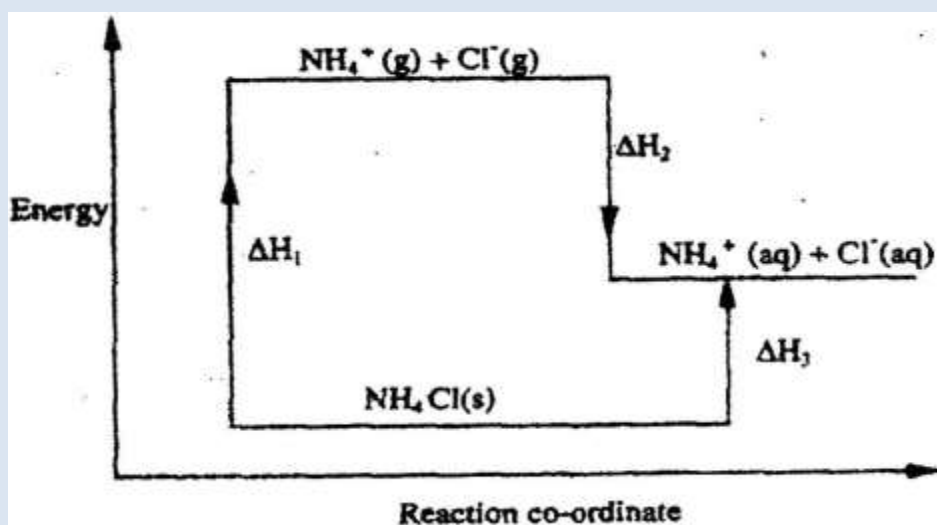


- a) What happens to the water molecules between points W and x? (1mk)
- b) In which part of the curve does a change of state occur? (1mk)
- c) Explain why the temperature does not rise between points X and Y. (1mk)
27. Study the flow chart below and answer the questions that follow.



- a) Name reagent z. (1mk)
- b) Describe the process which takes place in step 2. (1mk)
- c) Identify the white solid. (1mk)

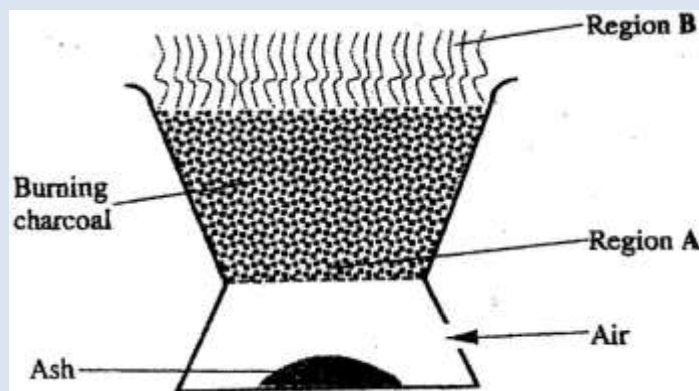
28. Study the diagram below and answer the questions that follow.



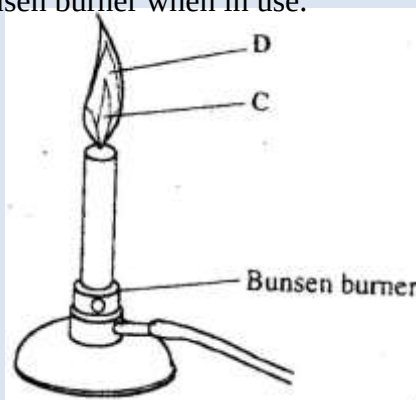
- a) What do ΔH_1 and ΔH_2 represent? (2mks)
- b) Write an expression to show the relationship between ΔH_1 , ΔH_2 and ΔH_3 . (1mk)

CHEMISTRY PAPER 233/1
KCSE 2007 QUESTIONS

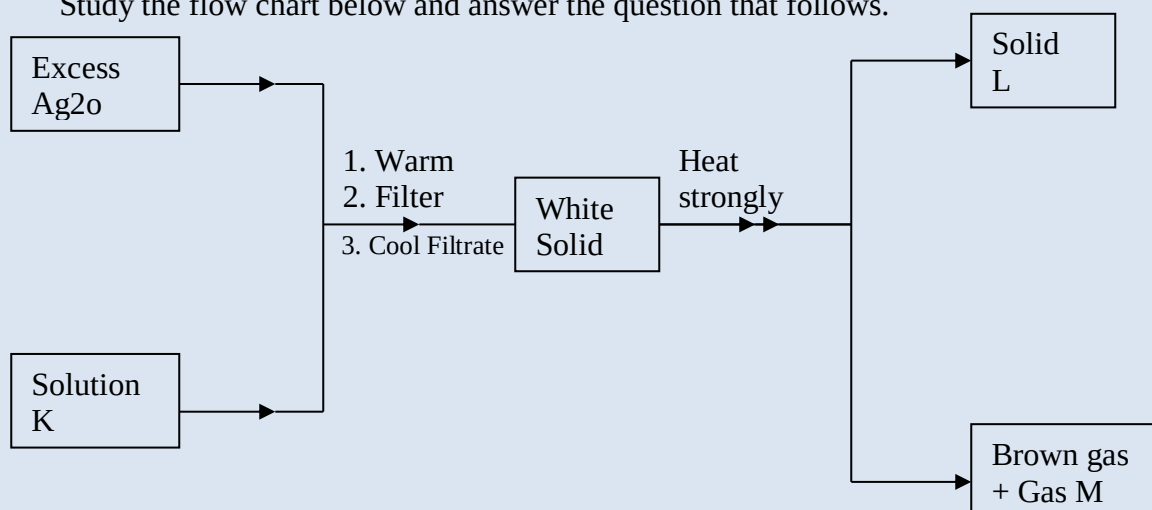
1. The diagram below shows a “Jiko” when in use. Study it and answer the questions that follow.



- a) Identify the gas formed at region A. (1mk)
- b) State and explain the observation made at region B. (2mks)
2. 15.0cm^3 of ethanoic acid (CH_3COOH) was dissolved in water to make 500cm^3 of solution. Calculate the concentration of the solution in moles per litre. (C=12.0; H=1.0; O=16.0; density of ethanoic acid is 1.05 g/cm^3) (3mks)
3. Both chlorine and iodine are halogens.
- a) What are halogens? (1mk)
- b) In terms of structure and bonding, explain why the boiling point of chlorine is lower than that of iodine. (2mks)
4. The diagram below shows a Bunsen burner when in use.



- Name the regions labeled C and D. (2mks)
5. When a student was stung by a nettle plant, a teacher applied an aqueous solution of ammonia to the affected area of the skin and the student was relieved of pain. Explain. (2mks)
6. In an experiment, a few drops of concentrated nitric acid were added to aqueous iron(II) sulphate in a test – tube. Excess sodium hydroxide solution was then added to the mixture.
- a) State the observations that were made when:
- i) Concentrated nitric acid was added to aqueous iron (II) sulphate (1mk)
- ii) Excess sodium hydroxide was added to the mixture. (1mk)
- b) Write an ionic equation for the reaction which occurred in (a) (ii) above. (1mk)
7. a) Use the information given below to draw a labeled diagram of an electrochemical cell that can be constructed to measure the electromotive force between G and J.
- $G^{2+}(aq) + 2e \rightleftharpoons G(s); E^{\theta} = -0.74 V$
- $J^{2+}(aq) + 2e \rightleftharpoons J(s); E^{\theta} = -0.14 V$
- b) Calculate the E^{θ} value for the cell constructed in (a) above. (1mk)
8. Explain why there is a general increase in the first ionization energies of the elements in period 3 of the periodic table from left to right. (2mks)
9. Study the flow chart below and answer the question that follows.

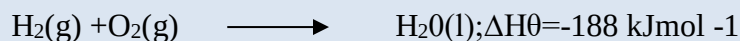
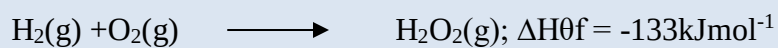


Identify:

(3mks)

- a) Solution K
- b) Solid L
- c) Gas M

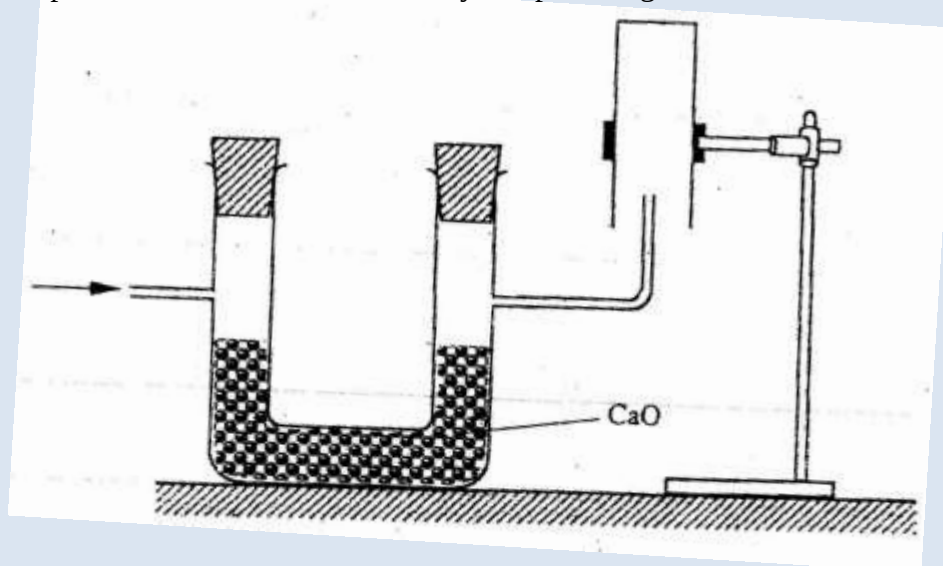
9. The thermo chemical equations for the formation of hydrogen peroxide under standard conditions are:



Write the thermo chemical equation for the molar heat of vaporization of hydrogen peroxide.

(2mks)

10. The set-up below was used to collect a dry sample of a gas.



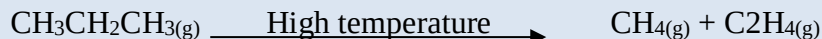
Give two reasons why the set-up cannot be used to collect carbon (IV) oxide gas. (2mks)

12. a) State the Charles law (1mk)
- b) The volume of a sample of nitrogen gas at a temperature of 291 K and 1.0×10^5 Pascal's was $3.5 \times 10^{-2} \text{ m}^3$. Calculate the temperature at which the volume of the gas would be $2.8 \times 10^{-2} \text{ m}^3$ at 1.0×10^5 Pascal. (2mks)
13. a) name the process that takes place when:

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- (i) Crystals of zinc nitrate change into solution when exposed to air (1mk)
(ii) An alcohol reacts with an organic acid in the presence of a catalyst to form a sweet smelling compound. (1mk)

b) Propane can be changed into methane and ethane as shown in the equation below;



Name the process undergone by propane. (1mk)

14. a) Distinguish between nuclear fission and nuclear fusion. (2mks)

b) Describe how solid wastes containing radioactive substances should be disposed of. (1mk)

15. a) Explain why permanent hardness in water cannot be removed by boiling. (2mks)

b) Name two methods that can be used to remove permanent hardness from water. (1mk)

16 The table below shows the tests that were carried out on solid N and the observations made.

I	Test	Observations
II	Dilute hydrochloric acid was added to solid N.	A colourless solution was formed.
III	To the colourless solution obtained in test II, excess sodium hydroxide solution was added.	A white precipitate was formed which dissolved to form a colourless solution.

Write the formula of the anion in;

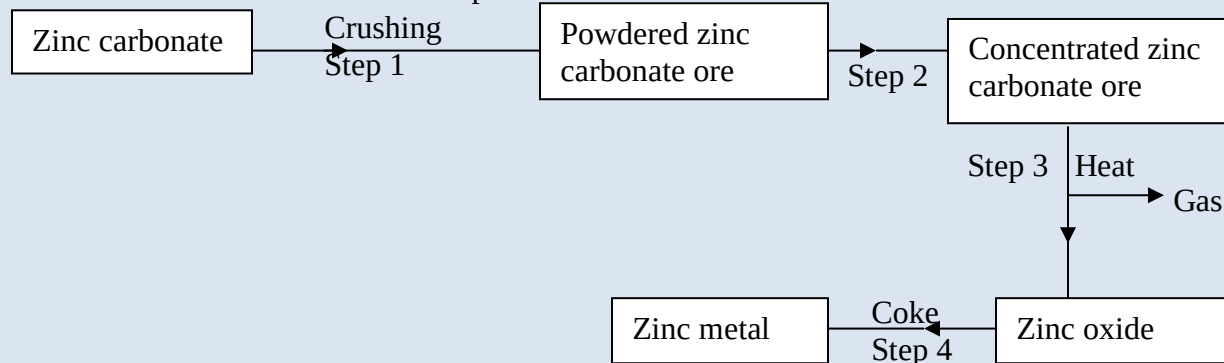
a) Solid N (1mk)

b) The colourless solution formed in test III (1mk)

17. The relative formula mass of a hydrocarbon is 58. Draw and name two possible structures of the hydrocarbon (C=12.0; H=1.0) (3mks)

18. starting with sodium metal, describe how a sample of crystals of sodium hydrogen carbonate may be prepared. (3mks)

- 19 The flow chart below shows steps used in the extraction of zinc from one of its ores.



- a) Name the process that is used in step 2 to concentrate the ore. (1mk)
 - b) Write an equation for the reaction which takes place in step 3. (1 mk)
 - c) Name one use of zinc other than galvanizing. (1mk)
20. An alcohol has the following composition by mass: hydrogen 13.5%, oxygen 21.6% and carbon 64.9%
- a) Determine the empirical formula of the alcohol($C=12.0$; $H=1.0$)= 16.0). (2mks)
21. a) When brine is electrolyzed using inert electrodes, chlorine gas is liberated at the anode instead of oxygen. Explain this observation. (2mks)
- b) Name the product formed at the cathode. (1 mk)
22. 6.84g of aluminium sulphate were dissolve in 150cm³ of water. Calculate the molar concentration of the sulphate ions in the solution. (Relative formula mass of aluminium sulphate is 342)
23. The table below shows the relative molecular masses and the boiling points of pentane and propan-1-ol

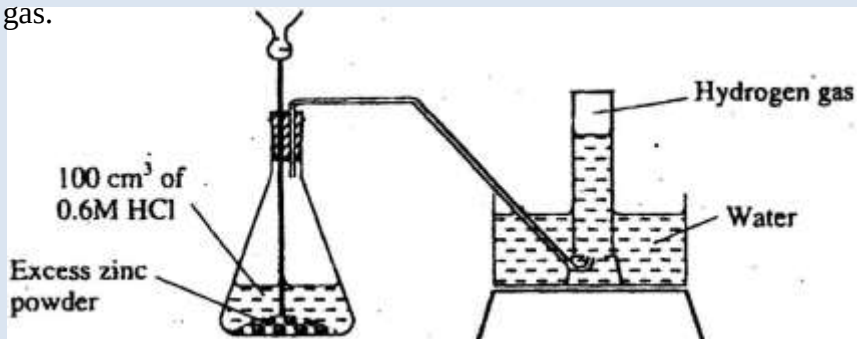
	Relative molecular mass	Boiling point(°C)
Pentane	72	36
Propan-1-ol	60	97

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Explain why the boiling point of propane 1-1-01 is higher than that of pentane. (2mks)

24. State and explain the observations made when excess ammonia gas reacts with chlorine gas (3mks)

25. The diagram below shows a student's set-up for the preparation and collection of hydrogen gas.



(a) How would the final volume of hydrogen gas produced be affected if 80cm³ of 0,75 M hydrochloric acid was used? (1mk)

(b) Give a reason why helium is increasingly being preferred to hydrogen in weather balloons.

26. The table below shows the number of valence electrons of the element P, Q and R.

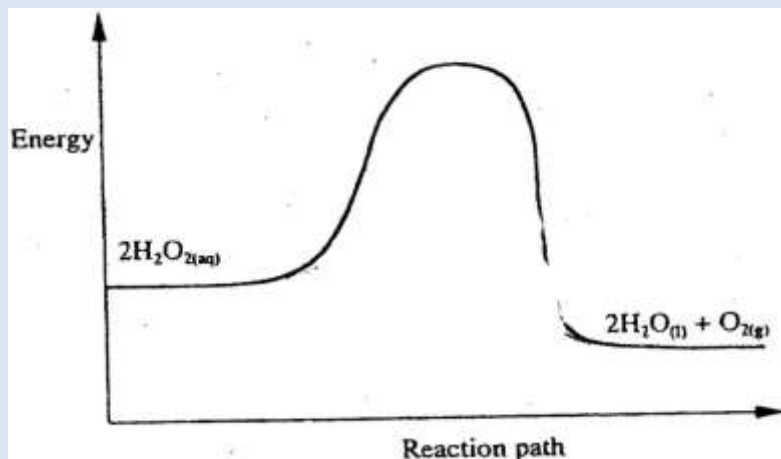
Element	P	Q	R
Number of valence electrons	3	5	2

a) Explain why p and R would not be expected to form a compound. (1mk)

b) Write an equation to show the effect of heat on the carbonate of R (1mk)

c) Write the formula for the most stable ion of q. (1mk)

27. The diagram below formula for the most stable ion of Q. (1mk)

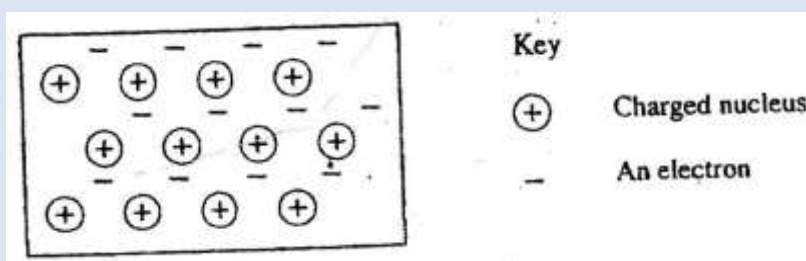


On the same axis, sketch the graph for the decomposition of hydrogen peroxide when manganese (IV) oxide is added. (2mks)

28. During the electrolysis of aqueous silver nitrate, a current of 5.0a was passed through the electrolysis for 3 hours.

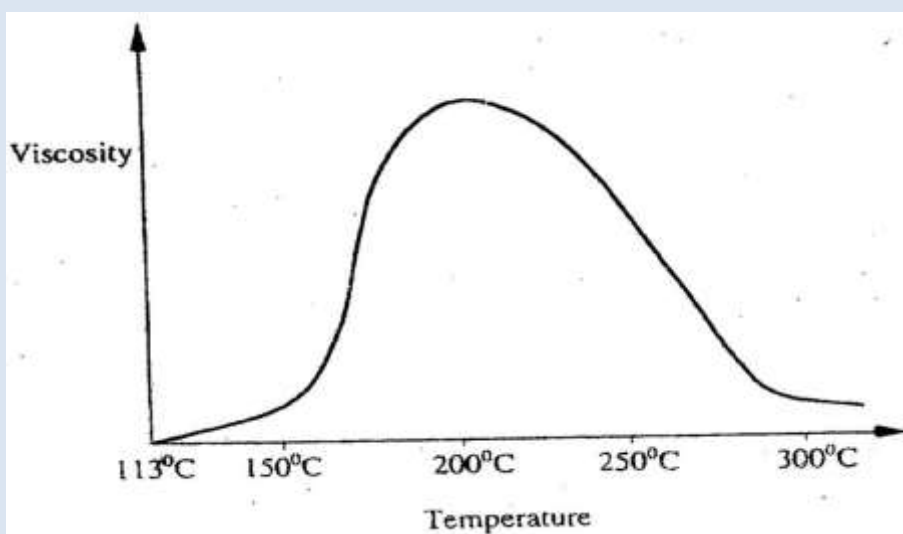
- a) Write the equation for reaction which took place at the anode. (1mk)
- b) Calculate the mass of silver deposited ($Ag = 108$; $IF=96500C$) (2mks)

29 The diagram below is a section of a model of the structure of element T.



- a) State the type of bonding that exists in T. (1mk)
- b) In which group of the period table does element T belong? Give a reason. (2mks)

30. Below is a sketch of a graph showing the change in viscosity? (Ease of flow) with temperature when solid sulphur is heated.



Describe what happens to the sulphur molecules when sulphur is heated from 150°C to about 200°C.

CHEMISTRY PAPER 233/1
KCSE 2008 QUESTIONS

1. A small crystal of potassium manganate (VII) was placed in a beaker water. The beaker was left standing for two days without shaking. State and explain the observations that were made. (2mks)

2. When a hydrated sample of calcium sulphate $\text{CaSO}_4 \cdot \text{XH}_2\text{O}$ was heated until all the water was lost, the following data recorded;

Mass of crucible = 30.296 g

Mass of crucible + hydrated salt = 33.111 g

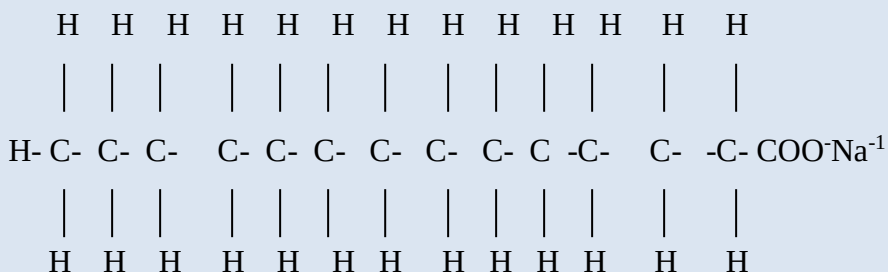
Mass of crucible + anhydrous salt = 32.781 g

Determine the empirical formula of the hydrated salt (Relative formula mass of $\text{CaSO}_4 = 136$, $\text{H}_2\text{O} = 18$). (3mks)

3. Complete the following table by filling in the missing test and observations (3mks)

No.	Gas	Test	Observation
I	Chlorine	Put a moist red litmus into the gas	
II	Sulphure (IV) oxide		Paper turns green
III	Butane	Add a drop of bromin water	

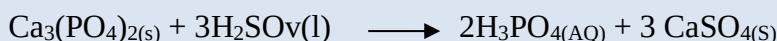
4. The structure of a detergent is



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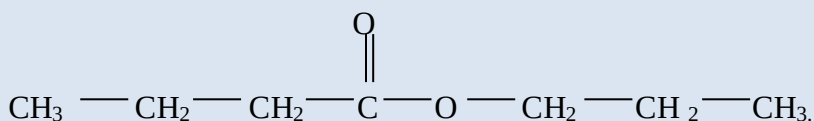
- a) Write the molecular formula of the detergent. (1mk)
- b) What type of detergent is represented by the formula? (1mk)
- c) When this type of detergent is used to wash linen in hard water, spots (mks) are left on the linen. Write the formula of the substance responsible for the spots (1mk)

5. Phosphoric acid is manufactured from calcium phosphate according to the following equation.



Calculate the mass in (Kg) of phosphoric acid that would be obtained if 155 Kg of calcium phosphate reacted completely with the acid
(Ca=40, P=31, S=32, O=16, H=1) (2mks)

6. The structure below represents a sweet smelling compound



Give the names of the two organic compounds that can be used to prepare this compound in the laboratory. (2mks)

7. a) What are isotopes? (1mk)
- b) Determine the number of neutrons in $^{18}_8\text{O}$. (1mk)
8. a) State the observation made at the end of the experiment when a mixture of iron powder and sulphur is heated in a test tube. (1mk)
- b) Write an equation for the reaction the product in (a) above and dilute hydrochloric acid. (1mk)
- c) When a mixture of iron powder and sulphur is heated, it glows more brightly than that of iron fillings and sulphur. Explain this observation (1mk)
9. Zinc reacts with both concentrated and dilute sulphuric (VI) acid. Write equations for two reactions. (2mk)
10. When magnesium was burnt in air, a solid mixture was formed. On addition of

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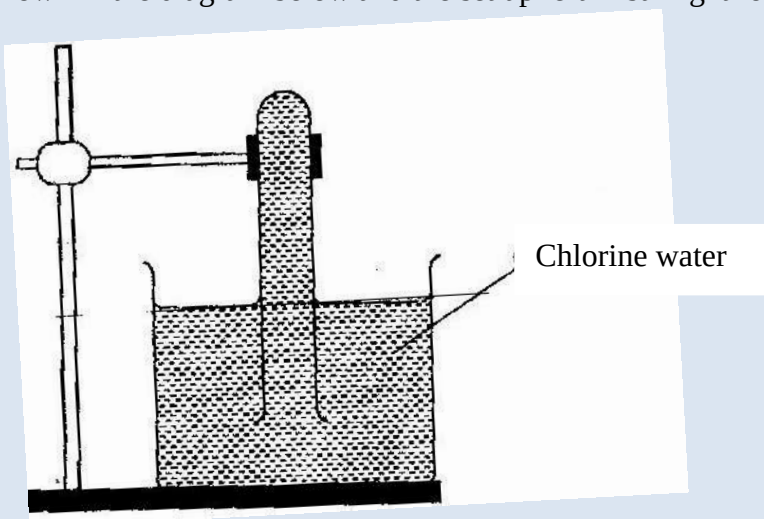
water to the mixture a gas which turned moist red litmus paper blue was evolved.
Explain these observations.

11. The table below gives atomic numbers of elements represented by the letters A, B, C and D.

Element	A	B	C	D
Atomic number	15	16	17	20

Use the information to answer the questions that follow.

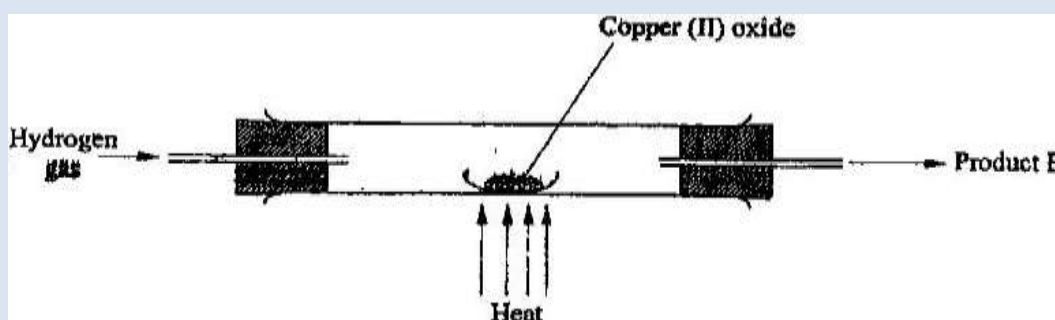
- a) Name the type of bonding that exists in the compound formed when A and D react (1mk)
- b) Select the letter which represents the best oxidizing agent. Give a reason for your answer. (2mks)
12. In an experiment, a test-tube full of chlorine water was inverted in chlorine water as shown in the diagram below and the set up left in sunlight for one day.



After one day, a gas was found to have collected in the test-tube

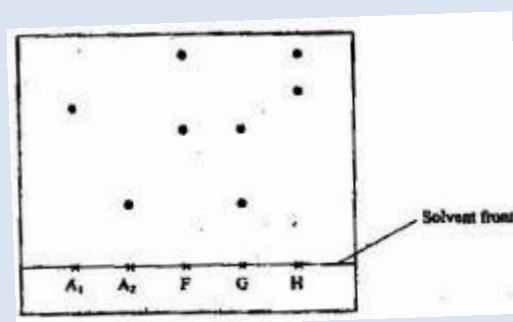
- a) Identify the gas.
- b) What will happen to the pH of the solution in the beaker after one day? Give an explanation. (2mks)

13. In a laboratory experiment hydrogen gas was passed over heated copper (II) oxide as shown the diagram below.



Describe a chemical test that can be used to identify the product e. (2mks)

14. Samples of urine from three participants F, G and H at an international sports meeting were spotted onto a chromatography paper alongside two from illegal drugs A_1 and A_2 .
A chromatogram was run using methanol. The figure below shows the chromatogram.



- Identify the athlete who had used an illegal drug. (1mk)
 - Which drug is more soluble in methanol? (1mk)
15. The table below gives the solubilities of substances J, K and L at different temperatures

Substance	Solubility in grammes per 100 g water at			
	0°C	20°C	40°C	60°C
J	0.334	0.16	0.097	0.0058

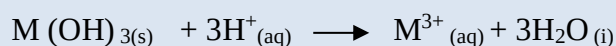
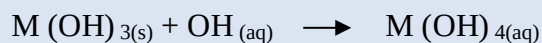
KCSE CHEM PAPER 1 PAST PAPERS MR CHEPKWONY

K	27.60	34.0	40.0	45.5
L	35.70	36.0	40.0	37.3

Select the substance which, when dissolved in water, heat is given out. Give a reason (2mks)

16. Starting with copper metal, describe how a sample of crystals of copper (II) chloride may be prepared in the laboratory. (3mks)

17. A compound whose general formula is $M(OH)_3$ reacts as shown by the equation below.



(a) What name is given to compounds which behave like $M(OH)_3$ in the two Reactions (1 mk)

(b) Name two elements whose hydroxides behave like that of M (2 mks)

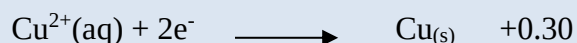
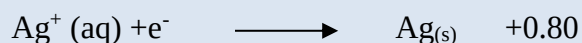
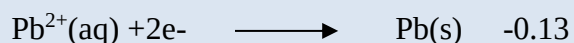
18. The grid below is part of the periodic table. Use it to answer the questions that follow, (the letters are not the actual symbols of the elements).

						R	S		
N	Q						T	U	
P									

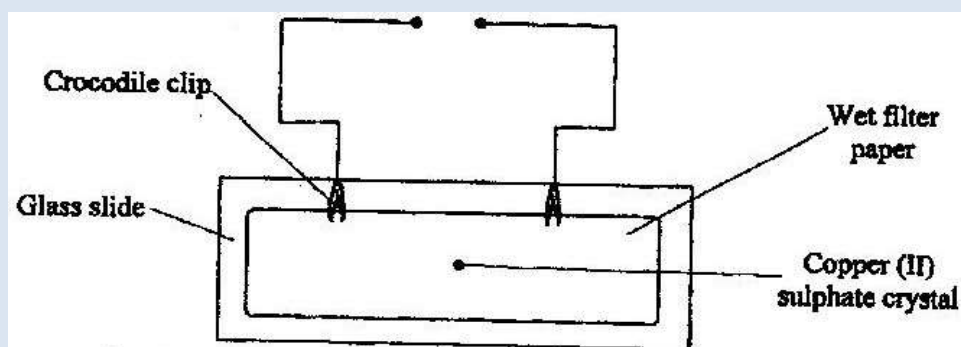
a) Indicate on the grid the position of an element represented by letter V whose atomic number is 14. (1mk)

b) Select a letter which reaction between Q and T. (1mk)

19. Select a letter which represents a mono atomic gas. (1mk)

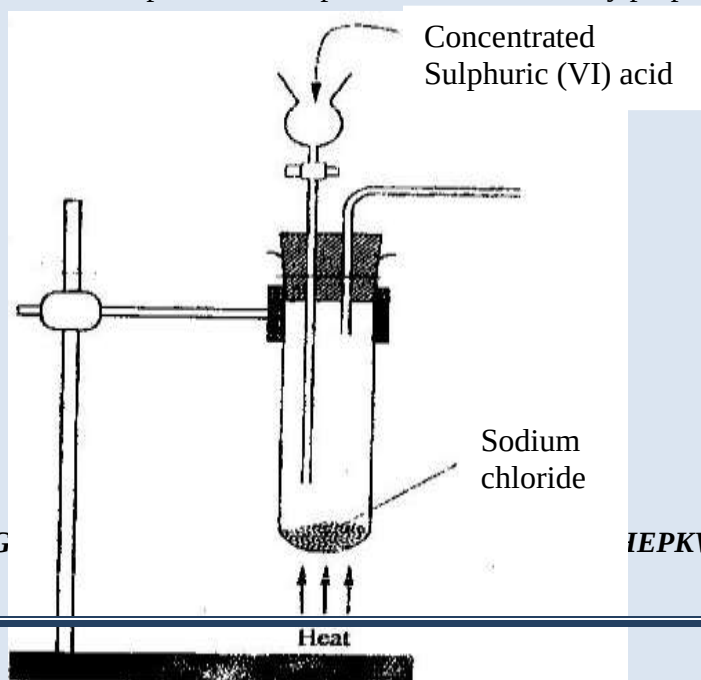


- a) Write the cell representation for the electrochemical cell that would give the highest E (1mk)
- b) State and explain the observations made when a copper rod is placed in a beaker containing silver nitrate solution. (2mks)
20. a) State the Graham's law diffusion. (1mk)
- b) The molar masses of gases W and X are 16.0 and 44.0 respectively. If the rate of diffusion of W through a porous material is $12\text{cm}^3\text{s}^{-1}$ calculate the rate of diffusion of X through the same material. (2mks)
21. The diagram below represents an experiment that was set up to investigate movement of ions during electrolysis.



When the circuit was completed, it was noticed that a blue colour spread towards the right.

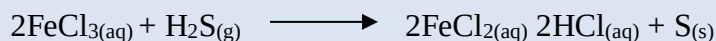
- a) Explain this observation (2mks)
- b) Write the equation for the reaction that occurred at the anode. (1mk)
22. The diagram below is part of a set up used in the laboratory preparation of a gas.



Complete the diagram to show how a dry sample of the gas can be collected.

(3mks)

23. In a closed system, aqueous iron (III) chloride reacts with sulphide gas as shown in the equation below.



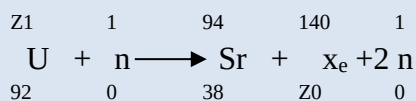
State and explain the observation that would be made if dilute hydrochloric acid is added to the system at equilibrium.

(2mks)

24. a) A radioactive substance emits three different particles.
Give the symbol of the particle with the highest mass.

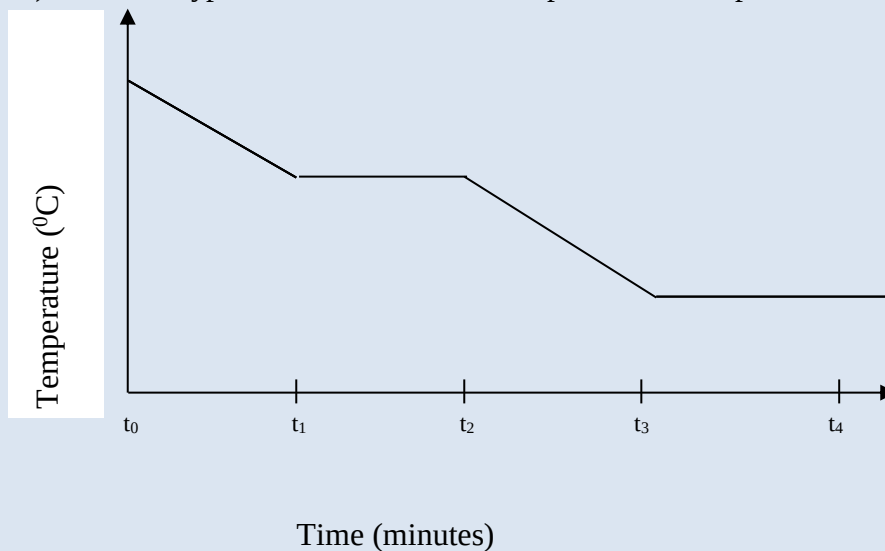
(1mk)

- b) (i) Find the values of Z_1 and Z_2 in the nuclear equation below



- iii) What type of nuclear reaction is represented in represented in b (i) above?

(1mk)



Give the name of the:

- a) Process taking place between t_0 and t_1 .

(1mk)

b) Energy change that occurs between t_3 and t_4

26. When solid B_1 was heated, a gas which formed a white precipitate when passed through lime water was produced. The residue was dissolved in dilute nitric (V) acid to form a colourless solution B_2 . When dilute hydrochloric acid was added to solution B_2 a white precipitate which dissolved on warming was formed.

a) Write the formula of the;
 I Cation in solid B_1 (1mk)
 II anion in solid B_1 (1mk)

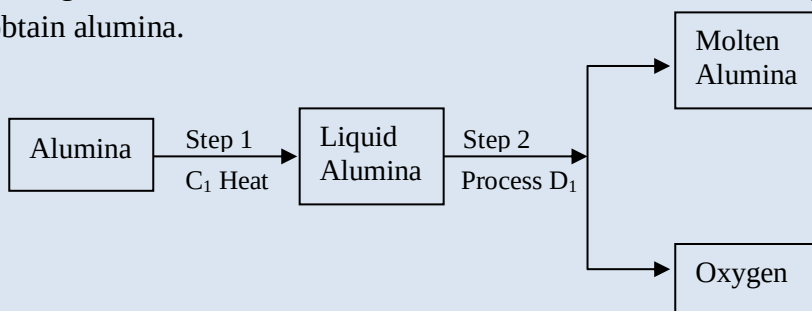
b) Write an ionic equation for the reaction between the residue and dilute nitric (V) acid. (1mk)

27. In an experiment to determine the percentage of magnesium hydroxide in an anti-acid, a solution containing 0.50 g of the anti-acid was neutralized by 23.0 cm^3 of 0.010M hydrochloric acid (Relative formula mass of magnesium hydroxide = 58)

a) Mass of magnesium hydroxide in the anti-acid; (2mks)

b) Percentage of magnesium hydroxide in the anti-acid (1mk)

28. During the extraction of aluminium from its ores; the ore is first purified to obtain alumina.



a) Name
 (i) Substance C_1 (1mk)

(ii) Process D_1 (1mk)

b) Give two reasons why aluminium is used extensively in the making of cooking pans. (1mk)

29. A certain mass of a metal E_1 reacted with excess dilute hydrochloric acid at 25°C .

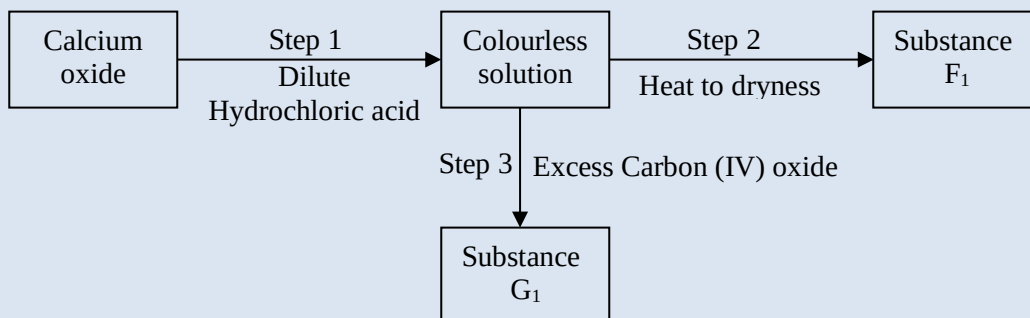
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The volume of hydrogen gas liberated was measured after every 30 seconds. The results were presented as shown in the graph below.

- a) Name one piece of apparatus that may have been used to measure the volume of gas liberated. (1mk)
- b) (i) On the same axis, sketch the curve that would be obtained if the experiment was repeated at 35°C. (1mk)
- (ii) Explain the shape of your curve in b(i) above. (1mk)

30. Crude oil contains sulphur. What would be the effect to the environment of using fuel containing sulphur? (1mk)

31. Study the flow chart below and answer the questions that follow.



- a) Give the name of the process that takes place in step 1. (1mk)
- b) Give;
- (i) The name of substance G₁
- (ii) One use of substance F₁

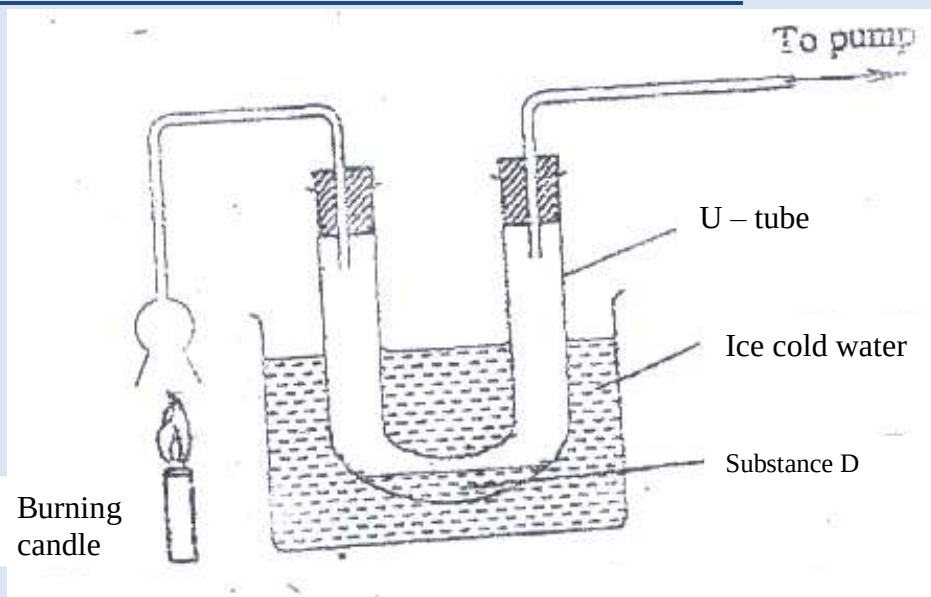
**CHEMISTRY PAPER 233/1
KCSE 2009 QUESTIONS**

1. The ionization energies for three elements A, B and C are shown in the table below:

Element	A	B	C
Ionisation energy (kJ/mole)	519	418	494

- a) What is meant by ionization energy? (1 mk)
b) Which element is the strongest reducing agent? Give a reason. (2 mks)
2. Hardness of water may be removed by either boiling or addition of chemicals
- a) write an equation to show how boiling removes hardness of water. (1 mk)
b) name two chemicals that are used to remove hardness of water. (2 mks)
3. The atomic number of sulphur is 16.
Write the electron arrangement of sulphur in the following: (2 mks)
- a) H_2S ;
b) SO_3^{2-}
4. An experiment was set up as shown in the diagram below:

To pump



- a) Identify substance D. (1 mk)
 - b) Describe how the other product of the burning candle could be prevented from getting into the environment. (2 mks)
5. In terms of structure and bonding, explain why the melting point of oxygen is much lower than that of sodium. (3 mks)
 6. An isotope of element E has 34 neutrons and its mass number is 64. E forms a cation with 28 electrons. Write the formula of the cation with 28 electrons. Write the formula of the cation indicating the mass and atomic numbers. (1mk)
 7. When aluminium oxide was electrolysed, 1800kg of aluminium metal were obtained.
 - a) Write equation for the formation of aluminium metal (1 mk)
 - b) Calculate the quantity of electricity in faradays used (Al=27) (2 mks)
 8. Using dots (.) and crosses (x) , show bonding in:
 - a) The compound formed when nitrogen reacts with fluorine

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(Atomic numbers F=9, N=7);

(1mk)

b) Sodium oxide.

(Atomic numbers Na= 11, O = 8)

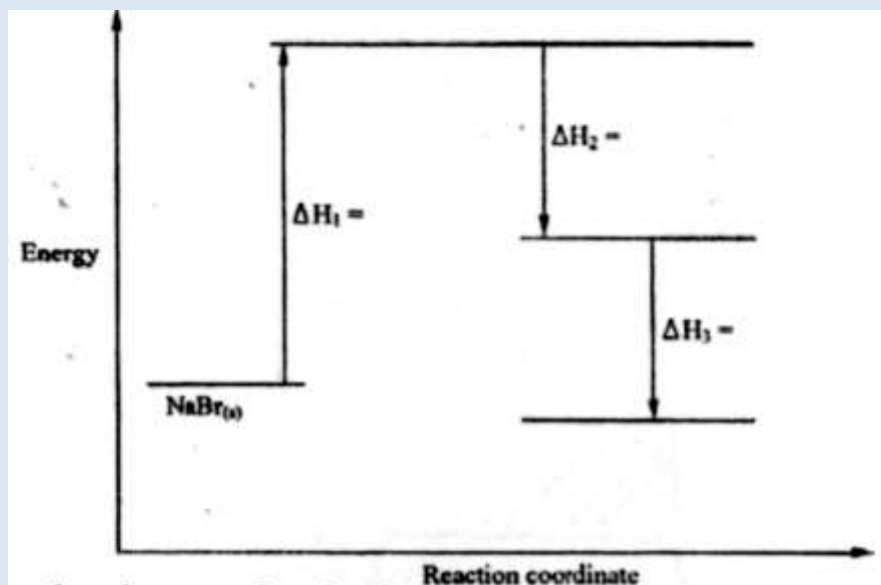
(1 mk)

9. a) what is meant by molar heat of solution?

(1 mk)

b) The lattice energy of sodium bromide and hydration energies of sodium and bromide ions are: 733,406 and 335 kJmol⁻¹ respectively.i) Complete the energy cycle diagram below by inserting the values of ΔH_1 , ΔH_2 , and ΔH_3

(1 ½ mks)



ii) Determine the molar heat of solution of solid sodium bromide.

(½ mk)

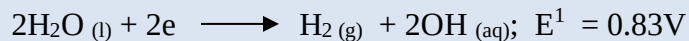
10. Hydrogen and oxygen can be obtained by electrolysis of acidified water. Using equations for the reactions at the electrodes, explain why the volume of hydrogen obtained is twice that of oxygen.

(2mks)

11. Starting with 50 cm³ of 2.8M sodium hydroxide, describe how a sample of pure sodium sulphate crystals can be prepared.

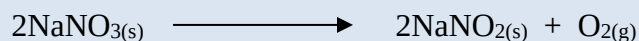
(3 mks)

12. The standard reduction potentials of two half –cells are:



Draw a labelled diagram of an electro chemical cell that can be constructed using the two half –cells (3 mks)

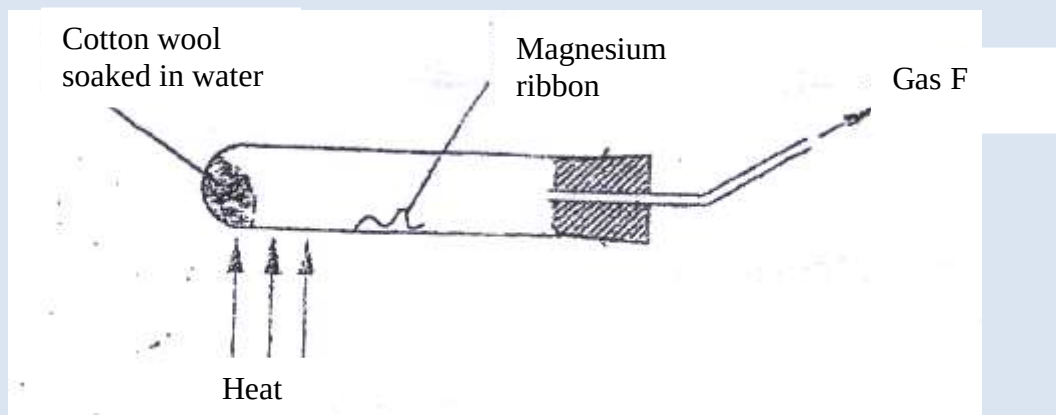
13. When 8.53 of sodium nitrate were heated in an open test-tube the mass of oxygen gas produced was 0.83 g . Given the equation of the reaction as



Calculate the percentage of sodium nitrate that was converted to sodium nitrite (Na = 23.0 N = 14.0 O = 16.0) (3 mks)

14. a) Draw and name the structure of the compound formed when one mole of ethyne reacts with one mole of hydrogen bromide. (2mks)
- b) Draw the structures of the alkynes whose molecular formula is C_4H_6 (1 mk)

15. A student used the set up shown in the diagram below in order to study the reactions of some metals with steam. The experiment was carried out for ten minutes



- a) What observation would be made if gas F is ignited? (1 mk)

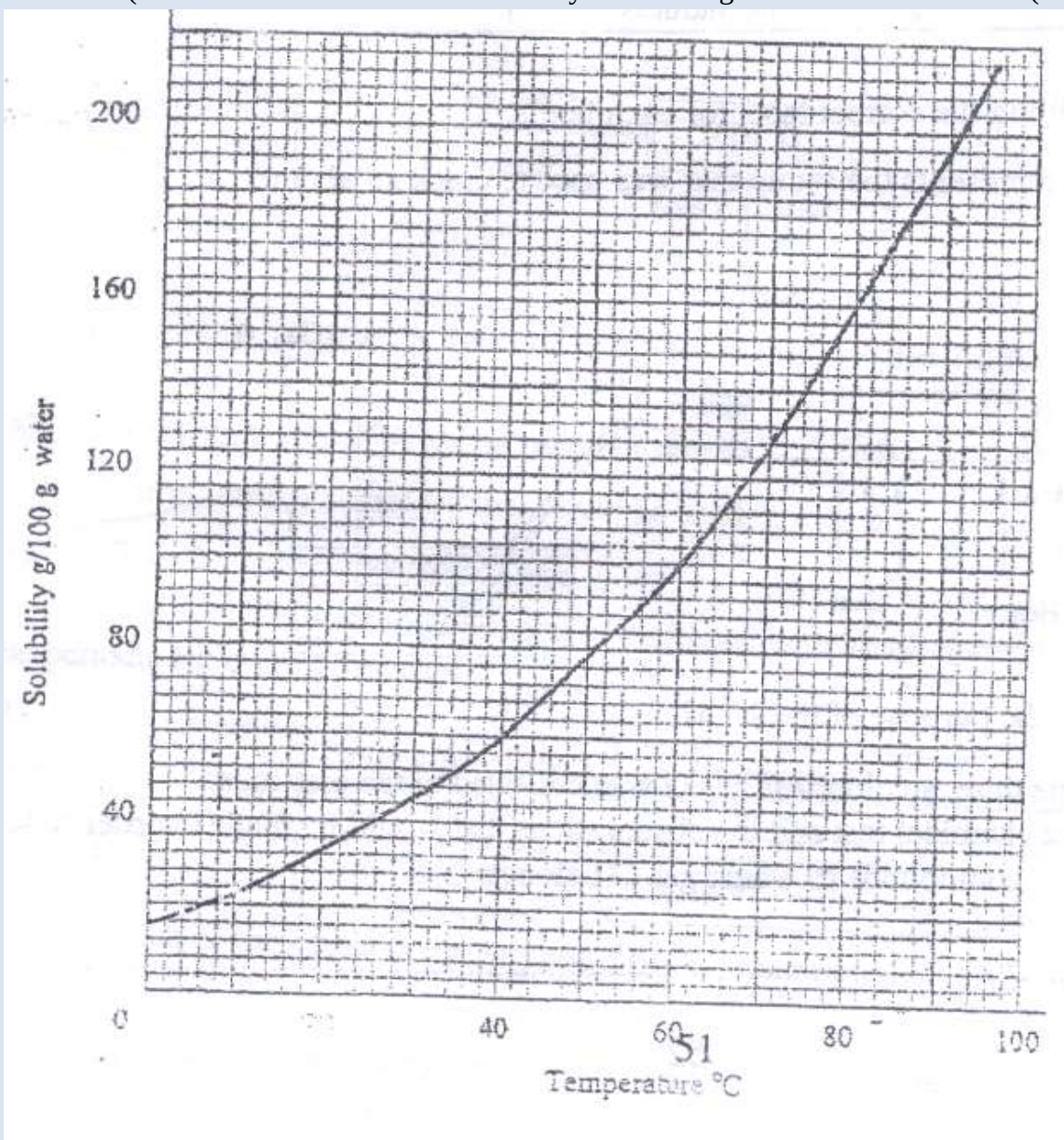
KCSE CHEM PAPER 1 PAST PAPERS MR CHEPKWONY

- b) When the experiment was repeated using iron powder instead of magnesium ribbon very little gas F was obtained.
- i) Give a reason for this observation (1 mk)
- ii) What change in the conditions of the experiment should the student have made in order to increase the volume of gas F Produced? (1mk)

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16. The solubility curve of potassium nitrate is shown below.

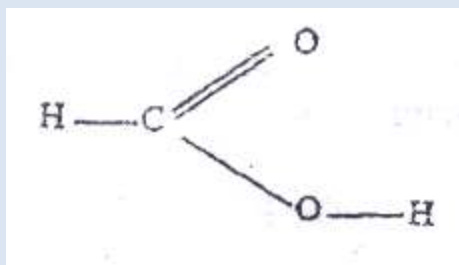
- a) Determine the solubility of potassium nitrate at 50°C (1 mk)
- b) Determine the molar concentration of saturated potassium nitrate at 50°C
(K=39.0 O= 16.0 N= 14.0 and density of water 1 g/cm^3). (2 mks)



Solubility g/100g water

Temperature $^{\circ}\text{C}$

17. The structure of methanoic acid is



What is the total number of electrons used for bonding in a molecule of methanoic acid? Give reasons.

(2 mks)

18. Bottles of sodium carbonate, sodium chloride and sugar have lost their labels. A student prepares and tests an aqueous solution of a sample from each bottle.

The results obtained are as shown in the table below.

Bottle	PH	Electrical Conductivity	Correct label
--------	----	-------------------------	---------------

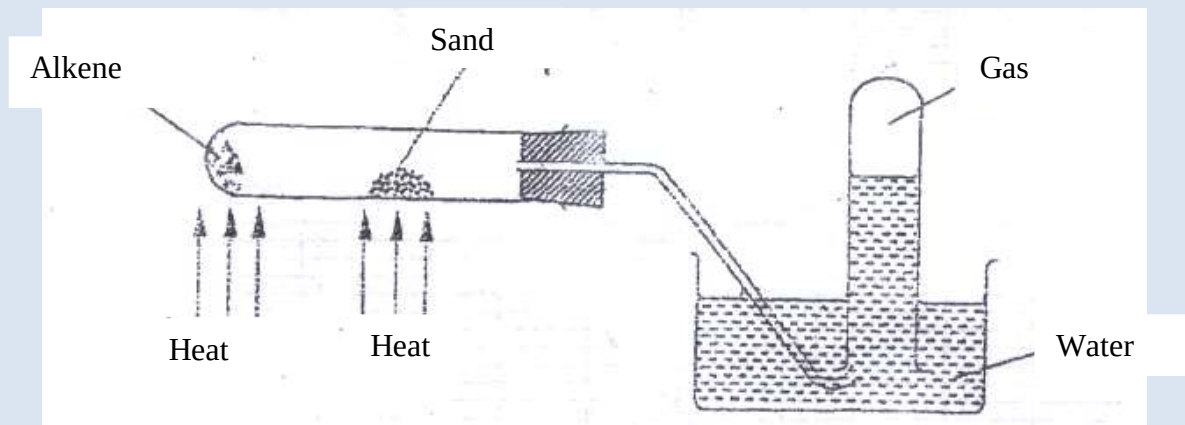
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1	7	Conducts	
2	7	Does not conduct	
3	10	conducts	

Complete the table by filling the correct label for each bottle.

(3 mks)

19. The figure below represents the set up that was used to crack an alkane.



a) What was the purpose of the sand?

(1 mk)

b) After some time, a colourless gas G collected in the test-tube. Describe a chemical test and the observations that would be made in order to identify the class of compounds to which gas G belongs.

(2 mks)

20. Classify the following processes as either chemical or physical.

(3 mks)

Process	Type of change
a) Heating copper (II) sulphate crystals	
b) Obtaining kerosene from crude oil	
c) Souring of milk	

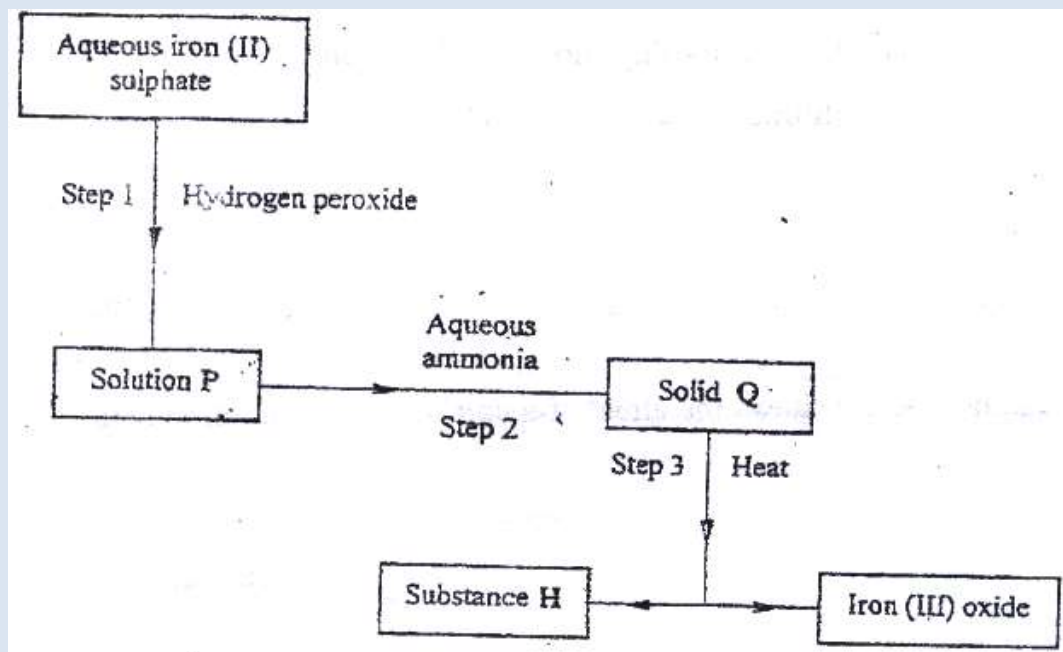
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21. Give the name of the product formed when magnesium reacts with phosphorus. (1 mk)
22. a student added very dilute sulphuric (VI) acid to four substances and recorded the observations shown in the table below.

Test	Substance	Gas given off
1	Sodium	Yes
2	Iron	No
3	Carbon	Yes
4	Copper	No

For which tests are the observations wrong? Explain (3 mks)

23. Use the flow chart below to answer the questions that follow.

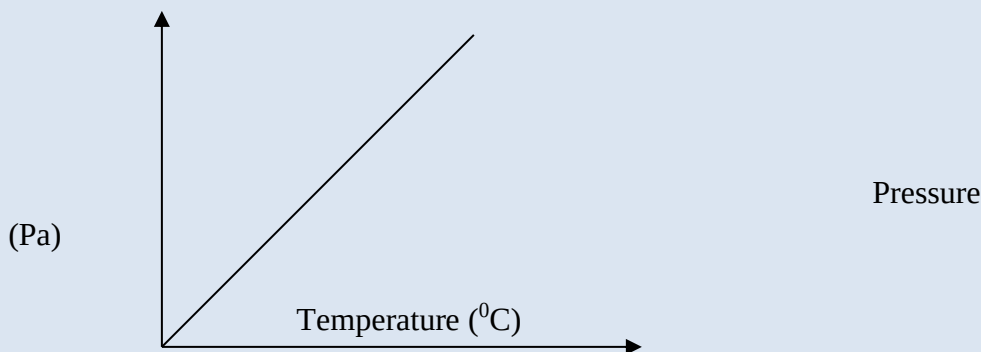


- a) What observation would be made in step I? (1 mk)
- b) Name another substance that could be used in step 2. (1 mk)
- c) Give the name of substance H. (1 mk)

24. The boiling points of some compounds of hydrogen with some elements in groups 4 and 6 of the periodic table are given below.

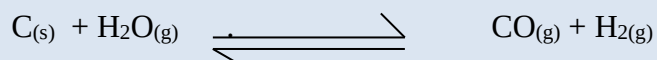
Compound	Boiling point ($^{\circ}\text{C}$)	Compound	Boiling point ($^{\circ}\text{C}$)
CH_4	-164.0	H_2O	100.0
SiH_4	-112.0	H_2S	-61.0

- a) Which of the compounds CH_4 and SiH_4 has stronger intermolecular forces? (1 mk)
- b) Explain why the boiling points of H_2O and H_2S show different trends from that of CH_4 and SiH_4 (2mks)
25. For each of the following reactions, state the observation and write the formula of the Compound responsible for the observation:
- a) Bromine water is added to aqueous potassium iodide; (1 ½ mks)
- b) Excess aqueous ammonia is added to copper (II) hydroxide (precipitate). (1 ½ mks)
26. The graph below shows the relationship between pressure and the temperature of a gas in a fixed volume container.



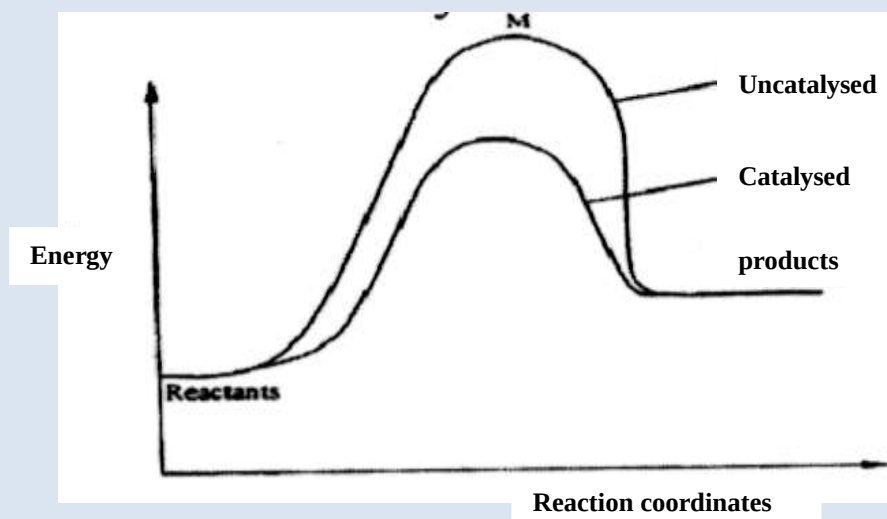
- a) State the relationship between pressure and temperature that can be deduced from the graph. (1 mk)
- b) Using kinetic theory, explain the relationship shown in the graph. (2 mks)

27. The following reaction is in equilibrium in a closed container



State giving reasons how an increase in pressure would affect the amount of hydrogen. (2 mks)

28. The energy level diagram below shows the effect of a catalyst on the reaction path.



- a) What does point M represent? (1 mk)

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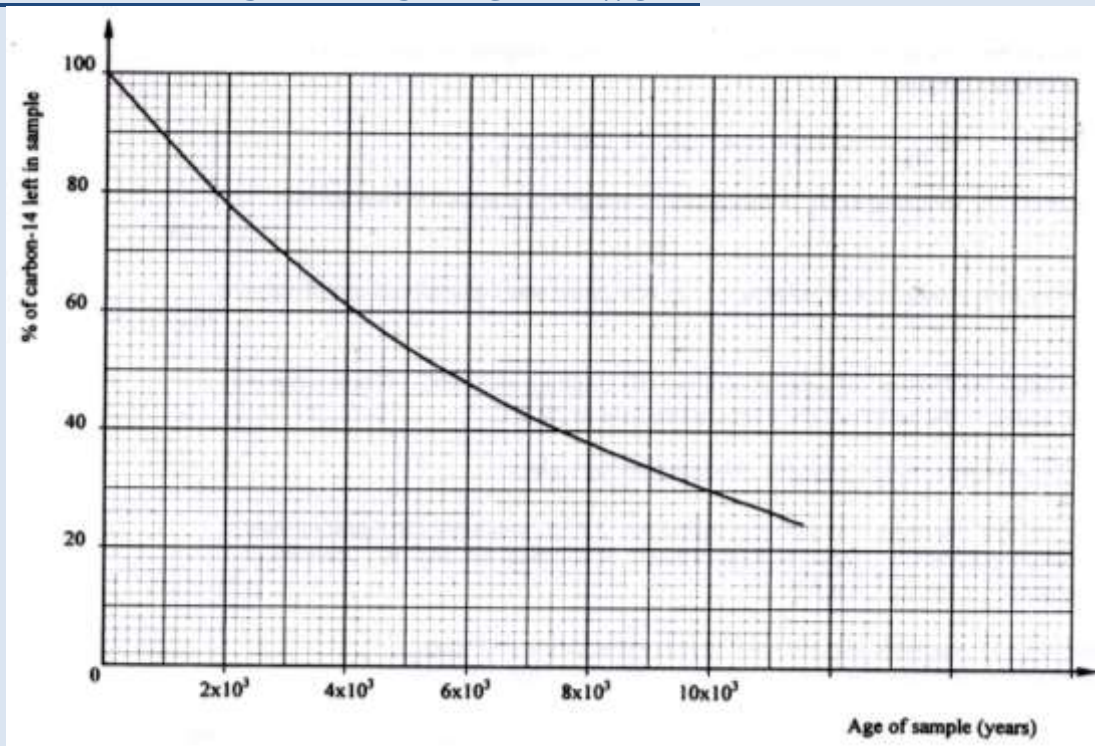
KCSE 2010 QUESTIONS

1.
 - a) Distinguish between a deliquescent and a hygroscopic substance. (2 mks)
 - b) Give **one** use of hygroscopic substance in the laboratory. (1 mk)
2.
 - a) What is meant by the terms: (2 mks)
 - a. element
 - b. atomic number
 - b) The formula for a chloride of titanium is TiCl_3 . What is the formula of its sulphate? (1 mk)
3. What is the name given to each of the following:
 - a) Ability of a metal to be made into a wire: (1 mk)
 - b) Minimum energy required for a chemical reaction to start; (1 mk)
 - c) Type of force that hold atoms of neon together? (1 mk)
4. Draw the structure and give the name of the three alkanes having molecular formula C_5H_{12} (3 mks)
5. Hydrated cobalt(II) chloride exists as pink crystals and anhydrous cobalt(II) chloride is a blue powder. Describe a laboratory experiment that can be used to show that the action of heat on hydrated cobalt(II) chloride is a reversible reaction. (3 mks)
6. Aluminium oxide reacts with both acids and bases.
 - a) Write an equation for the reaction between aluminium oxide and hydrochloric acid. (1 mk)
 - b) Using the equation in (a) above, calculate the number of moles of hydrochloric acid that would react completely with 153.0g of aluminium oxide. (2 mks)
(AL = 27.0, O = 16.0)

7. Complete the table below by writing the product formed at the electrodes during the electrolysis of the electrolytes given in the table. (3 mks)

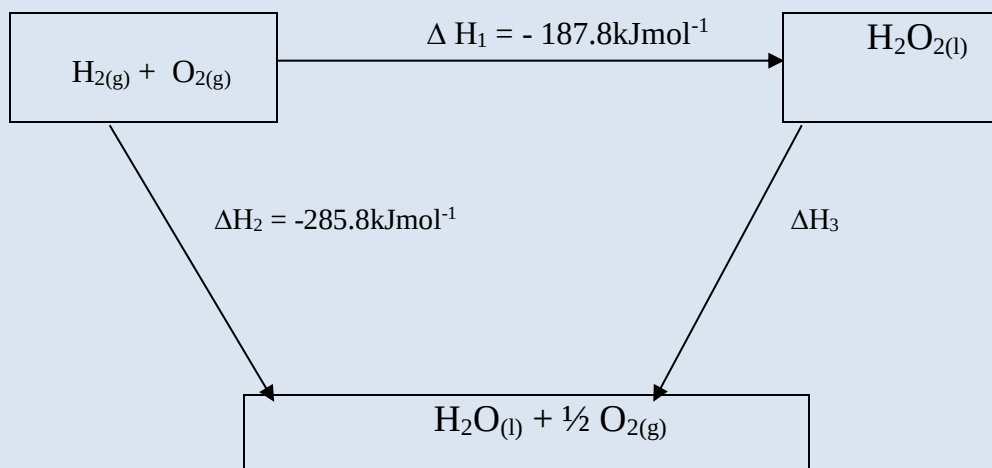
Electrolyte	Product at anode	Product at cathode
Aqueous sodium sulphate using inert electrodes	oxygen (1/2 mk)	Hydrogen (1/2 mk)
Aqueous copper(II) sulphate using copper electrodes	Copper ions (1 mk)	Copper metal (1 mk)

8. The pressure of nitrogen gas contained in a 1dm^3 cylinder at -196°C was 10^7 Pascals. Calculate the:
- a) Volume of the gas at 25°C and 10^5 Pascals. (1 ½ mks)
- b) Mass of nitrogen gas (Molar volume of gas is 24dm^3 , $N = 14.0$) (1 ½ mks)
9. Carbon -14, $^{14}_6\text{C}$, is used in carbon dating. It decays to form nitrogen, $^{14}_7\text{N}$. The graph below shows the amount of carbon -14 left in a sample against its age in years.



- a) Write a nuclear equation for the decay process of carbon -14. (1 mk)
- b) From the graph, determine the;
 - i) Half-life of carbon -14; (1 mk)
 - ii) Percentage of carbon -14 in a sample whose age is 1950 years. (1 mk)

10. The figure below shows an energy cycle.



- a) Give the name of the enthalpy change ΔH_1 . (1 mk)

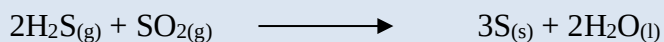
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b) Determine the value of ΔH_3 . (2 mks)

11. Hydrogen sulphide is a highly toxic and flammable gas. It is normally prepared in a fume chamber.

a) Name **two** reagents that can be used to prepare hydrogen sulphide in the laboratory. (1 mk)

b) One of the uses of hydrogen sulphide is to produce sulphur as shown in the following equation;



Identify the reducing agent in this reaction and give a reason for your answer. (1 mk)

c) Other than production of sulphuric(IV) acid, state **one** commercial use of sulphur. (1 mk)

12. A beaker contained 75.0cm^3 of aqueous copper (II) sulphate at 23.7°C . when scrap iron metal was added to the solution, the temperature rose to 29.3°C .

a) Write an ionic equation for the reaction that took place. (1 mk)

b) Given that the mass of copper deposited was 5.83g, calculate the molar enthalpy change in kJmol^{-1} . (specific heat capacity of solution = $4.2\text{Jg}^{-1}\text{K}^{-1}$, density of solution 1.0gcm^{-3} , $\text{Cu} = 63.5$) (2 mks)

13. Some animal and vegetable oils are used to make margarine and soap. Give the reagents and conditions necessary for converting the oils into:

a) Margarine (2 mks)

b) Soap (1 mk)

14. Using electrons in the outermost energy level, draw the dot (.) and cross (x) diagrams for the molecules H_2O and C_2H_4 . ($\text{H} = 1$, $\text{C} = 6$, $\text{O} = 8$) (2 mks)

i) H_2O

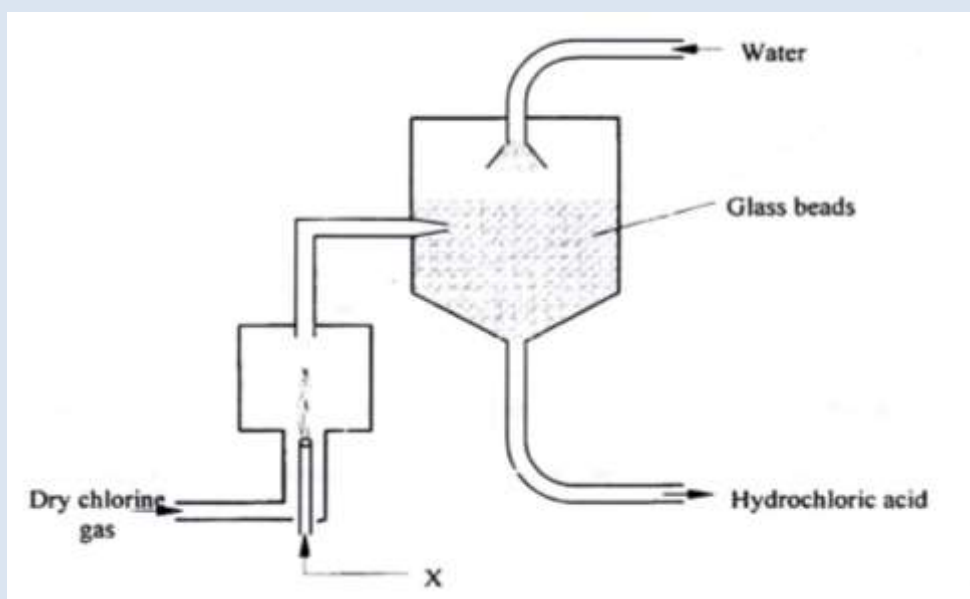
ii) C_2H_4

b) The formula of a complex ion is $\text{Zn}(\text{NH}_3)_4^{2+}$. Name the type of bond that is likely to exist between zinc and ammonia in the complex ion. (1 mk)

15. Carbon (II) oxide is described as a “silent killer”

a) State **one** physical property of carbon (II) oxide that makes it a “silent killer” (1 mk)

- b) State and explain **one** chemical property that makes carbon (II) oxide poisonous to human beings (2 mks)
16. A sample of fertilizer is suspected to be calcium ammonium nitrate. Describe chemical tests for each of the following ions in the sample:
- a) Calcium ions; (2 mks)
 - b) Ammonium ions. (1 mk)
17. Analysis of a compound showed that it had the following composition: 69.42% carbon, 4.13% hydrogen and the rest oxygen.
- a) Determine the empirical formula of the compound. (C = 12.0, H = 1.0, O = 16.0) (2 mks)
 - b) If the mass of one mole of the compound is 242, determine its molecular formula (1 mk)
18. The diagram below represents set up for large scale manufacture of hydrochloric acid. Study it and answer the questions that follow.

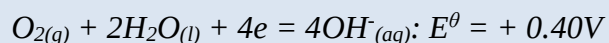
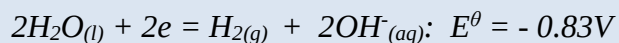


- a) Name substance X (1 mk)

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- b) What is the purpose of the glass beads? (1 mk)
- c) Give two uses of hydrochloric acid. (1 mk)

19. The half equations involved in a cell are:



- a) Write the overall equation for the electrochemical cell. (1 mk)
- b) Calculate the e.m.f. generated by a battery consisting of ten cells. (1 mk)
- c) State **one** environment advantage of using these cells in spacecrafts. (1 mk)

20. In an experiment to prepare nitrogen (I) oxide, ammonium nitrate was gently heated in a flask.

- a) Write the equation for the reaction that took place in the flask. (1 mk)
- b) State and explain how the gas was collected. (1 mk)
- c) A sample of the gas was tested with damp blue and red litmus papers. What observations were made?

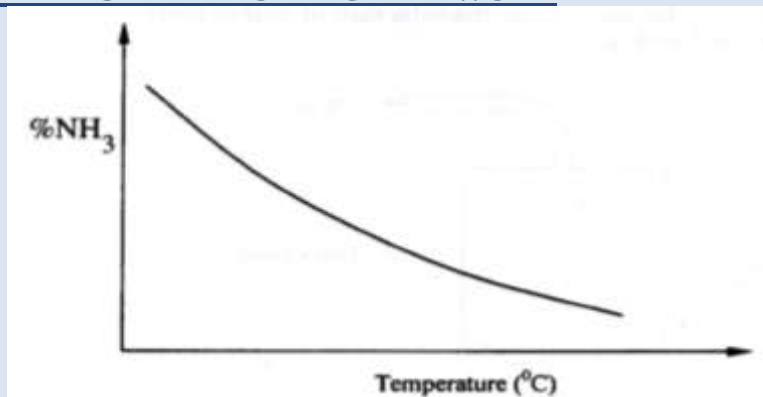
21. The use of CFCs has been linked to depletion of the ozone layer.

- a) What does CFC stand for? (1 mk)
- b) Explain the problem associated with the depletion of the ozone layer (1 mk)
- c) State another environment problem caused by CFCs (1 mk)

22. Nitrogen and hydrogen react to form ammonia gas as shown in the following equation:

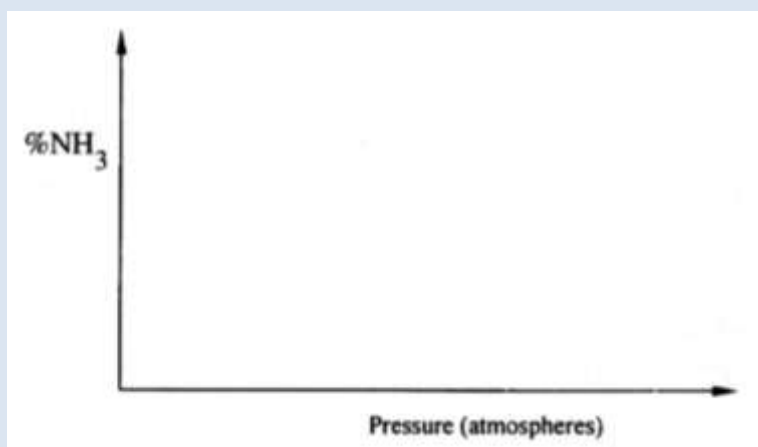


- a) The figure below shows how the percentage of ammonia gas in the equilibrium mixture change with temperature.

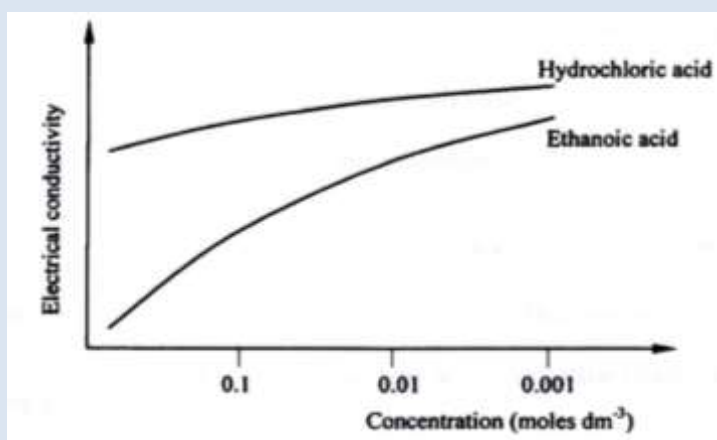


Explain why the percentage of ammonia gas change as shown in the figure. (2 mks)

- b) On the axes below, sketch a graph showing how the percentage of ammonia gas in equilibrium mixture changes with pressure.



23. The curves below shows how the electronic conductivity of hydrochloric and ethanoic acids vary with concentration.



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Explain why the electrical conductivity of 0.01M hydrochloric acids is higher than that of 0.01M ethanoic acid.

(2 mks)

24. Describe how a solid sample of the double salt, ammonium iron(II) sulphate, can be prepared using the following reagents; Aqueous ammonia, sulphuric(VI) acid and iron metal.

(3 mks)

25. A sample of river water was divided into three portions. The table below shows the test carried out on the portions and the observations made.

Test	Observation	Inference
To the first portion, 1cm ³ of soap solution was added	No lather formed	
The second portion was boiled, cooled and 1cm ³ of soap solution was added	No lather formed	
To the third portion, 3cm ³ of aqueous sodium carbonate was added, the mixture filtered and 1cm ³ of soap solution added to the filtrate.	Lather formed immediately	

Complete the table by filling in the inferences.

(3 mks)

26. A water trough, aqueous sodium hydroxide, burning candle, watch glass and a graduated gas jar were used in an experimental set up to determine the percentage of active part of air. Draw a labeled diagram of the set up at the end of the experiment.

(3 mks)

27. The atomic numbers of phosphorus, sulphur and potassium are 15, 16 and 19 respectively. The formulae of their ions are P³⁻, S²⁻ and K⁺. These ions have the same number of electrons.

a) Write the electron arrangement for the ions.

(1 mk)

b) Arrange the ions in the order of increasing ionic radius starting with the smallest. Give a reason for the order.

(2 mks)

CHEMISTRY PAPER 233/1
K.C.S.E. 2011 QUESTION

1.
 - (a) What name is given to the process by which alcohol is formed from a carbohydrate? (1 mk)
 - (b) Explain why the solubility of ethane in water is lower than that of ethanol. (2 mks)
2. Complete the nuclear equation below:

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(a) $^{131}_{53}\text{I} \rightarrow ^{131}_{54}\text{Xe} + \dots\dots\dots$ (1mk)

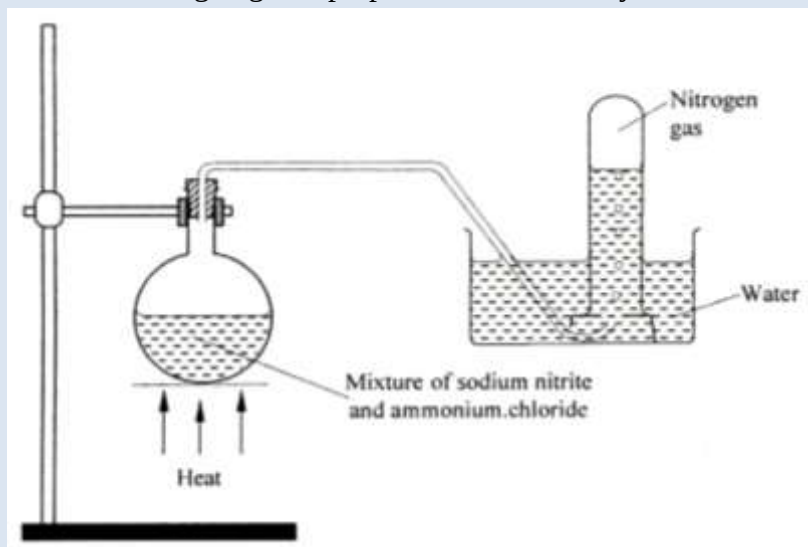
(b) The half-life of $^{131}_{53}\text{I}$ is 8 days

Determine the mass of $^{131}_{53}\text{I}$ remaining if 50 grammes decayed for 40 days. (1 mk)

(c) Give one harmful effect of radioisotopes. (1mk)

3. A mixture contains ammonium chloride, copper (II) oxide and sodium chloride. Describe how each of the substances can be obtained from the mixture. (3mks)

4. The set up below shows how nitrogen gas is prepared in a laboratory.



(a) Describe how nitrogen gas is formed in the flask. (2mk)

(b). Nitrogen is inert, state one use of the gas based on this property. (1 mk)

5. The diagram below represents part of the periodic table. Use it to answer the questions that follow.

M				Q			
T	V		W				

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--	--	--	--	--	--	--	--	--

- a) Write the electronic arrangement for the stable ion formed by W. (1mk)
- (b) Write an equation for the reaction between V and Q. (1mk)
- (c) How do the ionisation energies of the elements M and T compare?
Explain.

6. A certain mass of gas occupies 0.15dm^3 at 293K and $98,648.5\text{Pa}$. Calculate its Volume at $101,325\text{Pa}$ and 273K . (2 mks)

7. When lead(II) nitrate is heated, one of the products is a brown gas.

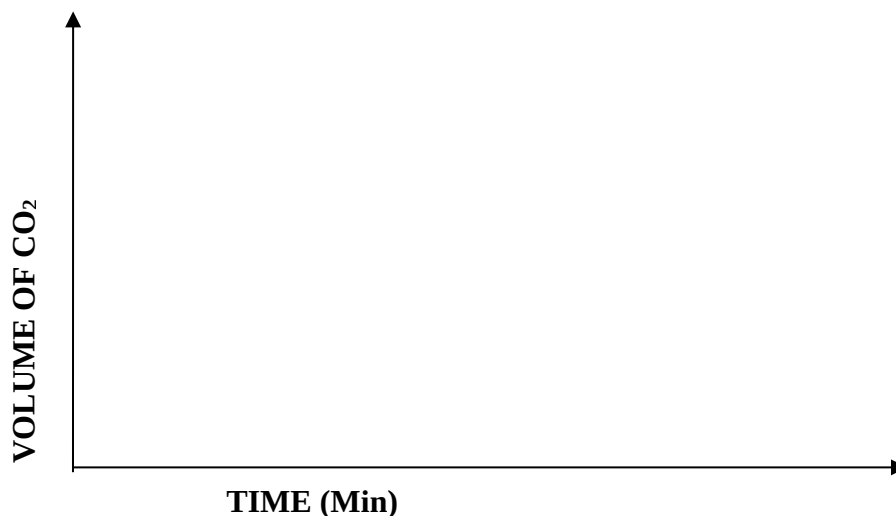
(a) Write the equation of the reaction that occurs . (1 mk)

(b) If 0.290dm^3 of the brown gas was produced, calculate the mass of the lead(II) nitrate that was heated. (R .F.M of lead (II) nitrate = 331; Molar gas volume = 24dm^3). (2 mks)

8. (a) What is meant by a strong acid? (1 mk)

(b) In an experiment 40cm^3 of 0.5M hydrochloric acid was reacted with excess sodium carbonate and the volume of carbon (IV) oxide produced recorded with time. In another experiment, the same volume and concentration of ethanoic acid was also reacted with excess sodium carbonate and the volume of carbon (IV) oxide produced recorded with time.

On the grid below, sketch and label the curves if the volumes of carbon (IV) oxide were plotted against time. (2 mks)



9. State two reasons why hydrogen is not commonly used as a fuel. (2mk)

10. During a class experiment, chlorine gas was bubbled into a solution of potassium iodide.

(a) State the observations made. (1mk)

(b) Using an ionic equation, explain why the reaction is redox. (2mk)

11. Exhaust fumes of some cars contain carbon(II)oxide and other gases. (1mk)

(a) Explain how carbon (II) oxide is formed in the internal combustion engines.

(b) Name two gases other than carbon (II) oxide that are contained in exhaust fumes and are pollutants. (2 mks)

12. Sodium hydroxide can be prepared by the following methods; I and II.

I. Sodium metal $\xrightarrow{\text{Cold water}}$ Sodium hydroxide + Hydrogen

II. Concentrated sodium chloride $\xrightarrow{\text{process A}}$ Sodium hydroxide + Chlorine + Hydrogen

(a) Name the precaution that needs to be taken in method I. (1 mk)

(b) Give the name of process A. (1 mk)

(c) Give one use of sodium hydroxide, (1 mk)

13. Distinguish between the terms deliquescent and efflorescent as used in chemistry (2 mk)

14. Two organic compounds P and Q decolourise acidified potassium manganate (VII)

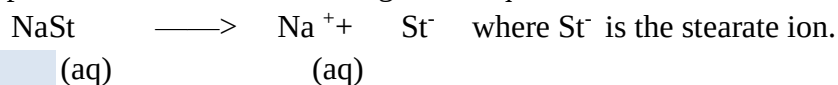
KCSE CHEM PAPER 1 PAST PAPERS MR CHEPKWONY

solution but only P reacts with sodium metal to give a colourless gas.

Which homologous series does compound P belong? Give a reason.

(2 mks)

15. Soap dissolves in water according to the equation below;



- (a) Write the formula of the scum formed when soap is used in hard water. (1 mk)

- (b) Write the ionic equation for the reaction that occurs when sodium carbonate is used to remove hardness in water. (1 mk)

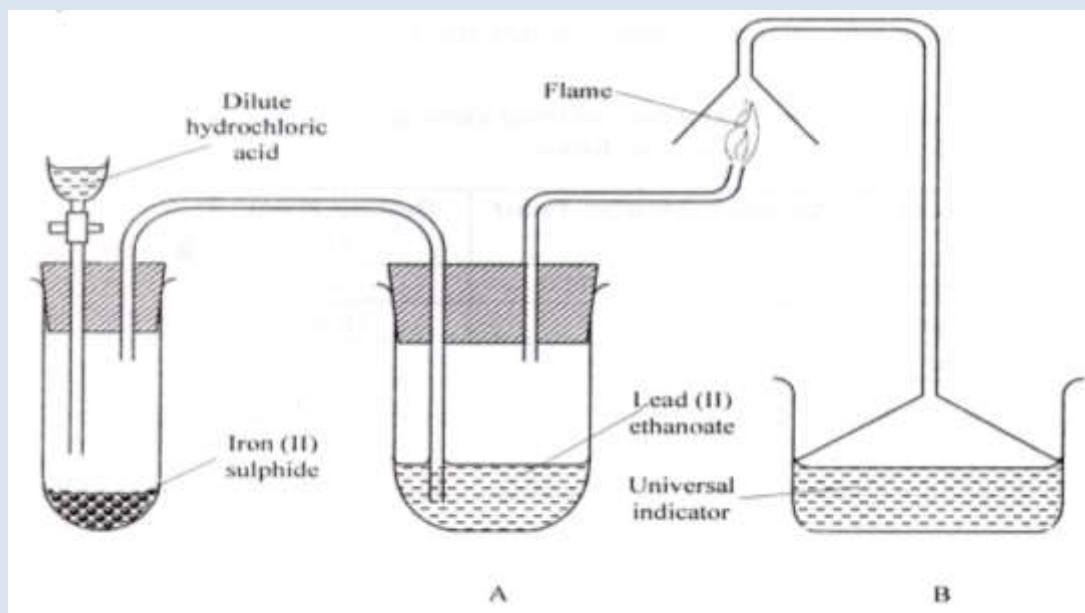
16. Ethanoic acid and ethanol react as shown in the equation below:

Other than warming, how would the state of equilibrium be established within a short time?

(1 mk)

17. The set up below was used to prepare a gas and study some of its properties.

Study it and answer the questions that follow:



- (a) State and explain the observations made in the:

- (i) tube labelled A;

(1 mk)

(ii) beaker labelled B.

(b) State one precaution that should be taken when carrying out this experiment.

(1 mk)

18. Under certain conditions, chlorine gas reacts with sodium hydroxide to form sodium hypochlorite.

(a) Name the conditions under which sodium hydroxide reacts with chlorine to form sodium hypochlorite.

(1 mk)

(b) State two uses of sodium hypochlorite.

(2 mks)

19. 50kg of ammonium sulphate $(\text{NH}_4)_2\text{SO}_4$ and 30kg of urea $\text{CO}(\text{NH}_2)_2$ fertilizers were applied in two equal sizes of plots A and B to enrich their nitrogen content. Show by working, which plot was more enriched with nitrogen. (N = 14; S = 32; O = 16; C = 12; H = 1)

(3 mks)

20. Describe how the P^{H} of anti-acid (Actal) powder can be determined in the laboratory

(2 mks)

21. Graphite is one of the allotropes of carbon.

(a) Name one other element which exhibits allotropy.

(1 mk)

(b) Explain why graphite is used in the making of pencil leads.

(2 mks)

22. The table below gives some properties three elements in group (VII) of the periodic table. Study it and answer the questions that follow:

Element	Atomic No.	Melting Point ($^{\circ}\text{C}$)	Boiling Point ($^{\circ}\text{C}$)
Chlorine	17	101	-34.7
Bromine	35	-7	58.8
Iodine	53	114	184

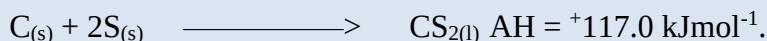
(a) Which element is in liquid form at room temperature? Give a reason.

(1 mk)

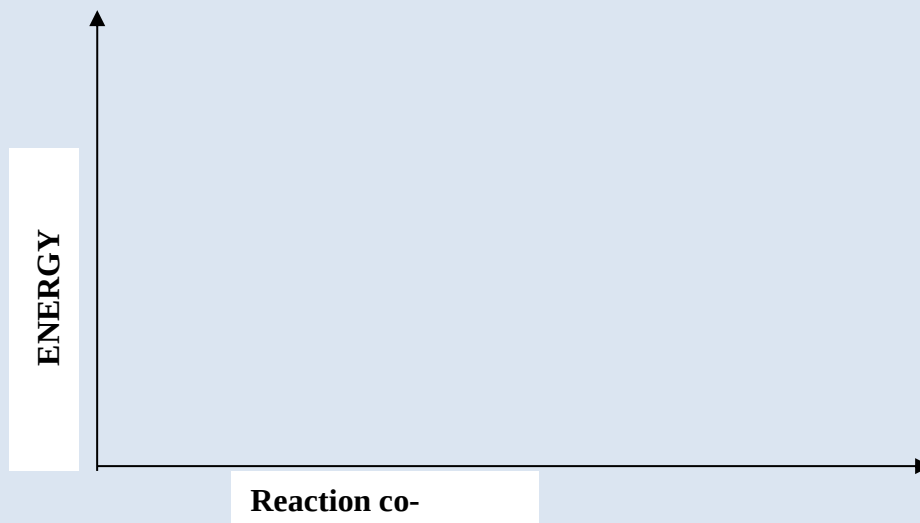
(b) Explain why the boiling point of iodine is much higher than that of chlorine.

(2 mks)

23. The thermal chemical reaction between carbon and sulphur is as shown by the equation below:



On the grid below, sketch and label the energy level diagram for the reaction. (2 mks)



24. The table below gives the number of electrons, protons and neutrons in substances X, Y and Z. Study it and answer the questions that follow.

Substance	Electrons	Protons	Neutrons
X	10	10	10
Y	10	8	10
Z	8	8	8

- (a) Which letter represents an ion? (1 mk)
- (b) Which of the substances are isotopes? Give a reason. (2mk)
- 25 (a) State the Gay Lussac's Law. (1 mk)
- (b) 10cm^3 of a gaseous hydrocarbon, C_2H_x required 30cm^3 of oxygen for complete combustion. If steam and 20cm^3 of carbon (IV) oxide were produced, what is the value of X? (2 mks)

26. The data given below was recorded when Metal M was completely burnt in air.
M is not the actual symbol of the metal. (R.A.M; M = 56, O ==16)

Mass of empty crucible and lid = 10.240g

Mass of crucible, lid and metal M = 10.352g

Mass crucible, lid and metal oxide = 10.400g

- (a) Determine the mass of:

(i) Metal M

($\frac{1}{2}$ mk)

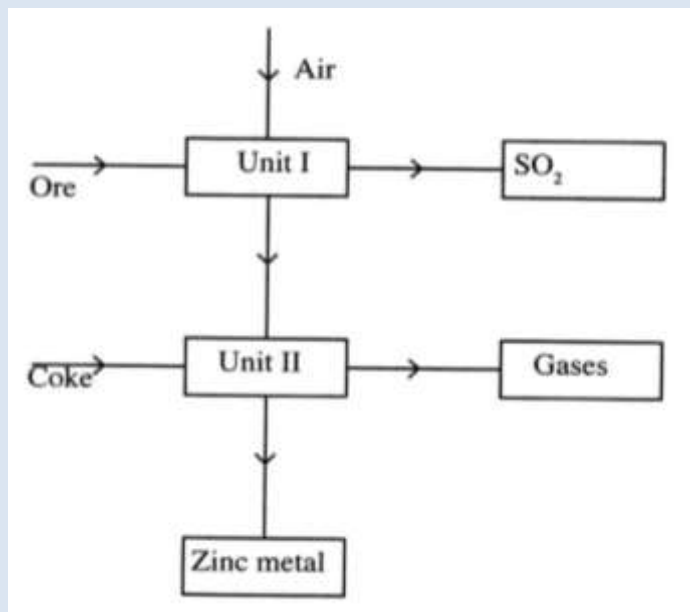
(ii) oxygen.

($\frac{1}{2}$ mk)

- (b) Determine the empirical formula of the metal oxide.

(2 mks)

27. The flow chart below shows some processes involved in the extraction of zinc metal:



- a) Name one ore from which zinc is extracted.

(1 mk)

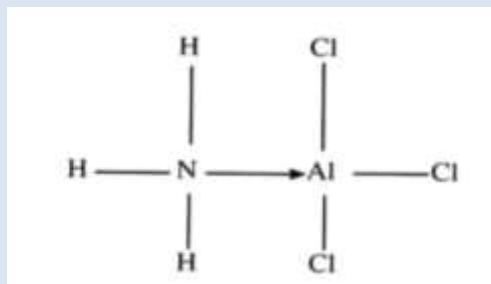
- b) Write the equation of the reaction taking place in unit II

(1 mk)

- c) Name two uses of zinc metal.

(1 mk)

28. The diagram below shows the bonding between aluminium chloride and ammonia.

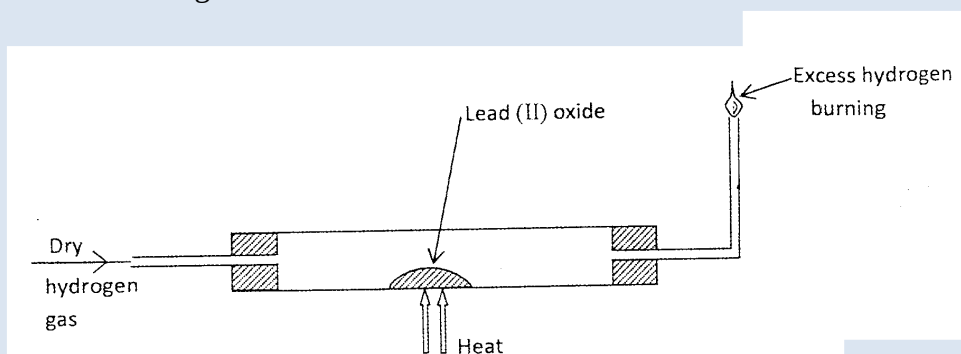


- (a) Name the types of bonds that exist in the molecule. (1mk)
- (b) How many electrons are used for bonding in the molecule? (1 mk)
29. Explain why the following substances conduct an electric current.
- (a) Magnesium metal. (1mk)
- (b) Molten magnesium chloride. (1mk)
30. A sample of river water is suspected to contain zinc and sulphate ions.
- Describe how the presence of zinc ions and sulphate ions can be established. (3 mks)
31. What name is given to elements which appear in group (II) of the periodic table? (1mk)

CHEMISTRY PAPER 233/1

K.C.S.E. 2012 QUESTIONS

1. Charcoal is a fuel that is commonly used for cooking. When it burns it forms two oxides.
 - (a) Name the **two** oxides. (2 mks)
 - (b) State **one** use of any of the two oxides. (1 mk)
2. Iron (III) oxide was found to be contaminated with copper (II) sulphate. Describe how a pure sample of iron (III) oxide can be obtained. (3 mks)
3. In an experiment, dry hydrogen gas was passed over heated Lead (II) Oxide as shown in the diagram below.



- State and explain the observations made in the combustion tube. (3 mks)
4. The table below shows properties of some elements A, B, C and D which belong to the same period of the periodic table. The letters are not the actual symbols of the elements.

Element	A	B	C	D
Mp (°C)	1410	98	-101	660
Atomic radii (nm)	0.117	0.186	0.099	0.143

KCSE CHEM PAPER 1 PAST PAPERS MR CHEPKWONY

Electrical conductivity	Poor	Good	Non conductors	Good
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(a) Arrange the elements in the order they would appear in the period. Give a reason. (2 mks)

(b) Select the metallic element which is the better conductor of electricity. Give a reason. (1 mk)

5. A sample of water in a beaker was found to boil at 101.5°C at 1 atmospheric pressure.

Assuming that the thermometer was not faulty, explain this observation. (1 mk)

6. Study the information in the table below and answer the questions that follow:

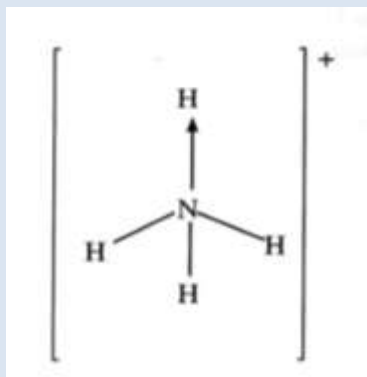
Salt	Solubility (g/100g water)	
	at 40°C	at 60°C
CuSO ₄	28	38
Pb(NO ₃) ₂	79	98

A mixture containing 35g of CuSO₄ and 78g of Pb(NO₃)₂ in 100g of water at 60°C was cooled to 40°C.

(a) Which salt crystallised out? Give a reason (2 mks)

(b) Calculate the mass of the salt that crystallised out. (1 mk)

7. Ammonium ion has the following structure:

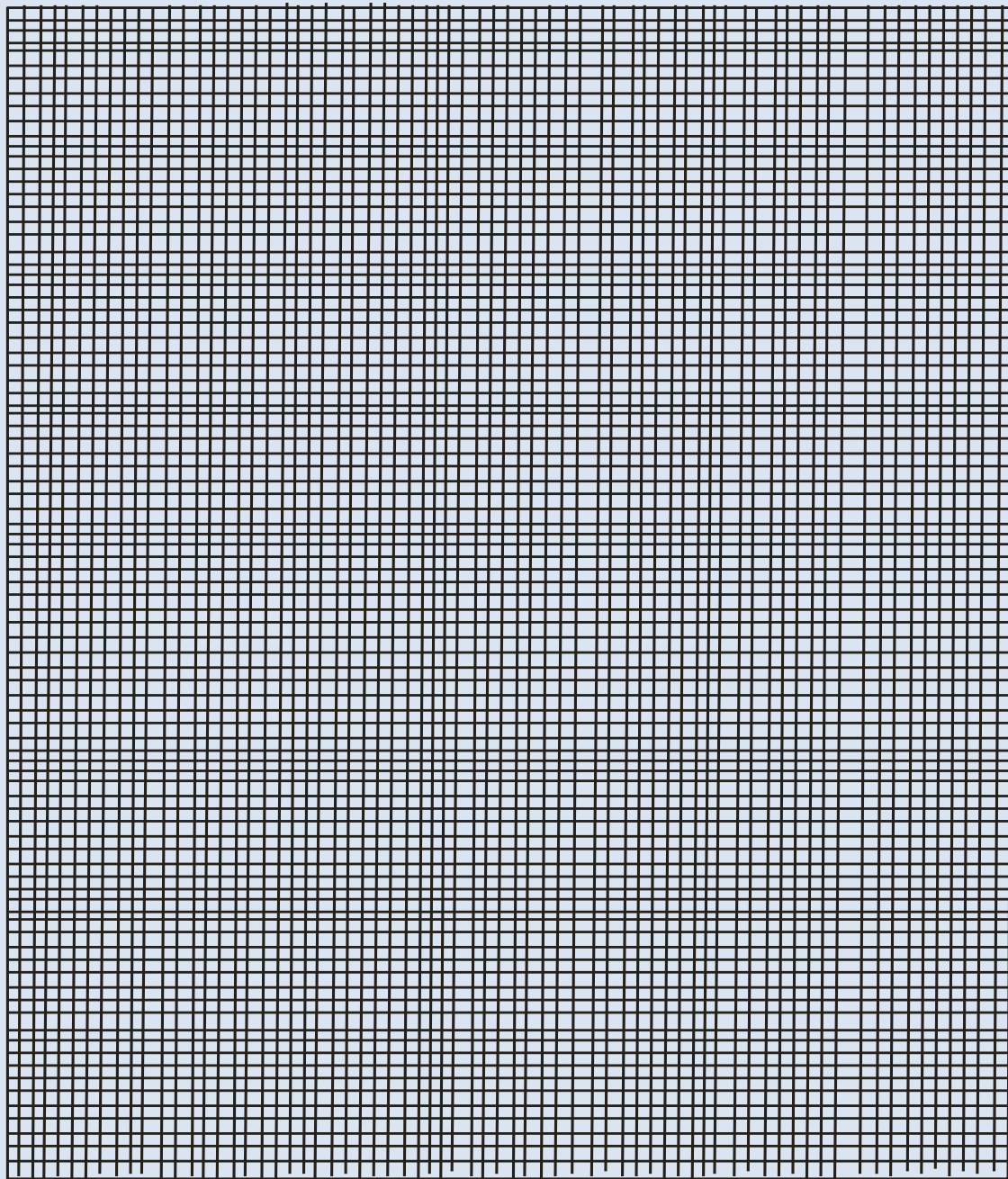


Label on the structure:

(a) covalent bond; (1 mk)

(b) coordinate (dative) bond. (1 mk)

8. 10cm^3 of concentrated sulphuric (VI) acid was diluted to 100cm^3 . 10cm^3 of the Resulting solution was neutralised by 36cm^3 of 0.1M sodium hydroxide solution. Determine the mass of sulphuric (VI) acid that was in the concentrated acid (S = 32.0; H = 1.0; O = 16.0). (3 mks)
9. 120g of iodine - 131 has a half life of 8 days and decays for 32 days. On the grid provided, plot a graph of the mass of iodine - 131 against time. (3 mks)

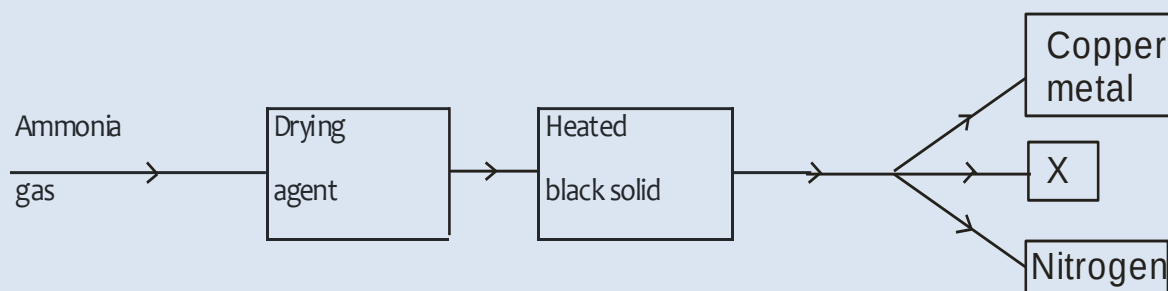


10. (a) Name **two** cations that are present in hard water. (1 mk)
- (b) Explain how the ion exchange resin softens hard water. (2 mks)
11. The empirical formula of A is CH_2Br . Given that 0.470g of A occupies a volume of 56cm^3 at 546K and 1 atmospheric pressure, determine its molecular formula.

(H = 1.0, C = 12.0, Br = 80.0, molar gas volume at STP = 22.4 dm³).

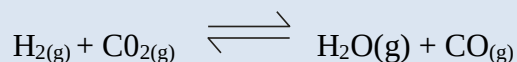
(3 mks)

12. Study the flow chart below and answer the questions that follow.



- (a) Name a suitable drying agent for ammonia. (1 mk)
- (b) Describe one chemical test for ammonia. (1 mk)
- (c) Name X. (1 mk)

13. A dynamic equilibrium is established when hydrogen and carbon (IV) oxide react as shown below:



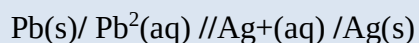
What is the effect of adding powdered iron catalyst on the position of the equilibrium?
Give a reason.

(2 mks)

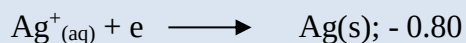
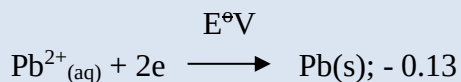
14. Distinguish between ionisation energy and electron affinity of an element.

(2 mks)

15. Below is a representation of an electrochemical cell.



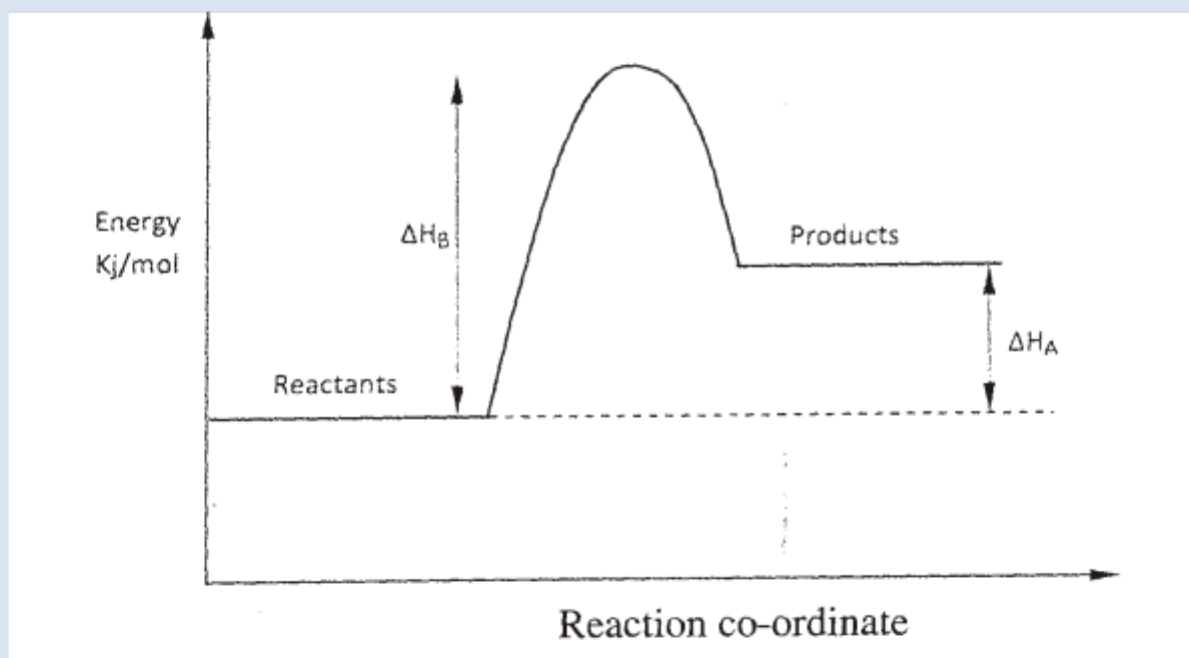
- (a) What does // represent? (1 mk)
- (b) Given the following:



Calculate the E.M.F of the electrochemical cell.

(2 mks)

16. Use the following information on substances S, T, V and hydrogen to answer the questions that follow:
- (i) T displaces V from a solution containing V ions.
 - (ii) Hydrogen reacts with the heated oxide of S but has no effect on heated oxide of V.
- (a) Arrange substances S, T, V and hydrogen in the order of increasing reactivity. (2 mks)
- (b) If T and V are divalent metals, write an ionic equation for the reaction in (i) above. (1 mk)
- 17 Study the energy level diagram below and answer the questions that follow.

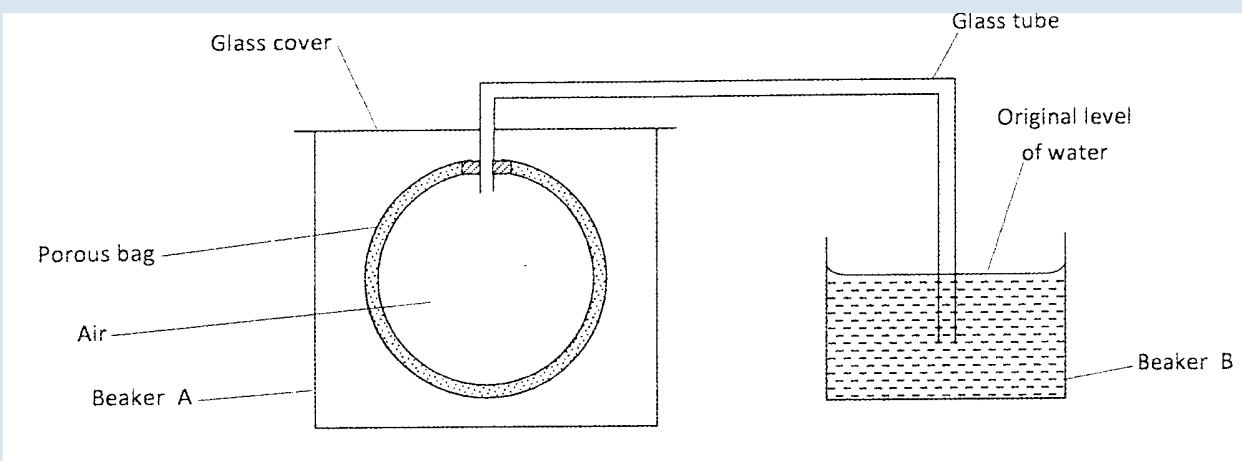


- (a) Give the name of ΔH_A (1 mk)
- (b) How can ΔH_B be reduced? Give a reason. (2 mks)

18. Acidified potassium manganate (VII) solution is decolourised when sulphur (IV) oxide is bubbled through it. The equation for the reaction is given below.



- (a) Which reactant is oxidised? Explain. (2 mks)
- (b) Other than the manufacture of sulphuric (VI) acid, state one other use of sulphur (IV) oxide. (1 mk)
19. The set up shown below was used to investigate a property of hydrogen gas.



- State and explain the observation that would be made in the glass tube if beaker A was filled with hydrogen gas. (3 mks)
20. Draw and name the isomers of pentane. (3 mks)
21. Give **two** uses of the polymer polystyrene. (1 mk)
22. Aluminium is both malleable and ductile,
(a) What is meant by?
(i) malleable; (1 mk)

(ii) ductible. (1 mk)

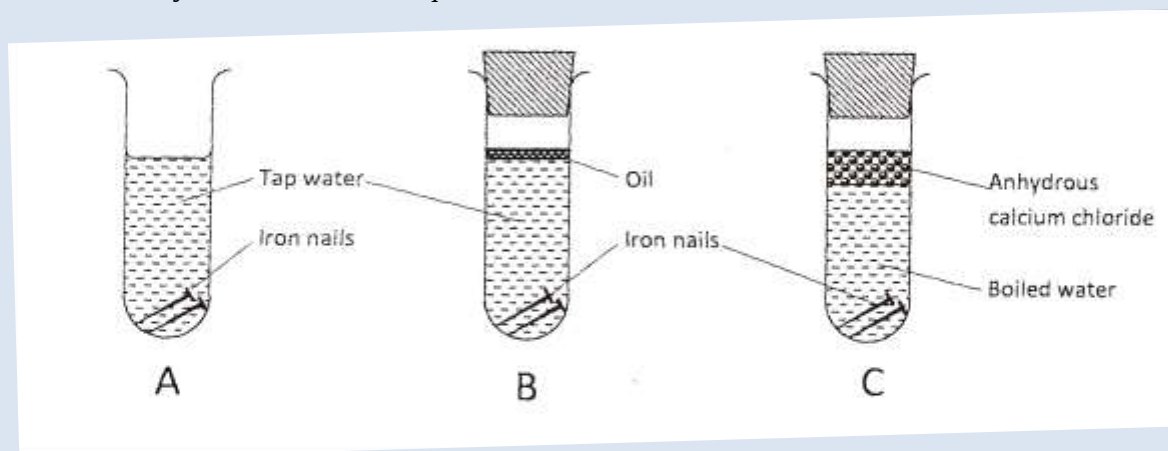
(b) State **one** use of aluminium based on:

(i) malleability (1mk)

(ii) ductility (1mk)

23. Describe how the percentage by mass of copper in copper carbonate can be determined. (3 mks)

24. The following set up of three test-tubes was used to investigate rusting of iron. Study it and answer the questions that follow.



(a) Give a reason why rusting did not occur in test-tube C. (1 mk)

(b) Aluminium is used to protect iron sheets from rusting. Explain **two** ways in which aluminium protects iron from rusting. (2 mks)

25. Describe how a solid sample of potassium sulphate can be prepared starting with 200cm^3 of 2M potassium hydroxide. (3 mks)

26. Describe **two** chemical tests that can be used to distinguish ethanol from ethanoic acid. (3 mks)

- 27 (a) The electronic arrangement of the ion of element Q is 2.8.8. If the formula of the ion is Q^3 , state the group and period to which Q belongs.

Group:

($\frac{1}{2}$ mk)

Period:

($\frac{1}{2}$ mk)

- (b) Helium, neon and argon belong to group 8 of the periodic table. Give:

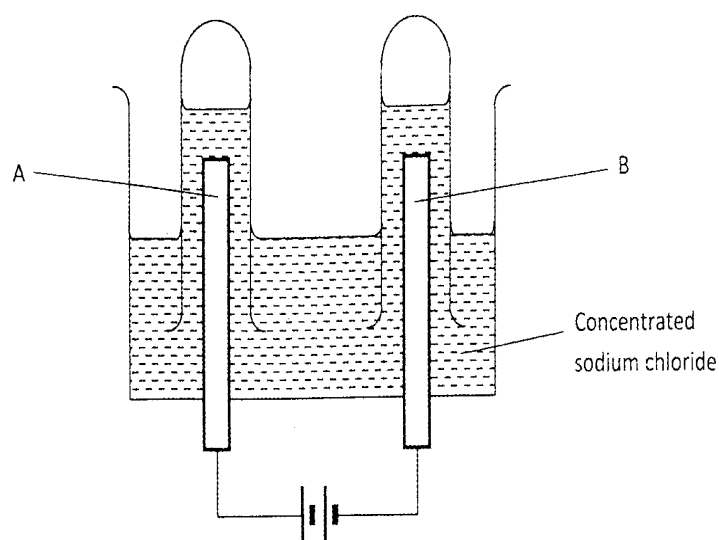
(i) the general name of these elements;

(1 mk)

(ii) one use of these elements.

(1 mk)

- 28 The apparatus shown in the diagram below were used to investigate the products formed when concentrated sodium chloride was electrolysed using inert electrodes.



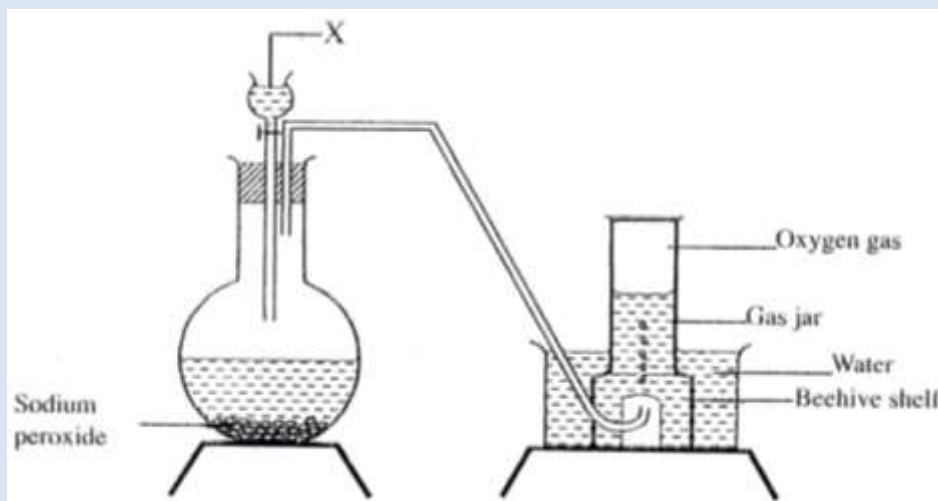
- (a) Write the equation for the reaction that takes place at electrode A.

(1 mk)

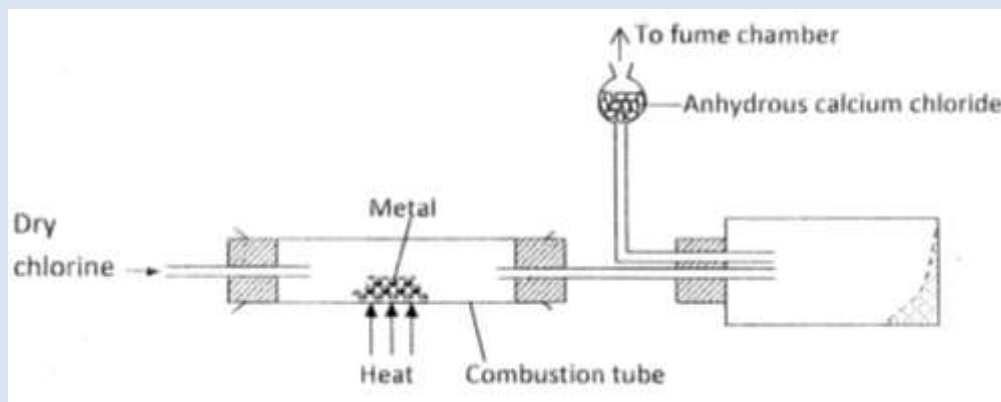
- (b) If the concentrated sodium chloride was replaced with dilute sodium chloride, what product would be formed at electrode A? Explain. (2 mks)
29. a) State and explain what would happen if a dry blue litmus paper was dropped in a gas. (1mk)
- b) By using only dilute hydrochloric acid, describe how a student can distinguish between barium sulphite from barium sulphate. (2mks)

CHEMISTRY PAPER 1
KCSE 2013 THEORY

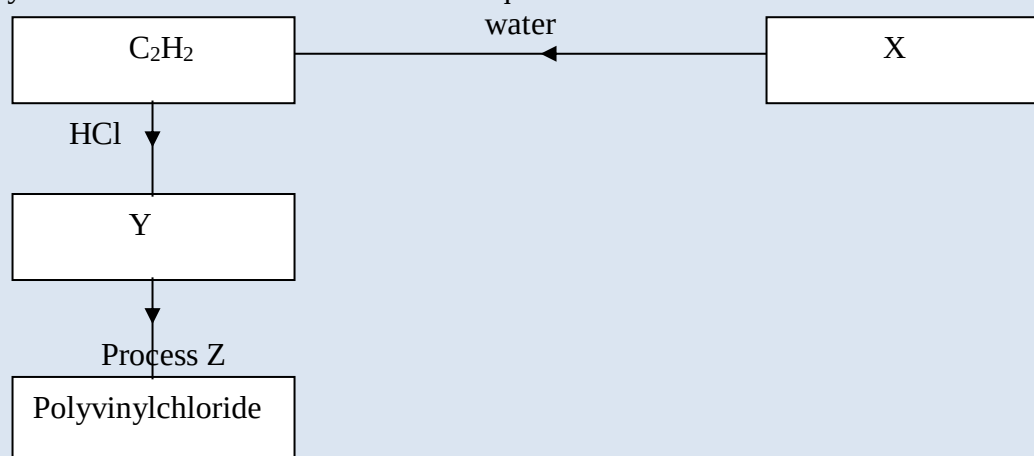
1. The set up below can be used to prepare oxygen gas. Study it and answer the questions that follow.



- (a) Identify X (1 mk)
- (b) What property of oxygen makes it possible for it to be collected as shown in the above set up? (1 mk)
- (c) State **two** uses of oxygen (1 mk)
2. Write equations to show the effect of heat on each of the following:
- (a) Sodium hydrogen carbonate (1 mk)
- (b) Silver nitrate (1 mk)
- (c) Anhydrous iron (II) sulphate (1 mk)
3. Describe an experiment procedure that can be used to extract oil from nut seeds (2 mks)
4. In terms of structure and bonding, explain the following observations:
- (a) The melting point of aluminum is higher than that of sodium (1½ mk)
- (b) Melting point of chlorine is lower than that of sulphur (1½ mk)
5. The diagram below illustrates a method of preparing salts by direct synthesis



- (a) This method can be used to prepare either aluminum chloride or iron (III) chloride. Explain why it cannot be used to prepare sodium chloride. (2 mks)
- (b) Describe how a sample of sodium chloride can be prepared in the laboratory by direct synthesis. (2 mks)
6. (a) A student electroplated a spoon with copper metal. Write an equation for the process that took place at the cathode. (1 mk)
- (b) Calculate the time in minutes required to deposit 1.184g of copper if a current of 2 amperes was used. (1 Faraday = 96500 coulombs, Cu=63.5). (2 mks)
7. Study the flowchart below and answer the questions that follow:



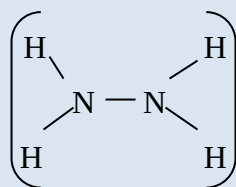
- (a) Identify:
- (i) X (1 mk)
 - (ii) Y (1 mk)
- (b) State two uses of polyvinylchloride (1 mk)

8. Draw a labeled diagram to illustrate how alpha, beta and gamma radiations can be distinguished from each other. (3 mks)
9. Aqueous hydrogen chloride reacts with potassium manganate (VII) to produce chlorine gas, while a solution of hydrogen chloride in methylbenzene has no effect on potassium manganate (VII). Explain this observation (2 mks)
10. The table below gives the solubility's of substances T and U at 10⁰c and 40⁰c.

Substance	Solubility g/100g water	
	10(⁰ c)	40(⁰ c)
T	40	65
U	15	17

When an aqueous mixture containing 55g of T and 12g of U at 80⁰c was cooled to 10⁰c, crystals formed.

- a) Identify the crystals formed (1 mk)
- b) Determined the mass of the crystals formed (1 mk)
- c) Name the method used to obtain the crystals (1 mk)
11. Hydrazine gas,



burns in oxygen to form nitrogen gas and steam.

- a) Write an equation for the reaction (1 mk)
- b) Using the bond energies given below, calculate the enthalpy change for the reaction in (a) above (2 mks)

Bond	Bond energy (kj per mole)
$\text{N} \equiv \text{N}$	944
$\text{N} - \text{N}$	163

N - H	388
O = O	496
H - O	463

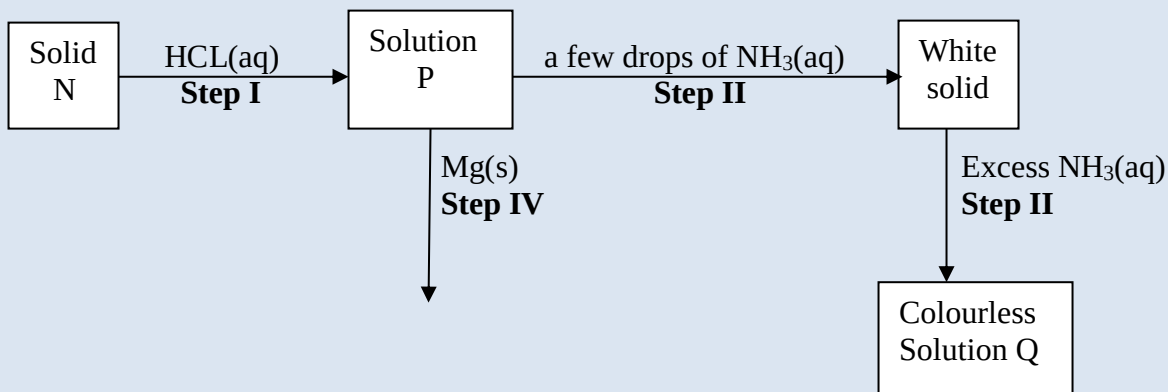
12. a) What would be observed if sulphur (IV) oxide is bubbled through acidified potassium manganate (VII) (1 mk)

(b) In an experiment, sulphur (IV) oxide was dissolved in water to form solution L.

(i) What would be observed if a few drops of barium nitrate solution were **immediately** added to solution L? (1 mk)

(ii) Write an ionic equation for the reaction that occurred between solution L and aqueous barium nitrate in (b) (i) above. (1 mk)

13. The scheme below shows some reaction sequence starting with solid N. Study it and answer the questions that follow.



a) Write the formula of the complex ion in solution Q. (1 mk)

b) Write an equation for the reaction in step IV. (1 mk)

14. (a) State the Charles' law (1 mk)

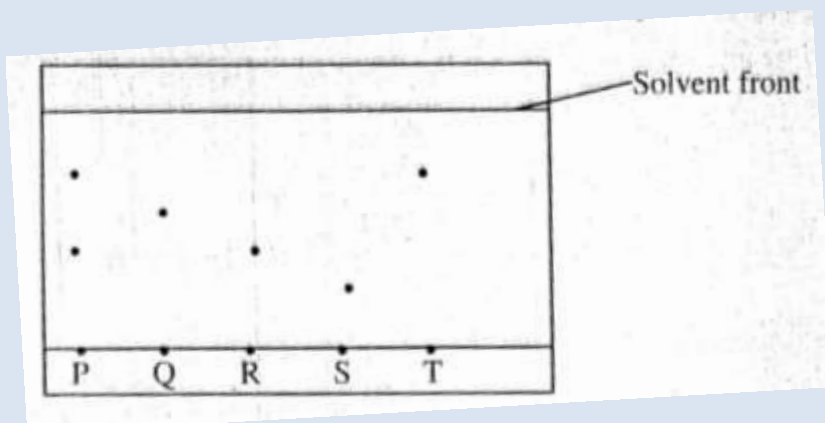
(b) A certain mass of gas occupies 146 dm³ at 291 K and 98.31 kPa.

What will be its temperature if its volume is reduced to 133 dm³ at

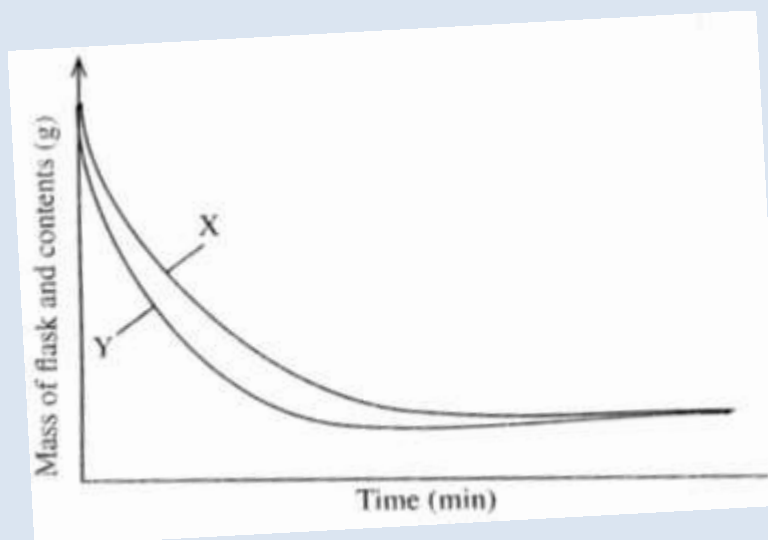
101.325 kPa?

(2 mks)

15. The chromatogram below was obtained from a contaminated food sample P. Contaminants Q, R, S and T are suspected to be in P. Use it to answer the following questions.



- a) Identify the contaminants in mixture P. (1 mk)
- b) Which is the most soluble contaminant in P.? (1 mk)
16. The curves below represent the change in mass when equal masses of powdered zinc and zinc granules were reacted with excess 2M hydrochloric acid. Study them and answer the question below.



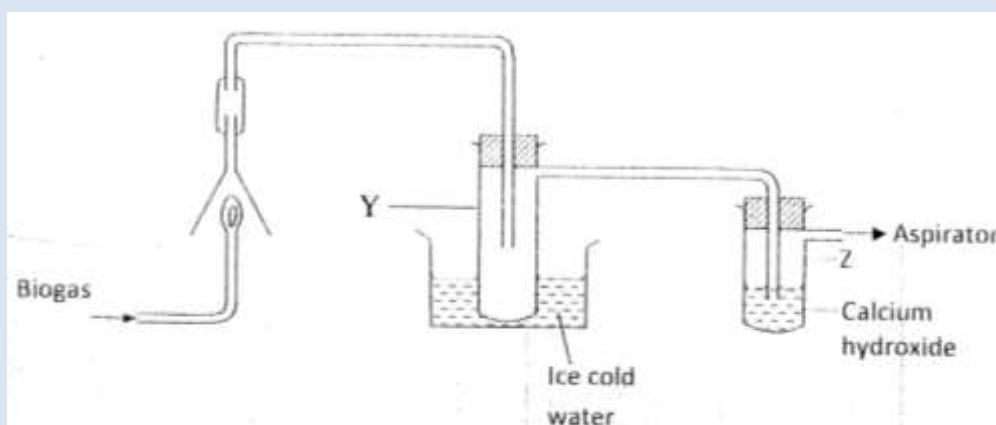
Which curve represents the reaction with zinc granules? Explain your answer. (3 mks)

17. When fuels burn in the internal combustion engine at high temperature, one of the products formed is nitrogen (II) oxide.

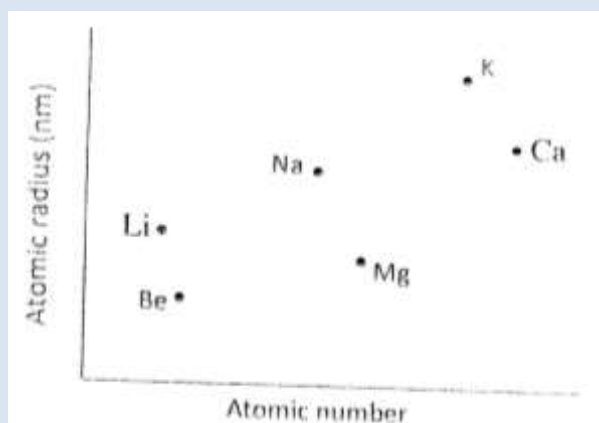
- Write the equation for the formation of nitrogen (II) oxide. (1 mk)
- Give a reason why nitrogen (II) oxide is not formed at room temperature. (1 mk)

- Describe how formation of nitrogen (II) oxide in the internal combustion engine leads to gaseous pollution. (1 mk)

18. The set-up below was used to investigate the products of burning biogas (methane). Study it and answer the questions that follow.



- a) What product will be formed in test- tube Y? (1 mk)
b) State and explain the observations which would be made in Z? (2 mks)
19. (a) Diamond and graphite are allotropes of carbon. What is meant by an allotrope? (1 mk)
(b) Explain why graphite can be used as a lubricant while diamond cannot. (1 mk)
20. The plots below were obtained when the atomic radii of some elements in groups I and II were plotted against atomic numbers.



Explain:

- a) The trend shown by Li, Na and K. (1 mk)
b) Why the atomic radii of elements Be, Mg, and Ca are lower than those of Li, Na and K. (2 mks)
21. On heating a pale green solid K, carbon (IV) oxide gas and a black solid M were formed. On reacting K with dilute hydrochloric acid, carbon (IV) oxide gas and green solution S were formed. When excess aqueous ammonia was added to solution S, a deep blue solution was formed.
- a) Identify the cation in solid K (1 mk)
b) Identify the **two** anions in solution S. (2 mks)
22. a) Name **two** ores from which copper is extracted (1 mk)
b) During extraction of copper metal, the ore is subjected to froth flotation. Give a reason why this process is necessary. (1 mk)

- c) Name **one** alloy of copper and state its use. (1 mk)
23. When 15 cm^3 of a gaseous hydrocarbon, P, was burnt in 100 cm^3 of oxygen, the resulting gaseous mixture occupied 70 cm^3 at room temperature and pressure. When the gaseous mixture was passed through potassium hydroxide solution, its volume decreased to 25 cm^3
- a) What volume of oxygen was used during the reaction? (1 mk)
- b) Determine the molecular formula of the hydrocarbon (2 mks)
24. A solution was made by dissolving 8.2g of calcium nitrate to give 2 litres of solution. (Ca= 40.0; N=14.0; O= 16.0)
Determine the concentration of nitrate ions in moles per litre. (3 mks)
25. State and explain what would happen if a dry red litmus paper was dropped in a gas jar of dry chlorine. (2 mks)
26. By using aqueous sodium chloride, describe how a student can distinguish calcium ions from lead ions. (2 mks)
27. A student investigated a property of acids M and N by reacting equal volumes of acid M and N of the same concentration with equal volumes of 2M potassium hydroxide. The results were recorded in the table below.

Acid	Rise in temperature(Δ)K
M	4
N	2

- a) Which of the acids is likely to be a weak acid? Explain. (2 mks)
- b) Write the equation for the reaction between ethanoic acid and potassium hydroxide. (1 mk)
28. A student investigated the effect of an electric current by passing it through some substances.
The student used inert electrodes, and connected a bulb to the circuit. The table below shows the substances used and their states.

Experiment	Substance	State
1	Potassium Carbonate	Solid
2	Copper II Sulphate	Solution
3	Sugar	Solution
4	Lead II iodide	Molten

- a) In which experiments did the bulb **not** light? (1 mk)
- b) Explain your answer in (a) above. (2 mks)
29. A sample of hydrogen gas was found to be a mixture of two isotopes ${}^1_1\text{H}$ and ${}^2_1\text{H}$. Determine the relative molecular masses of the molecules formed, when each of these isotopes is burnt in oxygen. (O =16.0) (2 mks)

${}^1_1\text{H}$

${}^2_1\text{H}$

CHEMISTRY PAPER (233/1)

KCSE 2014 QUESTIONS

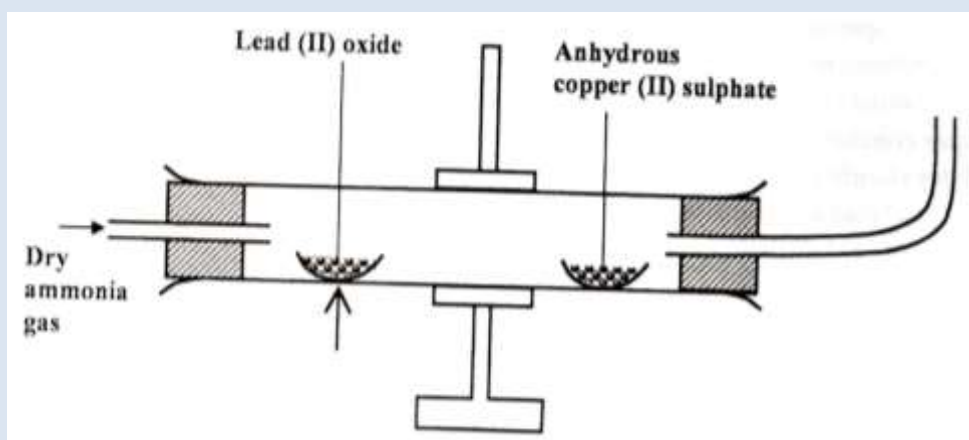
1. Explain how the hotness of a Bunsen burner flame can be increased (1 mk)

2. When dilute hydrochloric acid was reacted with solid B, a colourless gas which extinguished a burning splint was produced. When an aqueous solution of solid B was tested with a blue litmus paper, the paper turned red / pink.

(a) Identify the anion present in solid B. (1 mk)

(b) Write an ionic equation for the reaction between solid B and dilute hydrochloric acid. (1 mk)

3. Dry ammonia gas was passed over heated lead (II) oxide and the product passed over anhydrous Copper (II) sulphate as shown in the diagram below.



State:

(a) Two observations made in the combustion tube. (2 mks)

(b) The property of ammonia gas shown in this experiment (1 mk)

4. starting with zinc sulphate solution, describe how a sample of zinc oxide can be obtained (3 mks)

5. Explain how condition of electricity takes place in the following. (1 mk)

(a) Iron metal

(b) Molten lead (II) iodide (1 mk)

6. 100cm^3 of a sample of ethane gas diffuses through a porous pot in 100 seconds.

What is the molecular mass of gas Q if 1000 cm³ of the gas diffuses through the same porous not in 121 seconds under the same conditions? (C=12.0, H=1.0) (3 mks)

7. (a) Draw and name the isomers of butyne (2 mks)

(b) State **one** use of polystyrene. (1 mk)

8. Complete the nuclear reaction below

(a) $\frac{226}{88}\text{Q} \Rightarrow \frac{226}{88}\text{Q}$ (1 mk)

(b) State two uses of radioisotopes in health (2 mks)

9. The table below shows the relative molecular masses and boiling points of pentane and ethanoic acid

	Relative molecular mass	Boiling point (°C)
Pentane	72	36
Ethanoic acid	60	118

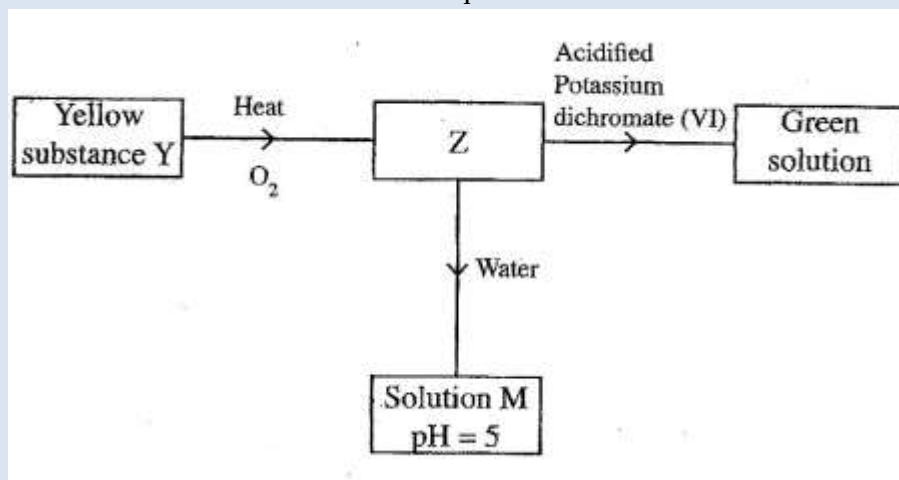
Explain the large difference in boiling point between ethanoic acid and pentane. (2 mks)

10. One of the ores of copper has formula, CuFeS₂.

(a) Describe how iron in the ore is removed during concentration of copper metal. (1 mk)

(b) State **two** environmental problems associated with extraction of copper metal. (2 mks)

11. Study the flow chart below and answer the questions that follow.



Identify Z and M.

(2 mks)

Z.....

M.....

12. The table below shows the pH values of solutions A, B, C and D

Solution	A	B	C	D
pH	2	7	11	14

Select solutions in which a sample of lead (II) hydroxide is likely to dissolve. Give reasons for each solution selected.

(3 mks)

13. 100cm^3 of 0.005 M sulphuric (VI) acid were placed in a flask and a small quantity of anhydrous sodium carbonate added. The mixture was boiled to expel all the carbon (IV) oxide.

25cm^3 of the resulting solution required 18cm^3 of 0.1 M sodium hydroxide solution to neutralize it. Calculate the mass of sodium carbonate added.

(Na = 23.0; O=16.0; C=12.0)

(3 mks)

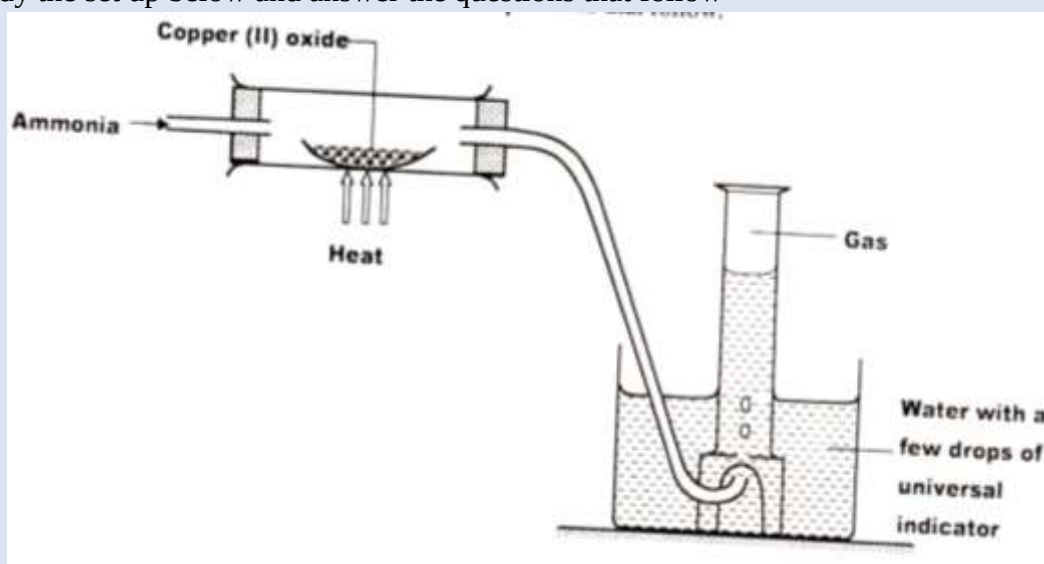
14. When 20cm^3 of 1 M sodium hydroxide was mixed with 20cm^3 of 1 M hydrochloric acid, the temperature rose by 6.7°C . Assuming the density of the solution is 1 g/cm^3

and the specific heat capacity of the solution is $4.2\text{Jg}^{-1}\text{k}^{-1}$.

(a) Calculate the molar heat of neutralization; (2 mks)

((b) When the experiment was repeated with 1 M ethanoic acid, the temperature changes was found to be lower than that with 1 M hydrochloric acid. Explain. (1 mk)

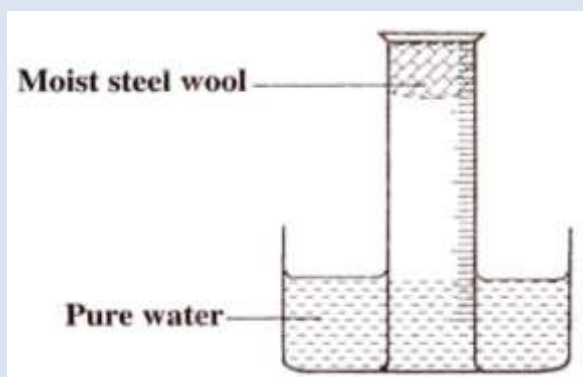
15. Study the set up below and answer the questions that follow



(a) Write an equation for the reaction between ammonia and copper (II) oxide. (1 mk)

(b) During the experiment, the colour of the contents in the water trough changed. State the colour change observed and give an explanation. (2 mks)

16. A measuring cylinder fitted with moist steel wool was inverted in a trough of water as shown in the diagram below



(a) State and explain the observations made on the;

i) Moist steel wool after four days.

(1 mk)

ii) Water level in the measuring cylinder after four days.

(1 mk)

(b) What would be the effect of using steel wool moistened with salty water?

(1 mk)

17. In an experiment on rates of reaction, potassium carbonate was reacted with dilute sulphuric (Vi) acid

(a) What would be the effect of an increase in the concentration of the acid on the rate of the reaction?

(1 mk)

(b) Explain why the rate of reaction is found to increase with temperature.

(2 mks)

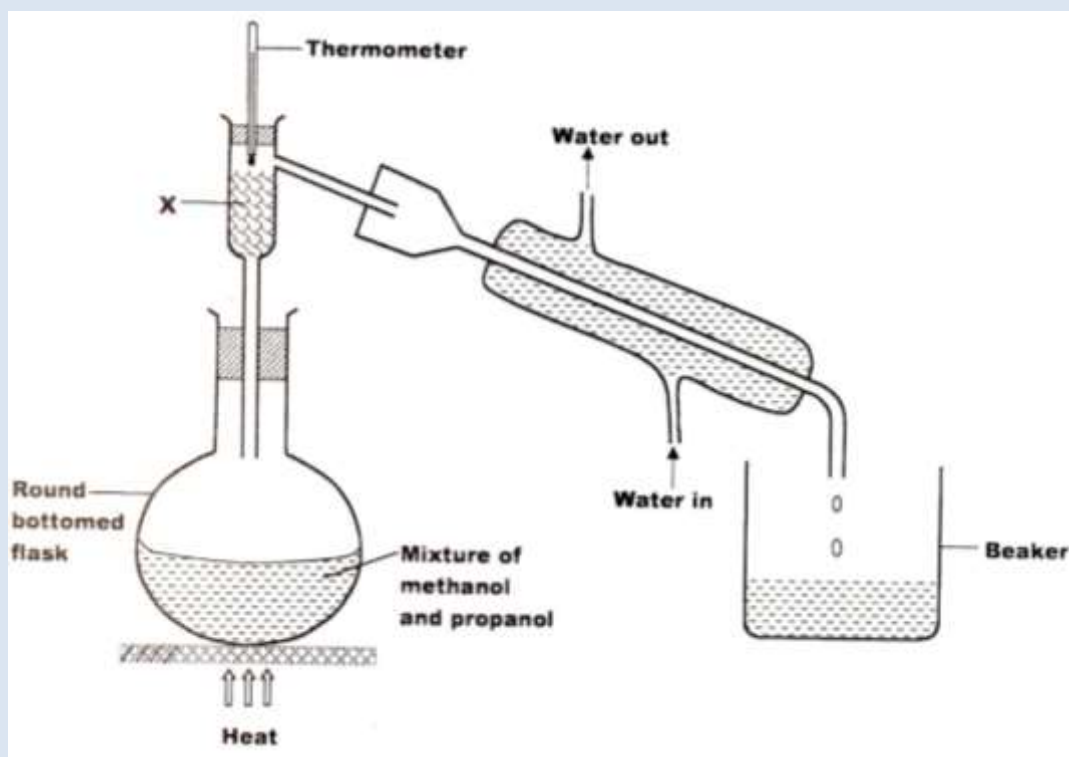
18. Use the part of the periodic table given below to answer the questions that follow
(Letters are not the actual symbols of the elements)

				N		P	
Q	M						R

a) Identify the element that forms giant covalent structures

(1 mk)

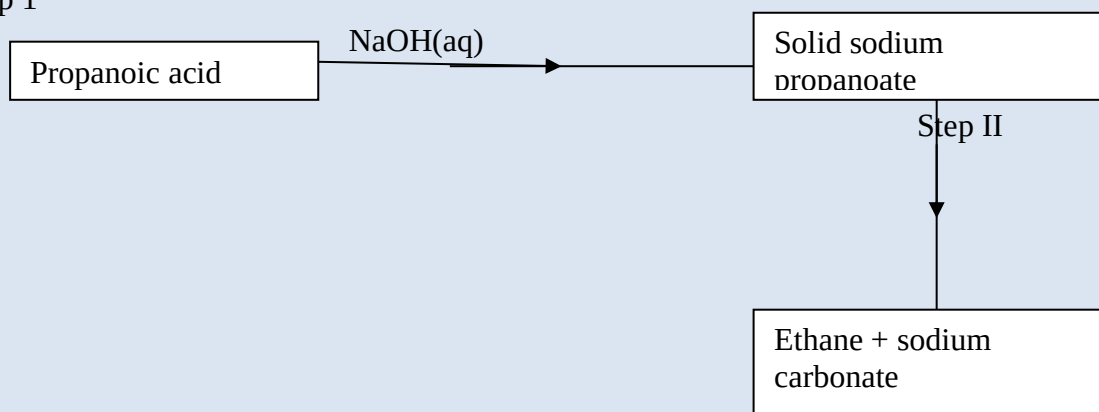
- b) Identify **one** element that does not form compounds (1 mk)
- c) Write the formula for the nitride of M (1 mk)
19. Draw a set up that can be used to separate a mixture of sand and iodine (3 mks)
20. In the contact process, during the production of sulphur (VI) oxide, a catalyst is used.
Give **two** reasons why vanadium (V) oxide is preferred to platinum. (2 mks)
21. Given that the atomic number of Y is 13 and that of Z is 9:
a) Write the electronic arrangement of Y and Z; (1 mk)
- b) Draw the dot (.) and cross(x) diagram for the compound formed by Y and Z
- (1 mk)
22. The set up below was used to separate a mixture of methanol and propanol. Study it
And answer the question that follow.



- State the function of X (1 mk)
- Which liquid will collect first in the beaker? give a reason. (2 mks)

23. Study the flow chart below and answer the question the follow.

Step 1



- Name the process in step I. (1 mk)
- Identify the reagent in step II. (1 mk)
- Give **one** use of ethane (1 mk)

24. (a) A student electrolyzed dilute sodium chloride solution using inert carbon electrodes. Name the products at:

i)Anode :

ii)Cathode: (2 mks)

(b) If the experiment was repeated using concentrated sodium chloride instead of dilute sodium chloride solution, write the half equation at the anode. (1 mk)

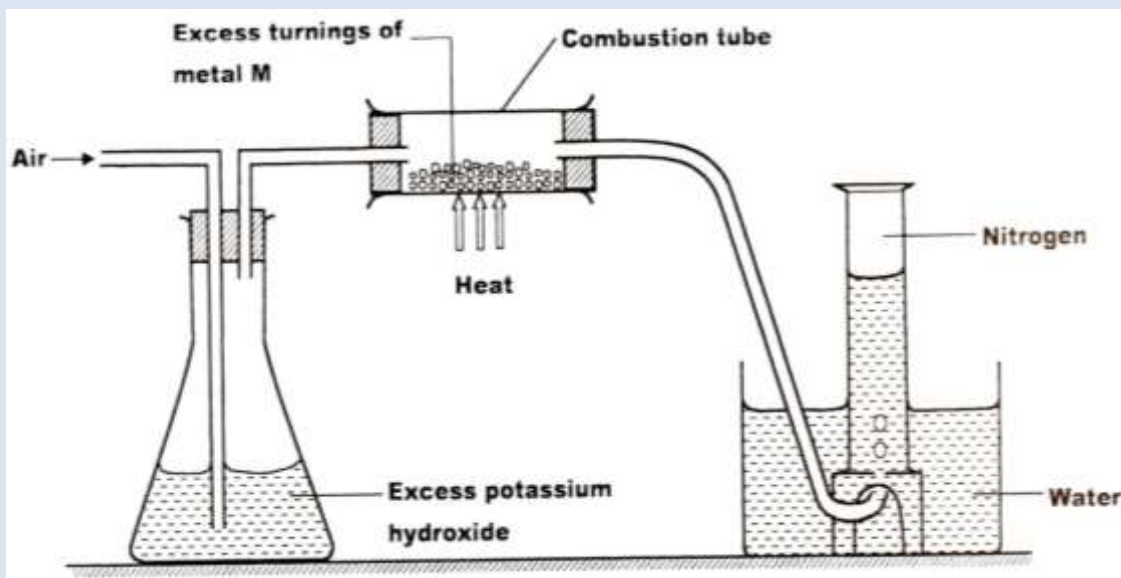
25. An organic compound had the following composition 37.21% carbon, 7.75% hydrogen and the rest chlorine. Determine the molecular formula of the compound, given that the molecular mass of the compound is 65. (C=12.0; H=1.0; CL=35.5) (3 mks)

26. Cotton is a natural polymer. State **one** advantage and **one** disadvantage of this polymer. (2 mks)

27. a) Name a suitable solvent for extracting an indicator from flowers: (1 mk)

b) Give a reason why the solvent named in (a) above is used (1 mk)

28. A student used the set up below to prepare a sample of nitrogen gas.



a) State the function of potassium hydroxide in the set up (1 mk)

b) Give a suitable metal M for use in the combustion tube (1 mk)

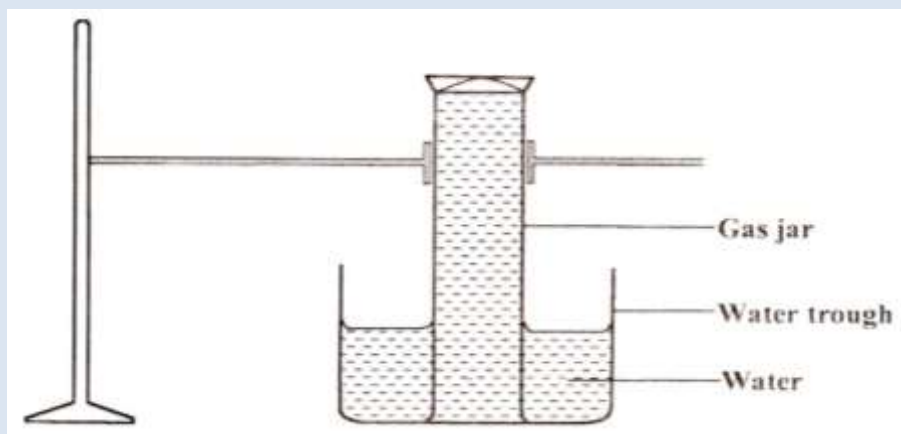
c) Give a reason why the nitrogen gas obtained is not pure. (1 mk)

29. a) What is meant by the term radical? (1 mk)

b) The table below contains atoms that form common radicals. Complete the table to show radicals formed from various atoms. (2 mks)

Element	N	S
H	NH_4^+	
O		

30. A gas jar full of chlorine water was inverted over water and allowed to stand for sometime



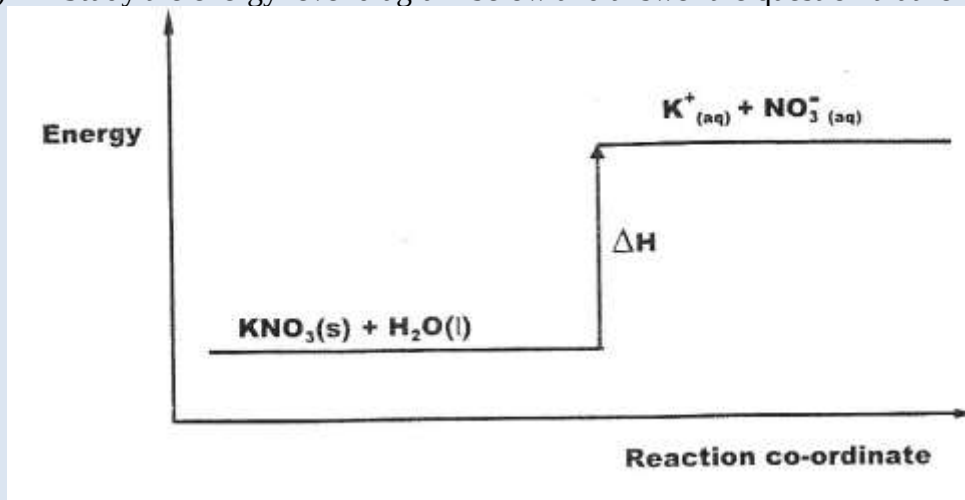
a) State and explain **two** observations made in the gas jar after some time. (2 mks)

b) Write the equation for the reaction between chlorine and hot concentrated potassium hydroxide (1 mk)

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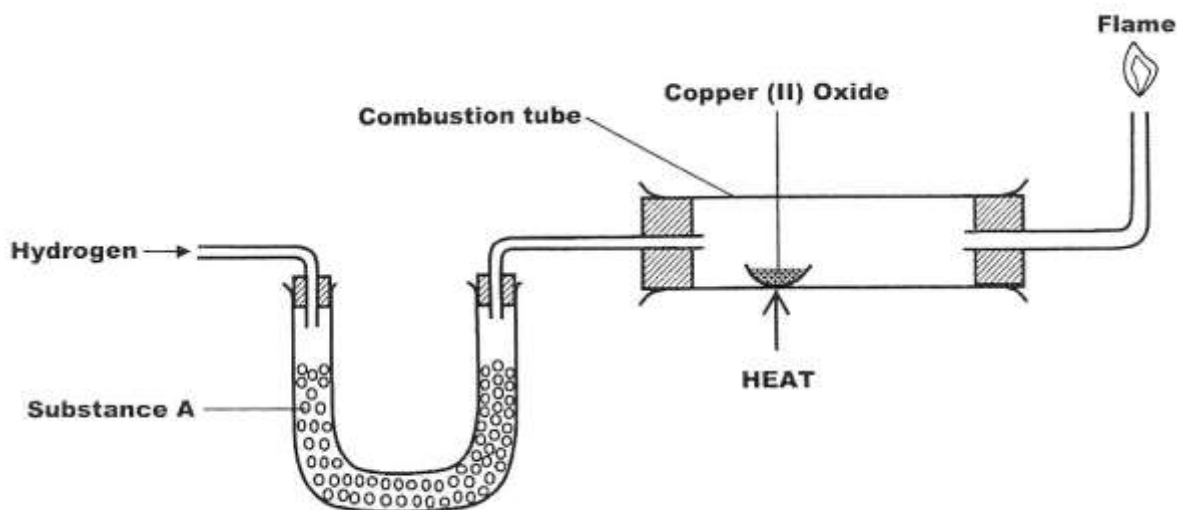
KCSE 2015 QUESTIONS

1. (a) Give the name of the first member of the alkene homologous series. (1 mk)
- (b) Describe a chemical test that can be used to distinguish butanol from butanoic acid (2 mks)
2. (a) Name the raw material from which sodium is extracted. (1 mk)
- (b) Give a reason why sodium is extracted using electrolysis. (1 mk)
- (c) Give two uses of sodium metal. (1 mk)
3. (a) What is meant by lattice energy?
- b) Study the energy level diagram below and answer the question that follows



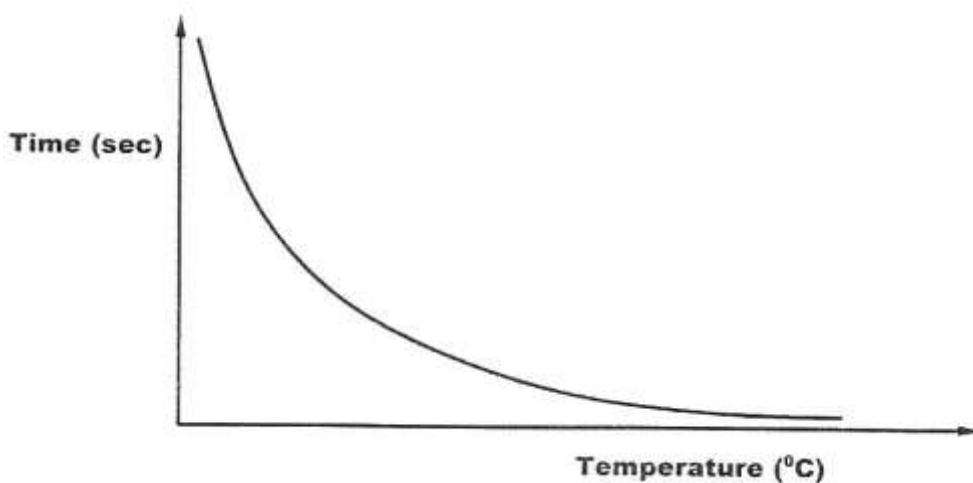
- What type of reaction is represented by the diagram? (1 mk)
4. a) State the Boyles Law (1 mk)
- b) A gas occupies 500cm^3 at 27°C and $100,000\text{ Pa}$, What will be its volume at 0°C and 101325 Pa ? (2 mks)

5. Calculate the mass of Zinc oxide that will just neutralize dilute nitric (V) acid containin12.6 g of nitric (V) acid in water. ($Z_n = 65.0$; $O = 16.0$, $H = 1.0$, $N = 14.0$). (2 mks)
6. Describe how sodium carbonate is used to remove water hardness (2 mks)
7. Hydrogen chloride gas can be prepared by reacting sodium chloride with an acid.
- (a) Write an equation for the reaction between sodium chloride and the acid. (1 mk)
 - (b) Give two chemical properties of hydrogen chloride gas (1 mk)
 - (c) State two uses of hydrogen chloride (1 mk)
8. When solid A was heated strongly, it gave off water and a solid residue. When water was added to the solid residue, the original solid A, was formed
- (a) What name is given to the process described? (1 mk)
 - (b) Give one example of solid A (1 mk)
9. The set up below was used to investigate the reaction between dry hydrogen gas and copper (II) oxide

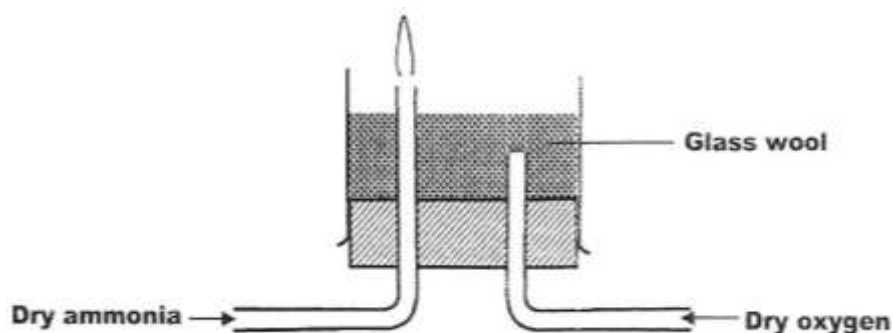


- a) Name substance A (1 mk)
- b) state the observation made in the combustion tube (1 mk)
- c) Explain the observation stated in (b) above. (1 mk)

10. The atomic number of an element, T is 15.
- (a) Write the electronic configuration of the ion T (1 mk)
- (b) Write the formula of an oxide of T. (1 mk)
11. Dilute sulphuric (VI) acid was electrolysed using platinum electrodes.
- Name the product formed at the anode and give a reason for your answer. (2 mks)
12. The curve shown below shows the variation of time against temperature for the reaction between sodium thiosulphate and hydrochloric acid.



- a) Write the equation for the reaction between sodium thiosulphate and dilute hydrochloric acid (1 mk)
- b) Explain the shape of the curve (2 mks)
13. Dry ammonia and dry oxygen were reacted as shown in the diagram below



a) What is the purpose of the glass wool? (1 mk)

b) What products would be formed if red hot platinum was introduced into a mixture of ammonia and oxygen? (1 mk)

14. The table below shows behaviour of metals R, X, Y and Z. Study it and answer the questions that follow:

Metal	Appearance on exposure to air	Reaction in water	Reaction with dilute hydrochloric acid
R	slowly tarnishes	Slow	Vigorous
X	Slowly turns white	Vigorous	Violent
Y	No change	Does not react	Does not react
Z	No change	No reaction	Reacts moderately

a) Arrange the metals in the order of reactivity starting with the most reactive (2 mks)

b) Name a metal which is likely to be (1 mk)

i X

ii Y

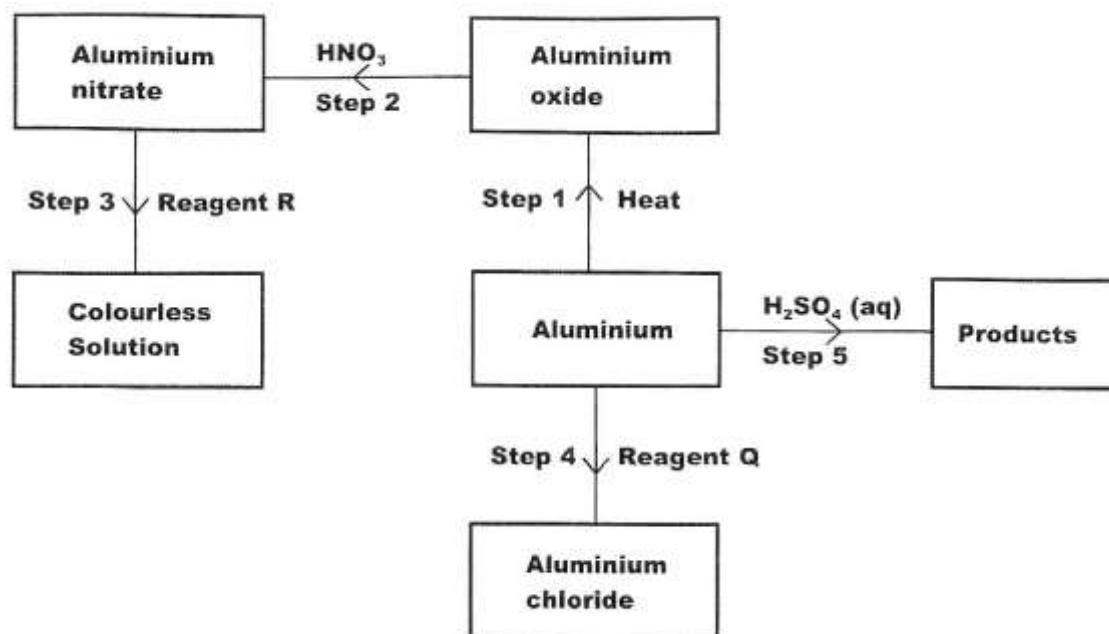
15. Given the following substances: wood ash, lemon juice and sodium chloride.

(a) Name one commercial indicator that can be used to show whether wood ash, lemon juice and sodium chloride are acidic, basic or neutral.

b) Classify the substances in 15(a) above as acids, bases or neutral

Acid	Base	Neutral

16. The flow chart below shows various reactions of aluminium metal. Study it and answer the questions that follow



- A i) Other than water, name another reagent that could be R (1 mk)
 ii) Write the formula of reagent Q (1 mk)
 b) Write an equation or the reaction in step 5. (1 mk)

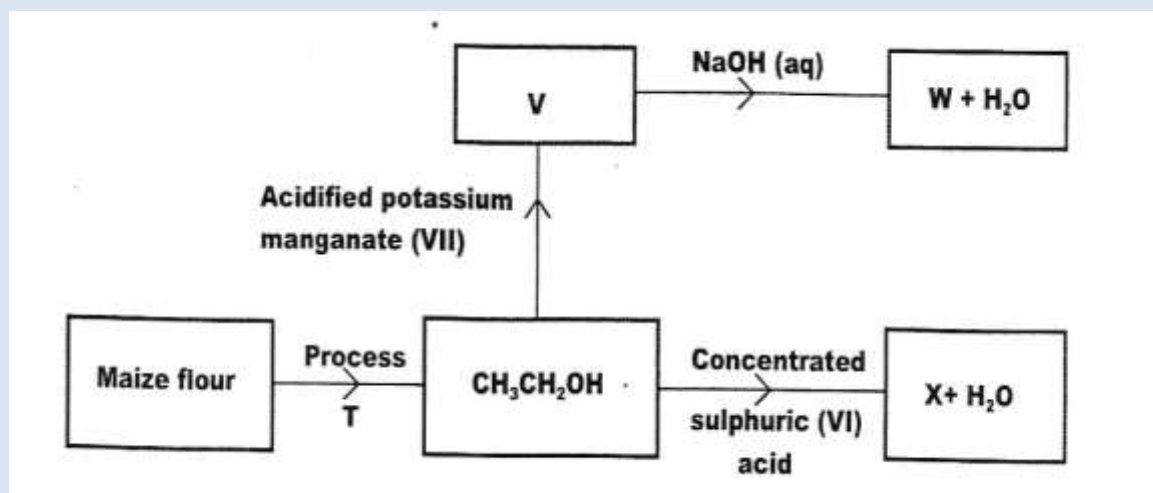
17. (a) One of the allotropes of sulphur is rhombic sulphur, name the other allotrope. (1 mk)
 (b) Concentrated sulphuric (VI) acid reacts with ethanol and copper. State the property of the acid shown in each case. (2 mks)
 (i) Ethanol
 (ii) Copper

18. Study the standard electrode potentials in the table below and answer the questions that follow.

	E^θ volts
$\text{Cu}^{2+}_{(\text{aq})} + 2\text{e} \rightarrow \text{Cu}_{(\text{s})};$	+ 0.34
$\text{Mg}^{2+}_{(\text{aq})} + 2\text{e} \rightarrow \text{Mg}_{(\text{s})};$	- 2.38
$\text{Ag}^{2+}_{(\text{aq})} + \text{e} \rightarrow \text{Ag}_{(\text{s})};$	+ 0.80
$\text{Ca}^{2+}_{(\text{aq})} + 2\text{e} \rightarrow \text{Ca}_{(\text{s})};$	- 2.87

- (a) Which of the metals is the strongest reducing agent? (1 mk)
- (b) What observations will be made if a silver coin was dropped into an aqueous solution of copper (II) sulphate? Explain. (2 mks)
19. A radioactive substance weighing M kg took 1900 years for the original mass to reduce to 15 kg. Given that the half life of the radioactive substance is 380 years; (2 mks)
- (a) Determine the original mass of the radioactive substance.
- (b) State two uses of radioactivity in medicine. (1 mk)
20. A crystal of iodine, heated gently in a test tube gave off a purple vapour.
- (a) Write the formula of the substance responsible for the purple vapour. (1 mk)
- (b) What type of bond is broken when the iodine crystal is heated gently? (1 mk)
- (c) State one use of iodine. (1 mk)
21. Describe how samples of lead (II) sulphate, ammonium chloride and sodium chloride can be obtained from a mixture of the three. (3 mks)

22. Study the flow chart below and use it to answer the questions that follows



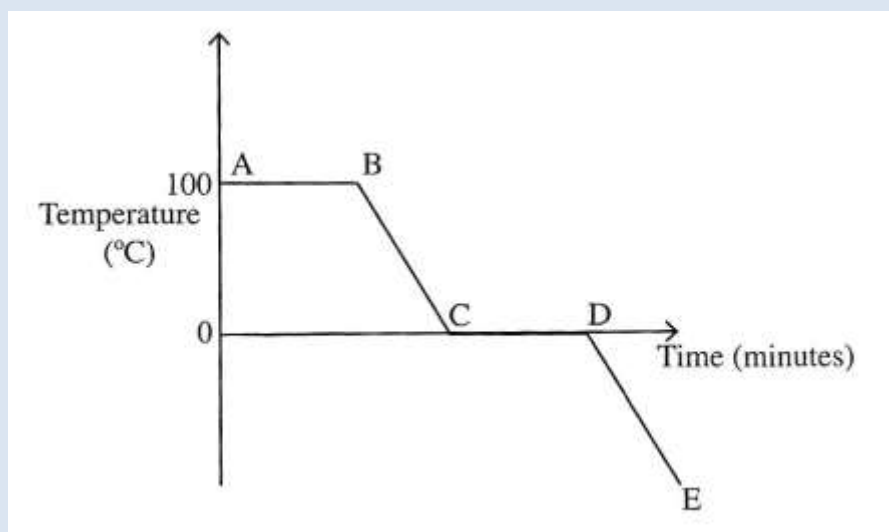
- Name process T (1 mk)
- Give the formula of W. (1 mk)
- State two uses of X (1 mk)

23. The table below is part of the periodic table. The letters are not the actual symbols of the elements. Study it and answer the questions that follows.

					C	D	E	F	
G	H						I		

- How do the ionic radii of E and I compare? Explain (2 mks)

24. The graph below is a cooling curve for water. Study it and answer the questions that follow



- (a) Explain what happens to the molecules of water in the region BC in terms of kinetic theory. (2 mks)
- (b) In what state is the water in the region DE?
25. Starting with barium nitrate solution, describe how a pure sample of barium carbonate can be prepared in the laboratory. (3 mks)
26. A hydrocarbon contains 14.5% of hydrogen. If the molar mass of the hydrocarbon is 56, determine the molecular formula of the hydrocarbon. (C = 12.0; H = 1.0) (3 mks)
27. (a) Describe how carbon (IV) oxide can be distinguished from Carbon II Oxide using calcium hydroxide solution. (2 mks)
- (b) What is the role of carbon (IV) oxide in fire extinguishing? (1 mk)

28. (a) State one source of alkanes. (1 mk)
- (b) Ethane gas was reacted with 1 mole of bromine gas. State one observation made during this reaction. (1 mk)

29. An electric current was passed through several substances and the results obtained recorded in the table below.

Substance	Physical state at room temperature	Conductivity	Products	
			Anode	Cathode
A	Liquid	Does not conduct	-	-
B	Solid	Conducts	-	-
C	Liquid	Conducts	Green gas	Grey solid
D	Liquid	Conducts	Brown gas	Grey solid
E	Liquid	Conducts	-	-

Which of these substance is likely to be

- a) Magnesium (1 mk)
- b) Hexane (1 mk)
- c) lead (II) bromide ? (1 mk)

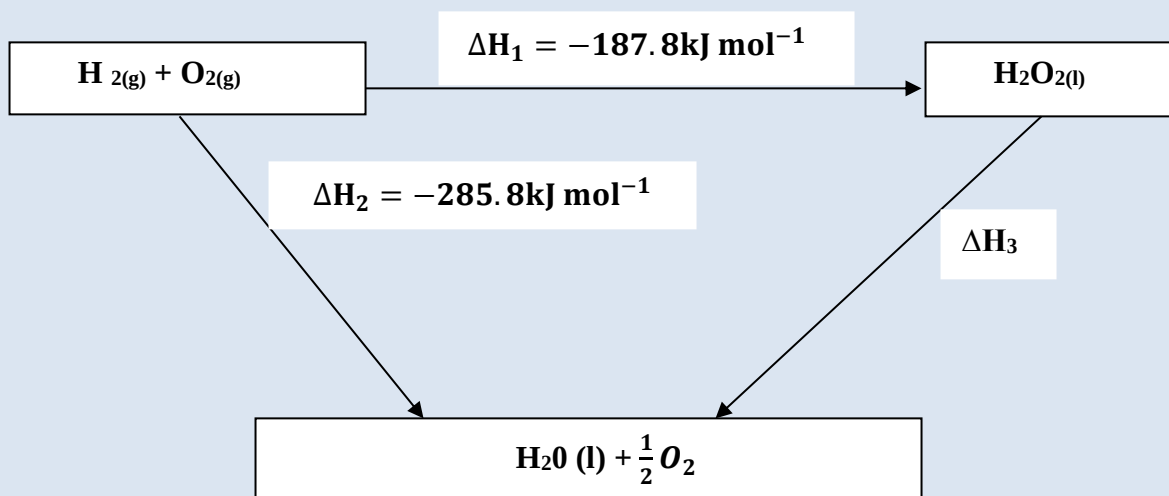
CHEMISTRY PAPER 233/1
K.C.S.E 2016 QUESTIONS

1. A student investigated the effect of an electric current by passing it through some substances. The student used inert electrode and connected a bulb to the circuit. The table below shows the substances used and their states.

experiment	substances	states
1	potassium carbonate	Solid
2	copper (II) Sulphate	Solution
3	sugar	Solution
4	Lead (II) Iodide	Molten

- a) In which experiment did the bulb not light? (1mk)
- b) Explain your answer in (a) above (2 mks)
2. An alkanol has the following composition by mass ; hydrogen 13.5% , oxygen 21.6% and carbon 64.9%
- a) Determine the empirical formula of the alkanol. (C=12.0, H=1.0, O=16) (2 mks)
- b) Given that the empirical formula and the molecular formula of the alkanol are the same, draw the structure of the alkanol (1 mk)

3. The figure below shows an energy cycle



a) Give the name of the enthalpy change ΔH_1 .

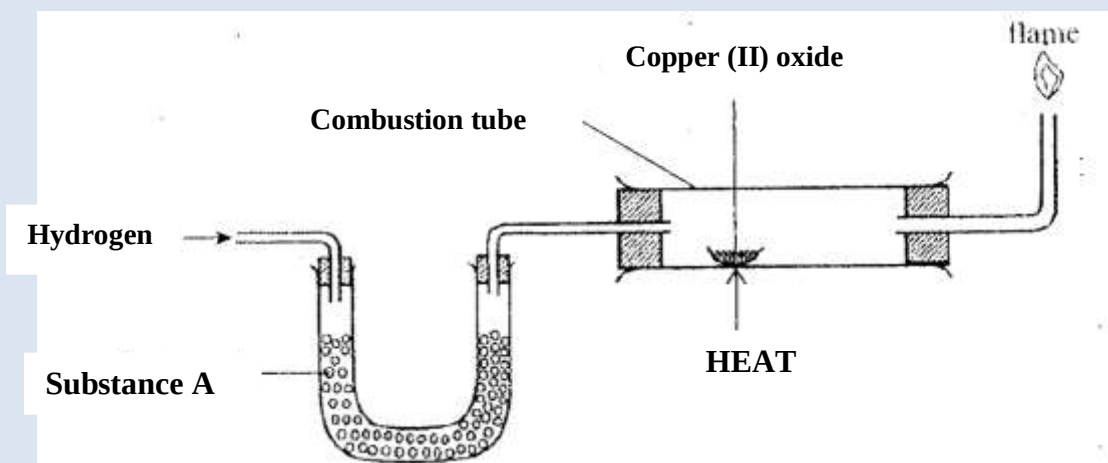
(1 mk)

b) Determine the value of ΔH_3 .

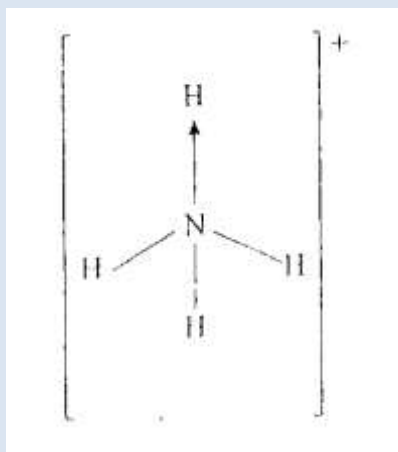
(2 Mks)

4. The set up below was used to investigate the reaction between dry hydrogen gas and copper (II) oxide

Flame



- Name substance A (1 mk)
 - State the observation made in the combustion tube (1 mk)
 - Explain the observation made in (b) above (1 mk)
5. Starting with sodium metal, describe how a sample of crystals of sodium hydrogen carbonate may be prepared (3 mks)
6. Ammonium ion has the following structure



- Label on the structure the
- Covalent bond (1 mk)
 - Coordinate(dative) bond (1 mk)
7. When 8.53g of sodium nitrate was heated in an open test tube, the mass of oxygen gas produced was 0.83g. Given the equation of the reaction as



Calculate the percentage of sodium nitrate that was converted to sodium nitrite

(Na = 23.0, N = 14.0, O = 16.0)

8. Aluminium is both malleable and ductile

a) What is meant by?

i) Malleable

($\frac{1}{2}$ mks)

ii) Ductile

($\frac{1}{2}$ mks)

b) State one use of aluminium based on

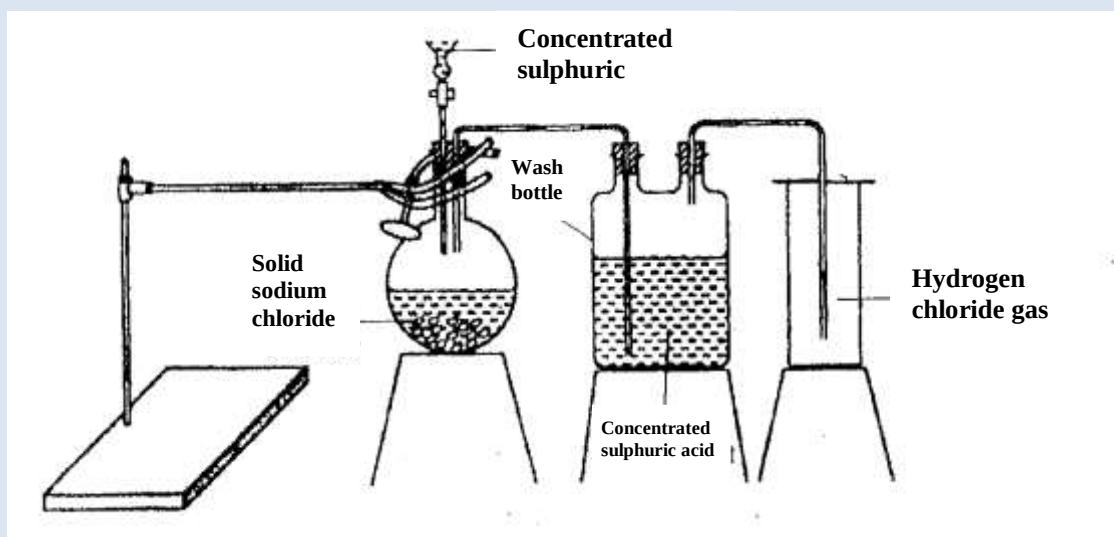
i) Malleable

($\frac{1}{2}$ mks)

ii) Ductile

($\frac{1}{2}$ mks)

9. The diagram below represents the set up that was used to prepare and collect hydrogen chloride gas in the laboratory



a) State the purpose of concentrated sulphuric acid in the wash bottle

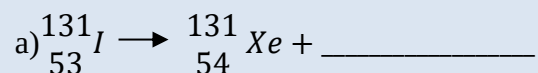
(1 mk)

- b) Write an equation for the reaction between dry hydrogen chloride gas and heated iron (1 mk)

10. Iron (III) oxide was found to be contaminated with copper (II) sulphate.

Describe how pure sample of iron (III) oxide can be obtained.

11. Complete the nuclear equation below



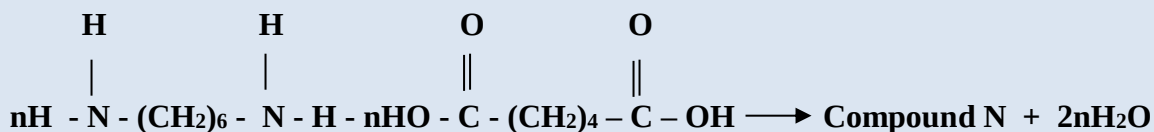
- b) The half life of ${}_{53}^{131}\text{I}$ is 8 days. Determine the mass of ${}_{53}^{131}\text{I}$ remaining if 50 grams decayed for 40 days.

- c) Give one harmful effect of radioisotopes.

12. During an experiment, chlorine gas was bubbled into a solution of potassium iodide

- a) State the observations made (1 mk)
b) Using an ionic equation, explain why the reaction is redox (2 mks)

13. a) Draw the structure of compound N formed in the following reaction (1 mk)

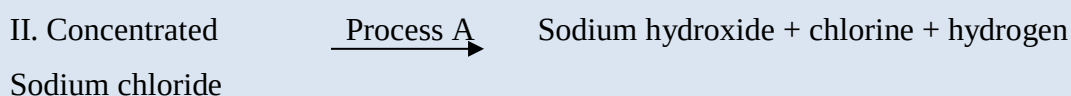
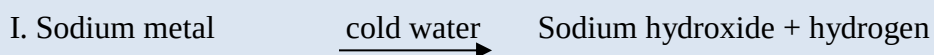


- b) Give one use of compound N (1 mk)

14. When fuel burns in the internal combustion engine at high temperature, one of the products formed is nitrogen (II) oxide

- a) Write the equation for the formation of nitrogen (II) oxide (1 mk)
- b) Give a reason why nitrogen (II) oxide is not formed at room temperature (1 mk)
- c) Describe how formation of nitrogen (II) oxide in the internal combustion engine lead to gaseous pollution (2 mks)

15. Sodium hydroxide can be prepared by the following methods; I and II

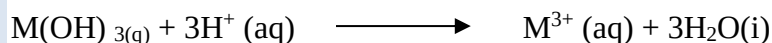
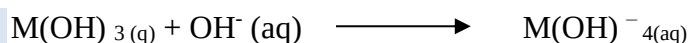


- a) Name one precaution that needs to be taken in method I (1 mk)
- b) Give the name of process A (1 mk)
- c) Give one use of sodium hydroxide (1 mk)

16. The atomic number of sulphur is 16. Write the electron arrangement of sulphur in the following ? (2 mks)

- a) H_2S
- b) SO_3^{2-}

17. A compound whose general formula is M(OH)_3 reacts as shown by the equation.



- (a) What name is given to compounds which behave like M(OH)_3 in the two reactions. (1 mk)
- (b) Name **two** elements whose hydroxides behave like that of M. (2 mks)

18. A water trough, aqueous sodium hydroxide, burning candle, watch glass and

a graduated jar were used in an experimental set up to determine the percentage of active part of air. Draw a labelled diagram of the set up at the end of the experiment. (3 mks)

19. In an experiment on rates of reaction, potassium carbonate was reacted with dilute sulphuric (VI) acid.

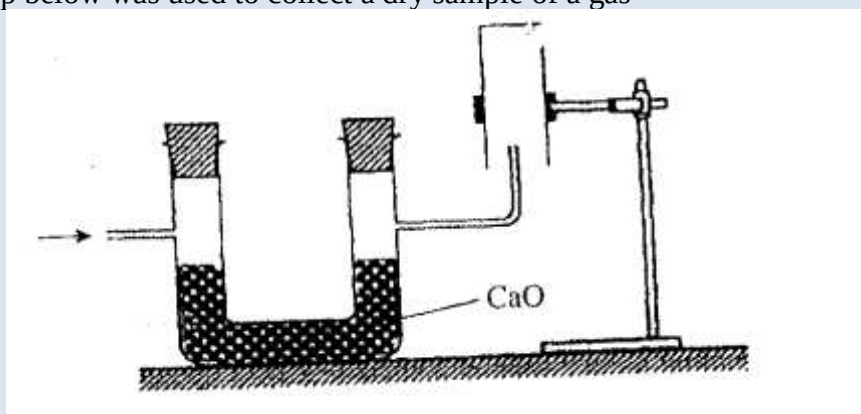
(a) What would be the effect of an increase in the concentration of the acid on the rate of the reaction? (1 mk)

(b) Explain why the rate of reaction is found to increase with temperature (1mk)

20. 60cm^3 of oxygen gas diffused through a porous partition in 50 seconds. How long would it take 60cm^3 sulphur (IV) oxide gas to diffuse through the same partition under the same condition (S=32.0, O=16.0)

21. Draw and name the isomers of pentane (3mks)

22. The set up below was used to collect a dry sample of a gas



Give two reasons why the set up cannot be used to collect carbon (IV) oxide gas (2 mks)

23. Given the following substances; wood ash, lemon juice and sodium chloride.

a) Name one commercial indicator that can be used to show whether wood ash,

lemon juice and sodium chloride are acidic, basic or neutral (1 mk)

b) Classify the substances in (a) above as acids bases or neutral (2mks)

acid	base	neutral

24. Describe how a solid sample of potassium sulphate can be prepared starting with 200cm³ of potassium hydroxide (3 mks)

25. Charcoal is a fuel that is commonly used for cooking. when it burns it forms two oxides

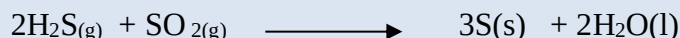
a) Name the two oxides (2 mks)

b) State one use of the two oxides (1 mk)

26. Hydrogen sulphide is a highly toxic and flammable gas. It is normally prepared in a fume chamber

a) Name two reagents that can be used to prepare hydrogen sulphide in the laborator (1 mk)

b) One of the use of hydrogen sulphide is to produce sulphur as shown in the following equation.



Identify the reducing agent in this reaction and give a reason for your answer (1mk)

c) Other than production of sulphuric (VI) acid , State one commercial use of sulphur (1 mk)

27. Describe an experimental procedure that can be used to extract oil from nut seeds. (2 mks)

28. A mixture contains ammonium chloride, copper (II) oxide and sodium chloride.

Describe how each of the substances can be obtained from the mixture (3 mks)

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29. When a student was stung by a nettle plant, a teacher applied an aqueous solution of ammonia to the affected area of the skin and the student was relieved of pain. Explain. (2 mks)

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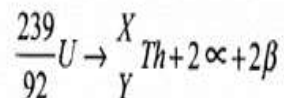
1. Table 1 shows the atomic numbers and the first ionisation energies of three elements. The letters are not actual symbols of the elements. Use it to answer the questions that follow.

Table 1

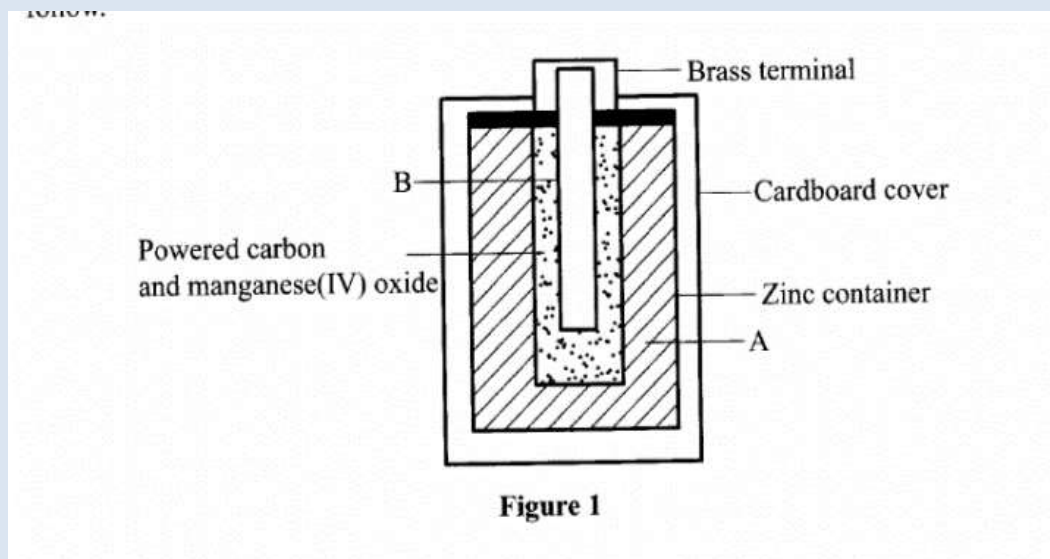
Element	Atomic number	First ionisation energy kJmol ⁻¹
A	3	519
B	11	494
C	19	418

- (a) Explain the trend in first ionisation energy from A to C. (2 mark)
(b) Write the electronic configuration for the ion of C. (1 mark)

2. Calculate the values of X and Y in the following nuclear equation.



3. The diagram in Figure 1 shows a section of a dry cell. Study it and answer the questions that follow.



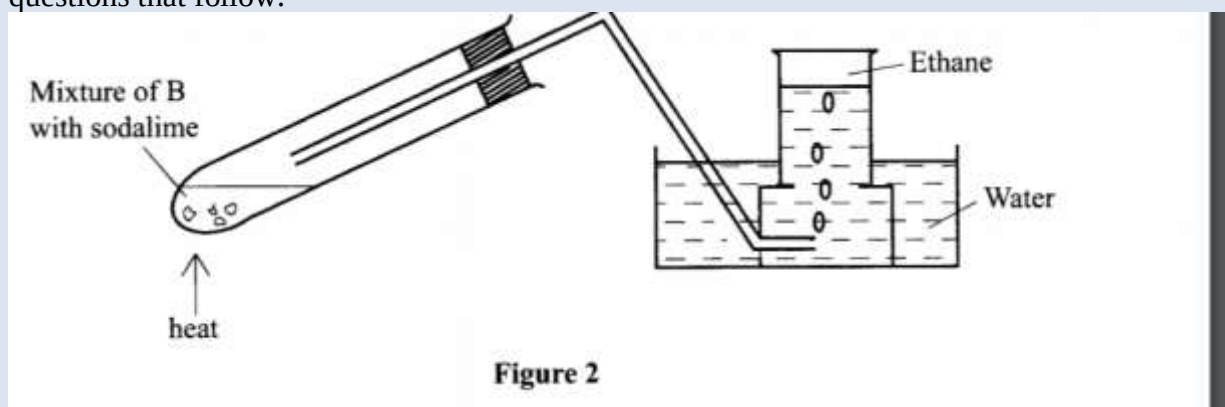
- (a) Name the part labelled B. (1 mark)
(b) The part labelled A is a paste. Give a reason why it is not used in dry form. (1 mark)
(c) What is the purpose of the zinc container? (1 mark)

4. The empirical formula of lead(II) oxide was determined by passing excess dry hydrogen gas over 6.69g of heated lead(II) oxide.

- (a) What was the purpose of using excess dry hydrogen gas? (2 marks)
(b) The mass of lead was found to be 6.21g.

Determine the empirical formula of the oxide. ($Pb = 207.0$ $O = 16.0$) (2 marks)

5. The set-up in Figure 2 was used to prepare a sample of ethane gas. Study it and answer the questions that follow.



6. (a) State Charles' Law. (1 mark)

(b) Explain why the pressure of a fixed mass of a gas increases, when the volume of the gas is reduced at constant temperature. (2 marks)

7. A sample of water is suspected to contain sulphate ions. Describe an experiment that can be carried out to determine the presence of sulphate ions. (3 marks)

8. (a) State one characteristic of a reaction where equilibrium has been attained. (1 mark)

(b) The following equation is in a state of equilibrium: $C \rightleftharpoons D$ Use it to sketch a graphical representation of concentration against time in seconds for the equilibrium. (2 marks)

9. Copper(II) ions react with excess aqueous ammonia to form a complex ion.

(a) (i) Write an equation for the reaction that forms the complex ion. (1 mark)

(ii) Name the complex ion. (1 mark)

(b) Explain why CH_4 is not acidic while HCl is acidic yet both compounds contain hydrogen. (1 mark)

10. 20 cm³ of ethanoic acid was diluted to 400 cm³ of solution. Calculate the concentration of the solution in moles per litre. (C = 12.0 ; H = 1.0 ; O = 16.0) (Density of ethanoic acid = 1.05 g/cm³) (3 marks)

11. An oxide of element K has the formula K_2O . (a) Determine the oxidation number of K. (1 mark)

(b) To which group of the periodic table does K belong? (1 mark)

12. Potassium nitrate liberates oxygen gas when heated. Draw a diagram of a set-up that shows heating of potassium nitrate and collection of oxygen gas. (3 marks)

13. Explain the observation made when chlorine gas is passed through a solution of potassium iodide. (3 marks)

14. Using the elements chlorine, calcium and phosphorus:

(a) Select elements that will form an oxide whose aqueous solution has a pH less than 7. (1 mark)

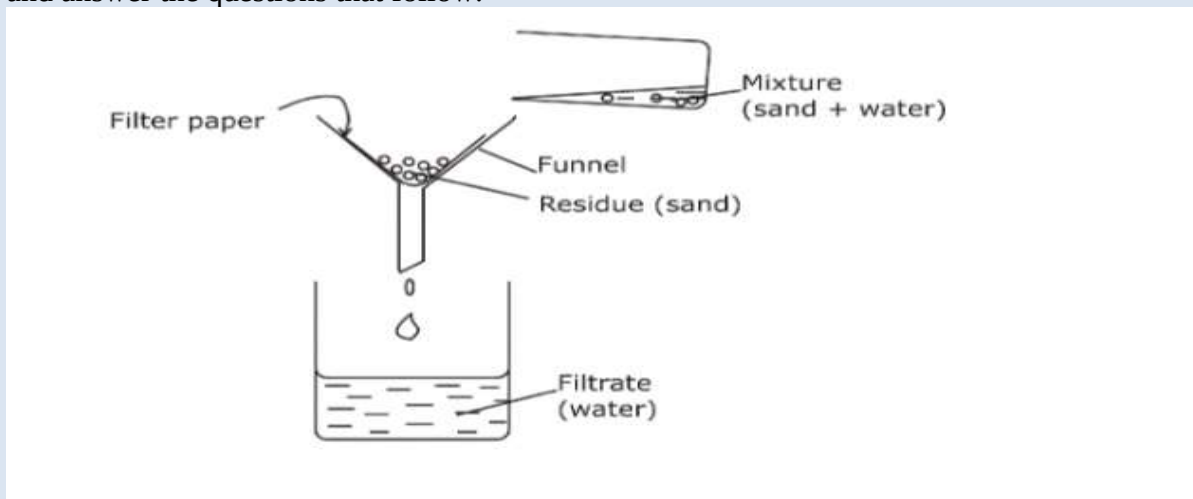
(b) Write an equation for the reaction between calcium oxide and dilute hydrochloric acid. (1 mark)

(c) Give one use of calcium oxide. (1 mark)

15. Starting with copper, describe how a pure sample of copper(II) carbonate can be prepared. (3 marks)

16. In an experiment, concentrated nitric(V) acid was reacted with iron(II) sulphate. State and explain the observations made. (2 marks)

17. The flow chart in Figure 3 shows the process of obtaining a sample of nitrogen gas. Study it and answer the questions that follow.



- (a) Identify X (1 mark)
- (b) Write an equation for the reaction with heated copper turnings. (1 mark)
- (c) Name an impurity in the sample of nitrogen gas. (1 mark)

18. The set-up in Figure 4 can be used to prepare nitrogen (II) oxide. Use it to answer the questions that follow.

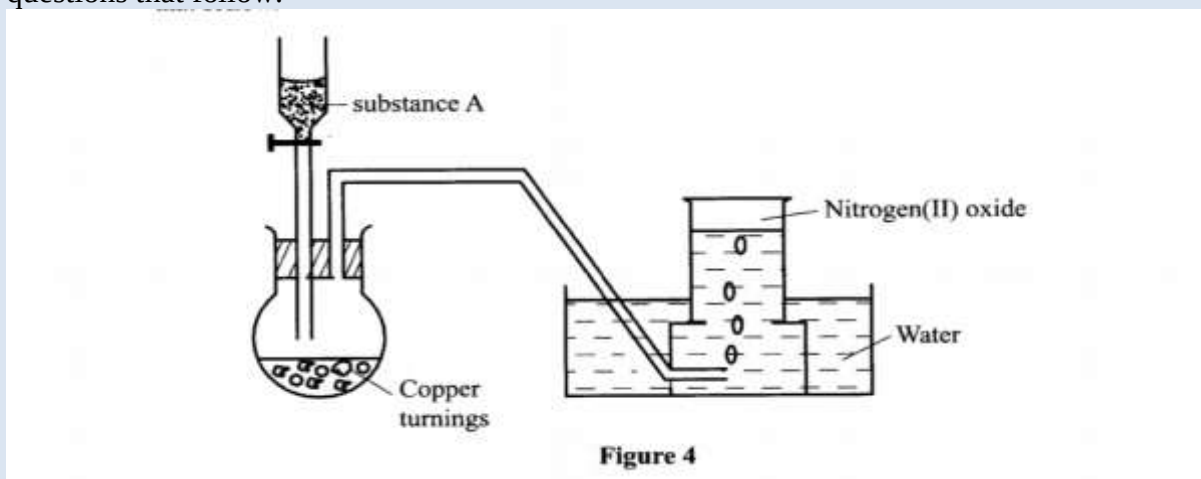


Figure 4

- (b) When the gas jar containing nitrogen (II) oxide is exposed to air, a brown color is observed. Explain. (1 mark)
- (c) Write an equation for the reaction which occurred in the flask. (1 mark)

19. The following procedure was used to investigate the temperature changes that occur when sodium hydroxide solution is added to dilute hydrochloric acid.

- (i) Place the acid in a glass beaker and record its temperature.
- (ii) Add a known volume of sodium hydroxide solution.
- (iii) Stir the mixture and record the highest temperature reached.
- (iv) Repeat steps
- (ii) and
- (iii) with different volumes of sodium hydroxide solution.

- (a) State two factors that must be kept constant in this experiment (1 mark)
- (b) Explain how the use of a polystyrene cup will affect the results. (1 mark)

20. Study the flow chart in Figure 5 and answer the questions that follow.

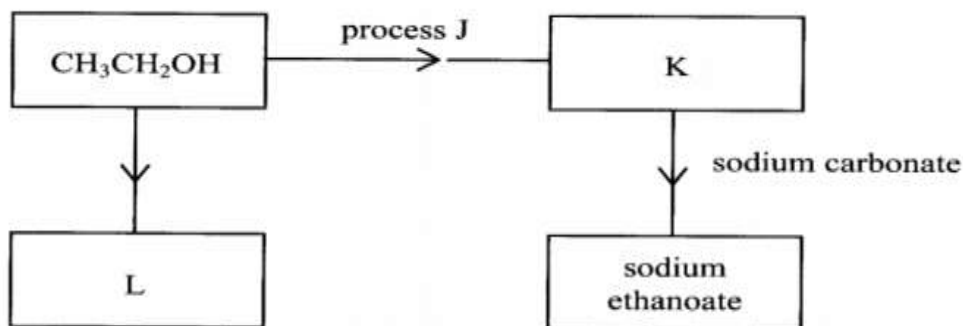


Figure 5

- (a) Identify substances K and L. K: (1 mark) L: (1 mark)
- (b) Name one reagent that can be used to carry out process J. (1 mark)

21. The atomic numbers of some elements P, Q, R and S are 6, 8, 12 and 17 respectively.

- (a) Draw the dot (•) and cross (X) diagrams for the compounds formed when:
 - (i) R and Q react (1 mark)
 - (ii) P and S react. (1 mark)
- (b) Explain why the melting point of the compound formed by P and S is lower than that formed by R and Q. (1 mark)

22. (a) What is an inert electrode? (1 mark)

(b) State the products formed when brine is electrolysed using inert electrodes. Anode: (1 mark)

Cathode: (1 mark)

23. Explain how a student can establish whether a liquid sample extracted from a plant is pure. (2 marks)

Anode (1 mark)

Cathode (1 mark)

23. Explain how a student can establish whether a liquid sample extracted from a plant is pure. (2 marks)

24. Figure 6 shows part of the periodic table. The letters are not the actual symbols of the elements. Stud it and answer the questions that follow.

H				K			
			L				
M				N		P	

Figure 6

(a) Write an equation for the reaction between M and K. (1 mark)

(b) Select the element which can form an ion with a charge of +3. (1 mark)

(c) An element J has atomic number 15. Indicate with a tick (✓), on the part of the periodic table the position of J. (1 mark)

25. In terms of structure and bonding, explain why graphite is used as a lubricant in machines. (3 marks)

26. (a) What is meant by the term bleaching? (1 mark)

(b) Write the formula of the bleaching agent formed when chlorine gas reacts with aqueous sodium hydroxide. (1 mark)

(c) State the role of chlorine in water treatment. (1 mark)

27. (a) Name two ores in which sodium occurs. (1 mark)

(b) During extraction of sodium using the down's process, calcium chloride is added to the ore. Give a reason for the addition of calcium chloride. (1 mark)

(c) State two uses of sodium. (I mark)

28. When an aqueous solution of compound X was mixed with a few drops of bromine water, the colour of the mixture remained yellow.

When another portion of solution X was reacted with acidified potassium dichromate(VI), the colour of the mixture changed from orange to green.

(a) What conclusion can be made from the use of:

(i) bromine water? (1 mark)

(ii) acidified potassium dichromate(VI)? (1 mark)

(b) Solution X was reacted with a piece of a metal and a colourless gas was produced. Describe a simple experiment to identify the gas. (1 mark)

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1. (a) Define a soluble base. (1 mark)

(b) Aqueous solutions of 2M ethanoic acid and 2M nitric(V) acid were tested for electrical conductivity. Which solution is a better conductor of electricity? Explain. (2 marks)

2. (a) Explain why it is not advisable to prepare a sample of carbon(IV) oxide using barium carbonate and dilute sulphuric(VI) acid. (2 marks)

(b) State a method that can be used to collect dry carbon(IV) oxide gas. Give a reason. (1 mark)

3. The following are formulae of organic compounds. Use the formulae to answer the questions that follow:

CH₃CH CH₂OH

CH₃ OOH

CH₃CH₂CH 2CH

CH₃CCCH₃ (a) Select:

(i) two compounds which when reacted together produce a sweet-smelling compound. (1 mark)

(ii) an unsaturated hydrocarbon.(1 mark)

(b) Name the compound selected in (a) (ii). (1 mark)

4. One of the allotropes of sulphur is rhombic sulphur.

(a) Name the other allotrope of sulphur. (1 mark)

(b) Draw a diagram to show the shape of the allotrope named in (a) above. (1 mark)

(c) Write an equation for the reaction between concentrated sulphuric(VI) acid and sulphur. (1 mark)

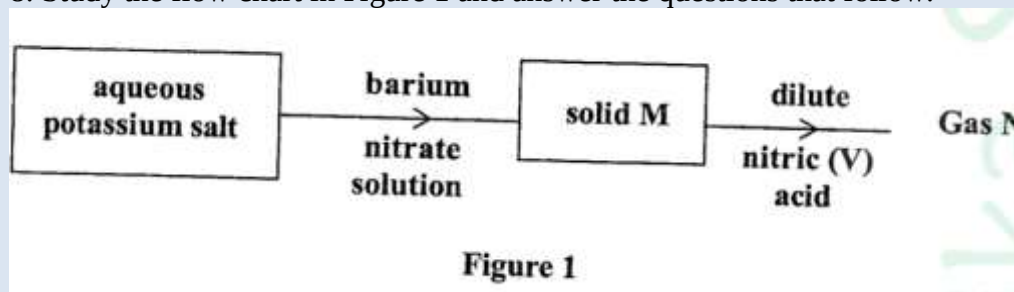
5. Describe an experiment to show that group one elements react with cold water to form alkaline solutions. (3 marks)

6. (a) State Graham's law of diffusion. (1 mark)

(b) Explain why a balloon filled with helium gas deflates faster than a balloon of the same size filled with argon gas. (2 marks)

7. 30.0 cm³ of aqueous sodium hydroxide containing 8.0 g per litre of sodium hydroxide were completely neutralised by 0.294 g of a dibasic acid. Determine the relative formula mass of the dibasic acid. (Na = 23.0 ; O = 16.0 ; H 1.0) (3 marks)

8. Study the flow chart in Figure 1 and answer the questions that follow.



Gas N forms a white suspension with aqueous calcium hydroxide. (1 mark)

(a) Name the anion present in the potassium salt. (1 mark)

(b) Write an ionic equation for the formation of solid M. (1 mark)

(c) Give one use of gas N. (1 mark)

9. An experiment was carried out to determine the presence of substances PQR and S in mixture T. The results obtained are as follows

(a) Name the method of separation illustrated in figure 2

(ii) a substance which is least soluble in the solvent used. 10. Using iron filings, describe an experiment that can be conducted to show that oxygen is present in air. (3 marks)

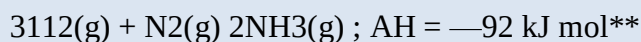
11. (a) Element U has atomic number 12 while element V has atomic number 16. How do the melting points of their oxides compare? Explain. (3 marks)

12. When ethene gas is compressed at a high temperature, a solid is formed.

(a) Give the name of the solid. (1 mark)

(b) Explain why it is not advisable to allow the solid to accumulate in the environment. (2 marks)

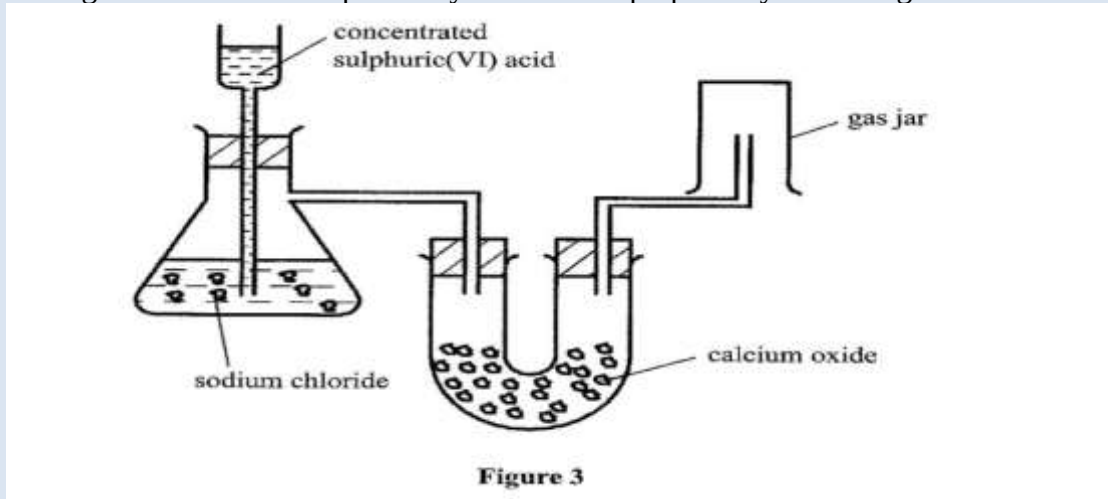
13. In the Haber process, nitrogen reacts with hydrogen according to the following equation.



(a) What would be the effect of adding a catalyst on the position of the equilibrium? (1 mark)

(b) Explain why it is not advisable to use temperatures higher than 773 K in the Haber process. (2 marks)

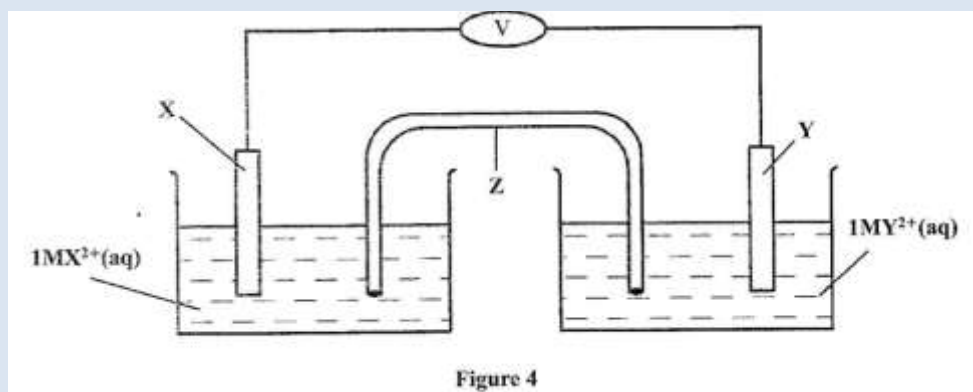
14. Figure 3 shows a set-up used by a student to prepare dry chlorine gas in the laboratory.



Identify three mistakes in the set-up, and give a reason for each. (3 marks)

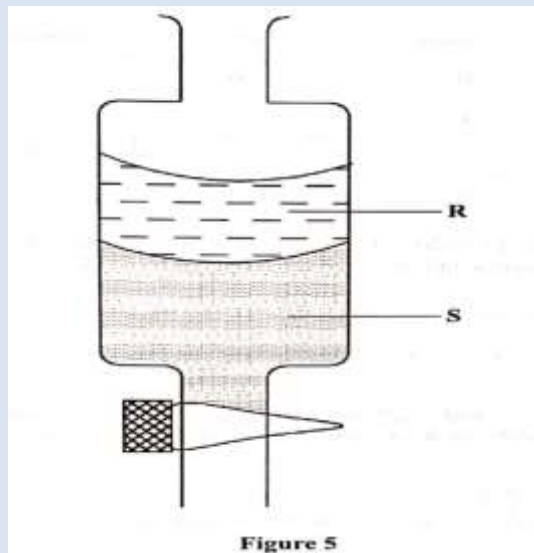
15. You are provided with solid potassium hydrogen carbonate. Describe how a solid sample of potassium nitrate can be prepared. (3 marks)

16. Metal X and Y have standard electrode potentials of -0.13 V and -0.76 V respectively. The metals were connected to form a cell as shown in Figure 4.



- (a) Name the part labelled Z.
- (b) State one function of the part labelled Z.
- (c) Calculate the e.m.f. of the cell. (1 mark)

17. Figure 5 represents a grid that is part of the periodic table. Study it and answer the questions that follow. The letters are not the actual symbols of the elements.

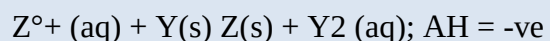


- (a) Write the electron arrangement of element C. (1 mark)
- (b) On the grid provided, show with a tick (I) the position of element D whose atomic number is 18. (1 mark)

(c) Element E is more reactive than A. Explain.

18. (a) Define molar heat of displacement. (1 mark)

(b) The following ionic equation represents the reaction between metal Y and an aqueous solution of Z^{2+} .



Draw an energy level diagram to represent the reaction.

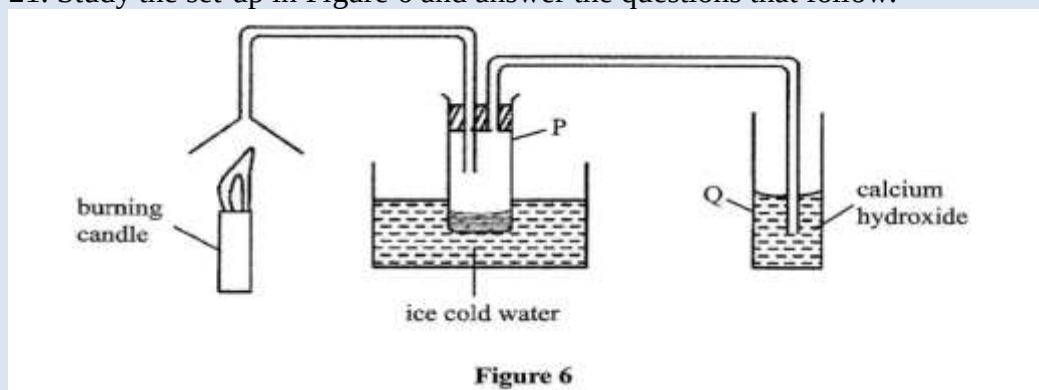
19. (a) Give the symbols of the two charged particles emitted by a radioactive isotope. (2 marks)

(b) An isotope $^{210}_{82}\text{Pb}$ disintegrates by emitting two beta particles. Determine the mass number and atomic number of the resulting nuclide.

20. (a) Zinc reacts with hydrochloric acid according to the following equation. $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$ Identify the reducing agent. Give a reason for the answer. (2 marks)

(b) Iron sheets are dipped in molten zinc to prevent rusting. Name this process. (1 mark)

21. Study the set-up in Figure 6 and answer the questions that follow.



(a) Name the substance that was collected in tube P. (1 mark)

(b) Write an equation for the reaction which occurs in tube Q in the first few minutes of the experiment. (1 mark)

(c) Give a suitable conclusion for the experiment in the set-up. (1 mark)

22. You are provided with the following: thermometer, boiling tube, beaker, Bunsen burner, pure substance X whose boiling point is about 80°C , water and any other apparatus that may be required. Draw a labelled diagram of the set-up that can be used to determine the melting point of X. (3 marks)

23. Explain why it is important to put off a non-luminous flame immediately after use. (2 marks)

24. (a) Name two ores of iron. (1 mark)

(b) Describe how the amount of iron in a sample of iron(III) oxide can be determined. (2 marks)

25. Explain why a solution of sodium chloride conducts electricity while that of sugar does not. (2 marks)

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26. Explain why commercial indicators are preferred to flower extracts as acid-base indicators. (2 marks)

27. $(\text{NH}_4)_2\text{HPO}_4$ is a fertiliser used by farmers to boost their crop production. (a) Calculate the mass of phosphorus in a 20 kg packet of $(\text{NH}_4)_2\text{HPO}_4$. (N = 14.0; H = 1.0; P = 31.0; O = 16.0) (2 marks)

(b) State one advantage of this fertilizer, $(\text{NH}_4)_2\text{HPO}_4$, over urea $\text{CO}(\text{NH}_2)_2$. (1 mark)

28. Distinguish between empirical and molecular formula of a compound

KCSE 2019 CHEMISTRY PAPER 1

1. An atom of element A has mass number 39 and 19 protons.

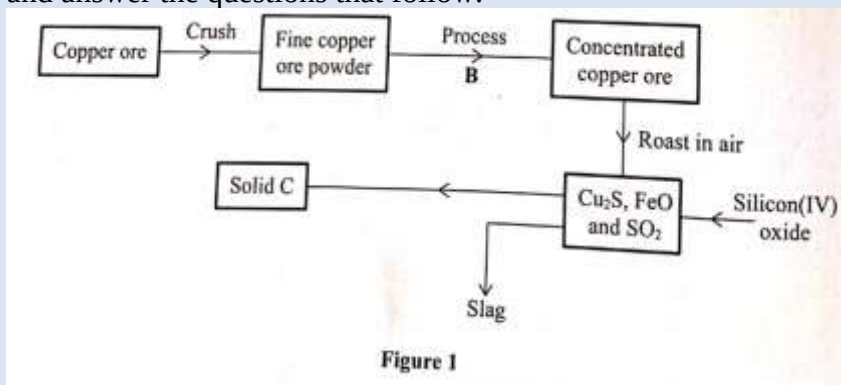
(a) Write the electron arrangements of the atom (1 mark)

(b) State the period and group to which element A belongs Group (½ mark)
Period (½ mark)

(c) State whether the element is a metal or a non-metal. (1 mark)

2. Describe how an increase in concentration increases the rate of a reaction. (2 mark)

3. The flow chart in Figure 1 represents some stages in the extraction of copper metal. Study it and answer the questions that follow.



(a) Identify :

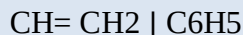
(i) The copper ore (1 mark)

(ii) Process B (½ mark)

(iii) Solid C (½ mark)

(b) Write an equation for the reaction that forms the slag. (1 mark)

4. A monomer has the following structure.

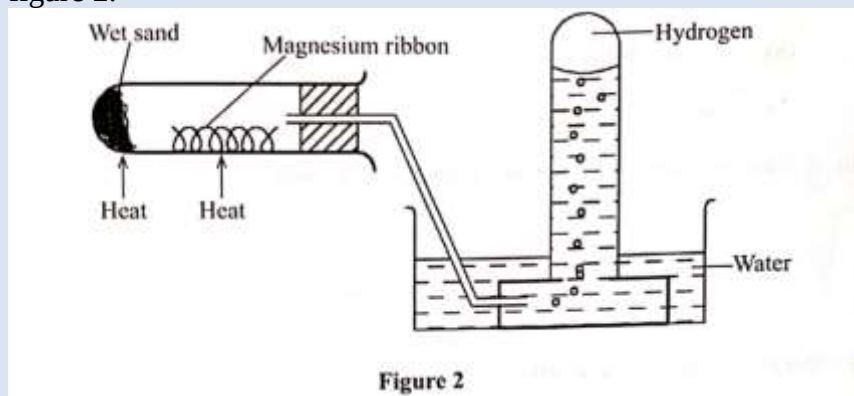


(a) Draw the structure of its polymer that contains three monomers.(1 mark)

(b) A sample of the polymer formed from the monomer has a molecular mass of 4992.

Determine the number of monomers that formed the polymer (C= 12; H= 1.0). (2 marks)

5. Hydrogen gas can be prepared by passing steam over heated magnesium ribbon as shown in the figure 2.



(a) Write an equation for the reaction that produces hydrogen gas. (1 mark)

(b) Explain why the delivery tube must be removed from beneath the water before heating is stopped. (1 mark)

(c) Explain why sodium metal is not suitable for this experiment. (1 mark)

6. A farmer intended to plant cabbages in his farm. He first tested the pH of the soil and found it to be 3.0.

If cabbages do well in alkaline soils, explain the advice that would be given to the farmer in order to realise a high yield.(2 marks)

7. A solution contains 40.3g of substance XOH per litre .250.0cm³ of this solution required 30.0cm³ of 0.3M sulphuric(VI)acid for complete neutralisation. (a) Calculate the number of moles of XOH that reacted. (½ mark)

(b) Determine the relative atomic mass of X. (1½ mark)

8. Table 1 shows the properties of two chlorides, D and E.

Table 1

Chloride (liquid)	Melting Points(°C)	Electrical Conductivity
D	1074	Good
E	203	Poor

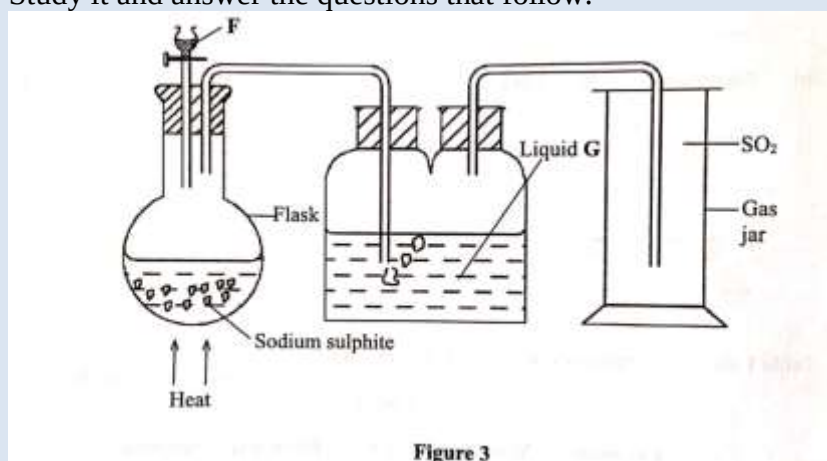
(a) State the type of bond present in:

(i) D.....(1 mark)

(ii) E.....(1 mark)

(b) Explain in terms of structure and bonding, the difference in electrical activity of the chlorides D and E. (1 mark)

9. Sulphur(IV) oxide is prepared in the laboratory using the set-up in Figure 3. Study it and answer the questions that follow.

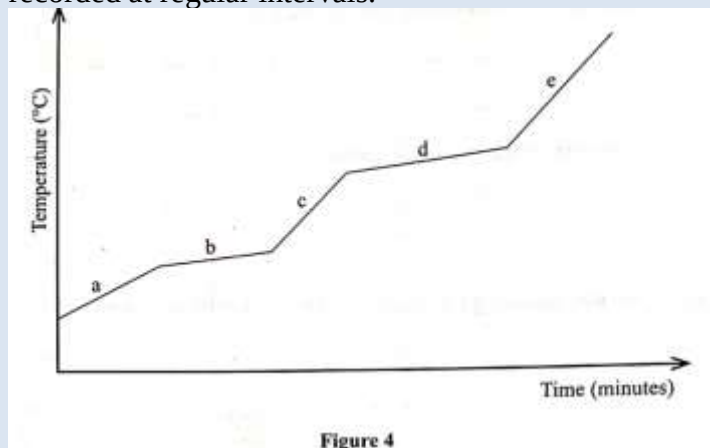


(a) Identify substance F. (1 mark)

(b) Write an equation for the reaction that takes place in the flask. (1 mark)

(c) State the purpose of liquid G. (1 mark)

10. The graph in Figure 4 was obtained when a certain substance was heated and its temperature recorded at regular intervals.



- (a) State the purity of the substance. (1 mark)
- (b) Explain the answer in (a). (1 mark)

11. Ethene is prepared in the laboratory by dehydration of ethanol.

- (a) Name a suitable dehydrating agent used in this process. (1 mark)
- (b) State the condition necessary for the reaction to occur. (1 mark)
- (C) Write an equation for the dehydration process. (1 mark)

112. A boiling tube filled with chlorine was inverted in a trough containing the same solution and the set-up left in sunlight for about 2 hours.

- (a) State the observation made in the boiling tube (1 mark)
- (b) Explain the observation made in (a) (1 mark)
- (c) Write an equation for the reaction that occurred in the boiling tube (1 mark)

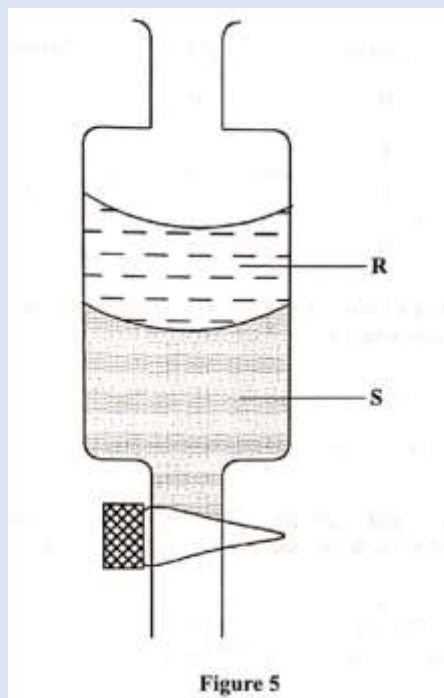
13. 5 g of calcium carbonate was strongly heated to a constant mass.

Calculate the mass of the solid residue formed (Ca = 40.0; C = 12.0; O = 16.0). (2 marks)

14. During laboratory preparation of oxygen, manganese(IV) oxide is added to reagent 11.

- (a) Name reagent H. (1 mark)
- (b) State the role of manganese(IV) oxide in this experiment. (1 mark)
- (c) Write the equation for the reaction that takes place. (1 mark)

15. Figure 5 shows an apparatus used to separate a mixture of water and hexene.



- (a) Name the apparatus in Figure 5. (1 mark)
- (b) State the principle by which the mixture of the two liquids is separated (1 mark)
- (c) Identify the liquids, R and S if the density of hexene is 0.66 g/cm^3 .
- (i) R (½ mark)
- (ii) S (½ mark)

16.(a) Complete the following table.(2 mark)

Solution	pH	Nature of solution
H	1.0	
I		Neutral
J		Weak acid
K	13.0	

(b) Explain why a solution of ammonia in methylbenzene has no effects on red litmus paper while in aqueous ammonia red litmus paper turns blue. (1 mark)

17. The heat of solution and hydration energy of potassium chloride is -17.2 kJ and -689 kJ respectively.

Calculate the lattice energy of potassium chloride. (2 marks)

18. Use the information in Table 2 to answer the questions that follow.

Bond Bond energy(KJ mol^{-1})

C-H	412
Cl-Cl	242
C-Cl	338
H-Cl	431

19. Given that the E^θ of $\text{Cu(s)}/\text{Cu}^{2+}(\text{aq})$ is $+0.34\text{V}$ and that of $\text{Zn(s)}/\text{Zn}^{2+}(\text{aq})$ is -0.76V , draw a labelled diagram of zinc and copper electrochemical cell. (3 marks)

20. During laboratory preparation of carbon(IV) oxide gas, substance L in a conical flask.

(a) Identify substance L. (1 mark)

(b) Write an equation that produces carbon(IV) oxide. (1 mark)

(c) State the observations made when the gas produced was bubbled through calcium hydroxide solution for a long time. (1 mark)

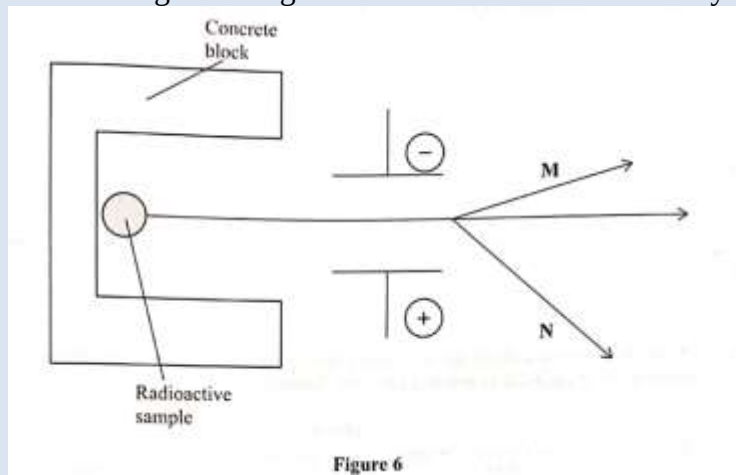
21. Study the information in Table 3 and use it to answer the questions that follow. (1 mark)

Elements	Na	Mg	Al	Si	P	S	Cl
Atomic Numbers	11	12	13	14	15	16	17
Atomic radii(nm)	0.157	0.136	0.125	0.117	0.110	0.104	0.099

(a) Explain the trend in atomic radii from sodium to chlorine. (1 mark)

(b) Explain how the chloride of aluminium differs from those of other metals in the period. (2 marks)

22. The diagram in figure 6 shows radiations emitted by a radioactive sample.



(a) Identify radiations:

(i) M (1 mark)

(ii) N (1 mark)

(b) Explain what would happen when a sheet of paper is placed in the path of the two radiations. (1 mark)

$^{16}_8\text{X}$ and $^{18}_8\text{X}$ are isotopes of element X.

They occur naturally in the ratio of 9:1 respectively.

Calculate the relative atomic mass of element X. (2 marks)

24. Starting with copper turnings, describe how a sample of copper (II) Sulphate crystals can be prepared in the laboratory. (2 marks)

25. Chemical tests were carried out on separate samples of water drawn from the same source. The observations made were recorded as shown in Table 4.

Test	Observation
(i) Addition of aqueous calcium chloride	No white precipitate
(ii) Addition of dilute sulphuric(VI) acid	No effervescence, colourless solution
(iii) Addition of a few drops of acidified barium nitrate	No white precipitate
(iv) Addition of aqueous ammonia	White precipitate dissolves

State the inferences made in reactions:

(i)..... (1 mark)

(ii)..... (1 mark)

(iii).....(1 mark)

26. 140 cm³ of nitrogen gas diffuses through a membrane in 70 seconds. How long will it take 200 cm³ of carbon(IV) oxide gas to diffuse through the same membrane under the same conditions of temperature and pressure (3 marks)

27. When burning magnesium ribbon is introduced into a gas jar full of nitrogen, it continues to burn producing a greenish yellow powder.

(a) Write an equation for the reaction between nitrogen and magnesium. (1 mark)

(b) Explain why magnesium continues to burn in nitrogen but sulphur does not. (2 marks)

(c) State one use of nitrogen. (1 mark)

28. Draw in the space provided a labelled diagram of the set-up of the apparatus that can be used to electrolyse molten lead(II) bromide. (3 marks)

29. Name an appropriate apparatus that is used to prepare standard solutions in the laboratory. (1 mark)

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